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Stochastic Processes I

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Lecture Notes

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1 Review of Elementary Probability

Overview. We establish the measure-theoretic probability conventions, distributional identities, and convergence results used throughout the notes. The review covers conditional probability, independence, random variables, expectations, joint laws, moment generating functions, and modes of convergence.

1.1 Probability Spaces and Measurability

Definition 1.1 (Probability space)**.** A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the sample space, $\mathcal{F}$ is a σ-algebra of subsets of Ω, and $\mathbb{P} : \mathcal{F} \to [0,1]$ is countably additive with $\mathbb{P}(\Omega) = 1$. Specifically, $\mathcal{F}$ contains Ω, is closed under complementation, and is closed under countable unions. A *random variable* is a measurable mapping $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

The distinction between a sample space and an event space is essential: not every subset of an arbitrary uncountable sample space need be measurable. In finite or countable elementary models, taking $\mathcal{F} = 2^{\Omega}$ is usually sufficient.

Proposition 1.1 (Continuity of probability measures)**.** *For events $A_n \in \mathcal{F}$, if $A_n \uparrow A$, then $\mathbb{P}(A_n) \uparrow \mathbb{P}(A)$. If $A_n \downarrow A$, then $\mathbb{P}(A_n) \downarrow \mathbb{P}(A)$.*

Proof. For the increasing case define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The B_n are disjoint and $A = \bigcup_{n \geq 1} B_n$. Hence, by countable additivity,

$$\mathbb{P}(A) = \sum_{k=1}^{\infty} \mathbb{P}(B_k) = \lim_{n \to \infty} \sum_{k=1}^{n} \mathbb{P}(B_k) = \lim_{n \to \infty} \mathbb{P}(A_n).$$

For decreasing events, $A_n^c \uparrow A^c$ and the first assertion yields

$$\mathbb{P}(A_n) = 1 - \mathbb{P}(A_n^c) \longrightarrow 1 - \mathbb{P}(A^c) = \mathbb{P}(A).$$

□

Definition 1.2 (Probability model)**.** A probability model consists of 3 essential components: a *sample space*, a collection of events, and a *probability function (measure).*

- **Sample Space:** For a random experiment in which all possible outcomes are known, the set of all possible outcomes is called the sample space (denoted by Ω).
- **Event:** An event is a measurable subset $A \in \mathcal{F}$ of the sample space Ω.
- **Probability Function:** For each event A of Ω, $P(A)$ is defined as the *probability of event* A, satisfying the following elementary probability conditions (with countable additivity understood):

(i) $0 \leq P(A) \leq 1$,

(ii) $P(\Omega) = 1$, or equivalently, $P(\varnothing) = 0$, where $\varnothing$ is the *null event*,

(iii) For $n \in \mathbb{Z}^+$ (in fact, $n = \infty$ as well),

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i)$$

if the sequence of events $\{A_i\}_{i=1}^n$ is *mutually exclusive* (i.e., $A_i \cap A_j = \varnothing \; \forall i \neq j$).

As a result of conditions (ii) and (iii), and noting that A^c is the complement of A, it follows that

$$1 = P(\Omega) = P(A \cup A^c) = P(A) + P(A^c) \implies P(A^c) = 1 - P(A).$$

1.2 Conditional Probability

Definition 1.3. The *conditional probability* of event A given event B occurs is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)},$$

provided that $P(B) > 0$.

Remarks.

(1) When $B = \Omega$, $P(A|\Omega) = P(A \cap \Omega)/P(\Omega) = P(A)/1 = P(A)$, as one would expect.

(2) Rewriting the above formula,

$$P(A \cap B) = P(A|B)P(B),$$

which is often referred to as the basic "multiplication rule". For a sequence of events $\{A_i\}_{i=1}^n$, the generalized multiplication rule is given by

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2|A_1) \cdots P(A_n|A_1 \cap A_2 \cap \cdots \cap A_{n-1}).$$

Example 1.1.

Suppose that we roll a fair six-sided die once (i.e., $\Omega = \{1, 2, 3, 4, 5, 6\}$). Let A denote the event of rolling a number less than 4 (i.e., $A = \{1, 2, 3\}$), and let B denote the event of rolling an odd number (i.e., $B = \{1, 3, 5\}$). Given that the roll is odd, what is the probability that the number rolled is less than 4?

Solution.

$$\Omega = \{1, \ldots, 6\}, \quad A = \{1, 2, 3\}, \quad B = \{1, 3, 5\}.$$

$$P(A) = P(B) = \frac{3}{6} = \frac{1}{2} \quad \text{and moreover,}$$

$$P(A \cap B) = P(\{1, 2, 3\} \cap \{1, 3, 5\}) = P(\{1, 3\}) = \frac{2}{6} = \frac{1}{3}.$$

We use:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$
$$\therefore \quad P(A|B) = \frac{1/3}{1/2} = \frac{2}{3}.$$

1.3 Independence of Events

Definition 1.4. Two events A and B are *independent* if and only if (iff)

$$P(A \cap B) = P(A)P(B).$$

In general, if an experiment consists of a sequence of independent trials, and $A_1, A_2, \ldots, A_n$ are events such that A_i depends only on the ith trial, then $A_1, A_2, \ldots, A_n$ are independent events and

$$P\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} P(A_i).$$

1.4 Law of Total Probability

For $n \in \mathbb{Z}^+$ (and even $n = \infty$), suppose that

$$\Omega = \bigcup_{i=1}^{n} B_i,$$

where the sequence of events $\{B_i\}_{i=1}^{n}$ is *mutually exclusive*. Then,

$$\begin{aligned} P(A) &= P(A \cap \Omega) \\ &= P\left(A \cap \bigcup_{i=1}^{n} B_i\right) \\ &= P\left(\bigcup_{i=1}^{n} (A \cap B_i)\right) \\ &= \sum_{i=1}^{n} P(A \cap B_i) \\ &= \sum_{i=1}^{n} P(A|B_i)P(B_i), \end{aligned}$$

where the second last equality follows from the fact that the sequence of events $\{A \cap B_i\}_{i=1}^{n}$ is also mutually exclusive.

Proposition 1.2 (Bayes' Formula)**.** *Under the same assumptions as given directly above,*

$$P(B_j|A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^{n} P(A|B_i)P(B_i)}.$$

1.5 Random Variables

Definition 1.5. A *random variable* (rv) X is a real-valued function which maps a sample space Ω onto a state space $S \subseteq \mathbb{R}$ (i.e., $X : \Omega \to S$).

- **Discrete type:** S consists of a finite or countable number of possible values. Important functions include

$$\begin{aligned} p(a) &= P(X = a) \qquad \text{(pmf)}, \\ F(a) &= P(X \leq a) = \sum_{x \leq a} p(x) \qquad \text{(cdf)}, \\ \bar{F}(a) &= P(X > a) = 1 - F(a) \qquad \text{(tpf)}, \end{aligned}$$

  where pmf stands for *probability mass function*, cdf stands for *cumulative distribution function*, and tpf stands for *tail probability function.*

Remark. If X takes on values in the set $S = \{a_1, a_2, a_3, \ldots\}$ where $a_1 < a_2 < a_3 < \cdots$ such that $p(a_i) > 0 \ \forall i$, then we can recover the pmf from knowledge of the cdf via

$$\begin{aligned} p(a_1) &= F(a_1), \\ p(a_i) &= F(a_i) - F(a_{i-1}), \quad i = 2, 3, 4, \ldots \end{aligned}$$

Special Discrete Distributions.

1. **Bernoulli:** If we consider a *Bernoulli trial*, which is a random trial with probability p of being a "success" (denoted by 1) and probability $1 - p$ of being a "failure" (denoted by 0), then X is Bernoulli (i.e., $X \sim \text{BERN}(p)$) with pmf

$$p(x) = p^x(1-p)^{1-x}, \quad x = 0, 1.$$

2. **Binomial:** If X denotes the number of successes in $n \in \mathbb{Z}^+$ independent Bernoulli trials, each with probability p of being a success, then X is Binomial (i.e., $X \sim \text{BIN}(n, p)$) with pmf

$$p(x) = \binom{n}{x} p^x(1-p)^{n-x}, \quad x = 0, 1, \ldots, n,$$

   where

$$\binom{n}{x} = \frac{n!}{(n-x)!x!} = \frac{n(n-1)\cdots(n-x+1)}{x!}$$

   is the number of distinct groups of x objects chosen from a set of n objects.

Remarks.

(1) A $\text{BIN}(1, p)$ distribution simplifies to become the $\text{BERN}(p)$ distribution.
(2) The binomial pmf is even defined for $n = 0$, in which case $p(x) = 1$ for $x = 0$. Such a distribution is said to be degenerate at 0.
(3) Note that $\binom{n}{x} = 0$ if $n, x \in \mathbb{N}$ with $n < x$.

3. **Negative Binomial:** If X denotes the number of Bernoulli trials (each with success probability p) required to observe $k \in \mathbb{Z}^+$ successes, then X is Negative Binomial (i.e., $X \sim \text{NB}_t(k, p)$) with pmf

$$p(x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \quad x = k, k+1, k+2, \ldots$$

Remarks.

(1) In the above pmf, $\binom{x-1}{k-1}$ appears rather than $\binom{x}{k}$ since the final trial must always be a success.
(2) Sometimes, a negative binomial distribution is alternatively defined as the number of failures observed to achieve k successes. If Y denotes such a rv and $X \sim \text{NB}_t(k, p)$, then we clearly have the relationship $X = Y + k$, which immediately leads to the following pmf for Y:

$$p_Y(y) = P(Y = y) = P(X = y + k) = \binom{y+k-1}{k-1} p^k (1-p)^y, \quad y = 0, 1, 2, \ldots$$

To refer to this negative binomial distribution, we will write $Y \sim \text{NB}_f(k, p)$.

4. **Geometric:** If $X \sim \text{NB}_t(1, p)$, then X is *Geometric* (i.e., $X \sim \text{GEO}_t(p)$) with pmf

$$p(x) = p(1-p)^{x-1}, \quad x = 1, 2, 3, \ldots$$

In other words, the geometric distribution models the number of Bernoulli trials required to observe the first success.

Remark. In a similar fashion, if $X \sim \text{NB}_f(1, p)$, then we obtain an alternative geometric distribution (denoted by $X \sim \text{GEO}_f(p)$) which models the number of failures observed prior to the first success.

5. **Discrete Uniform:** If X is equally likely to take on values in the (finite) set $\{a, a+1, \ldots, b\}$ where $a, b \in \mathbb{Z}$ with $a \leq b$, then X is Discrete Uniform (i.e., $X \sim \text{DU}(a, b)$) with pmf

$$p(x) = \frac{1}{b-a+1}, \quad x = a, a+1, \ldots, b.$$

6. **Hypergeometric:** If X denotes the number of success objects in n draws without replacement from a finite population of size N containing exactly r success objects, then X is Hypergeometric (i.e., $X \sim \text{HG}(N, r, n)$) with pmf

$$p(x) = \frac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}, \quad x = \max\{0, n-N+r\}, \ldots, \min\{n, r\}.$$

7. **Poisson:** A rv X is Poisson (i.e., $X \sim \text{POI}(\lambda)$) with parameter $\lambda > 0$ if its pmf is of the form

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

Remark. The pmf is even defined for $\lambda = 0$ (if we use the standard convention that $0^0 = 1$), in which case $p(x) = 1$ for $x = 0$ (i.e., X is degenerate at 0).

Example 1.2.

Show that when n is large and p is small, the $\text{BIN}(n, p)$ distribution may be approximated by a $\text{POI}(\lambda)$ distribution where $\lambda = np$.

Solution. Recall the setup and let $X \sim \text{BIN}(n, p)$ with $p = \lambda/n$.

$$\begin{aligned} P(X = x) &= \binom{n}{x} p^x (1-p)^{n-x} \\ &= \frac{n(n-1)\cdots(n-x+1)}{x!} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \underbrace{\frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-x+1}{n}}_{x \text{ terms}} \cdot \frac{\lambda^x}{x!} \cdot \underbrace{\left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}}_{\to e^{-\lambda} \cdot 1} \end{aligned}$$

Let $n \to \infty$,

$$\approx \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, \ldots$$

1.6 Continuous Random Variables

A random variable X takes on a continuum of possible values (which is uncountable) with cdf

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(y)\, dy,$$

where $f(x)$ denotes the *probability density function* (pdf) of X, which is a non-negative real-valued function that satisfies

$$P(X \in B) = \int_{x \in B} f(x)\, dx,$$

where B is a set of real numbers (e.g., an interval).

Remarks.

(1) If $F(x)$ (or the tpf $\bar{F}(x) = 1 - F(x)$) is known, we can recover the pdf using the relation

$$f(x) = \frac{d}{dx}F(x) = F'(x) = -\bar{F}'(x),$$

which holds by the Fundamental Theorem of Calculus.

(2) When working with pdfs in general, it is usually not necessary to be precise about specifying whether or not a range of numbers includes the endpoints. This is quite different from the situation we encounter with discrete rvs. Throughout these notes, however, we will adopt the convention of not including the endpoints when specifying the range of values for pdfs.

1.7 Special Continuous Distributions

1. **Uniform:** A rv X is Uniform on the real interval (a, b) (i.e., $X \sim U(a, b)$) if it has pdf

$$f(x) = \frac{1}{b-a}, \quad a < x < b,$$

where $a, b \in \mathbb{R}$ with $a < b$.
Remark. The choice of name is because X takes on values in (a, b) with all subintervals of a fixed length being equally likely.

2. **Beta:** A rv X is Beta with parameters $m \in \mathbb{Z}^+$ and $n \in \mathbb{Z}^+$ (i.e., $X \sim \text{Beta}(m, n)$) if it has pdf

$$f(x) = \frac{(m+n-1)!}{(m-1)!(n-1)!}x^{m-1}(1-x)^{n-1}, \quad 0 < x < 1.$$

Remark. A Beta$(1, 1)$ distribution simplifies to become the $U(0, 1)$ distribution.

3. **Erlang:** A rv X is Erlang with parameters $n \in \mathbb{Z}^+$ and $\lambda > 0$ (i.e., $X \sim \text{Erlang}(n, \lambda)$) if it has pdf

$$f(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!}, \quad x > 0.$$

Remark. The Erlang(n, λ) distribution is actually a special case of the more general Gamma distribution in which n is extended to be any positive real number.

4. **Exponential:** A rv X is Exponential with parameter $\lambda > 0$ (i.e., $X \sim \text{EXP}(\lambda)$) if it has pdf

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0.$$

Remark. An Erlang$(1, \lambda)$ distribution simplifies to become the EXP(λ) distribution.

1.8 Expectation

If $g(\cdot)$ is an arbitrary real-valued function, then

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x)p(x), & \text{if } X \text{ is a discrete rv,} \\ \int_{-\infty}^{\infty} g(x)f(x)\,dx, & \text{if } X \text{ is a continuous rv.} \end{cases}$$

1.9 Special choices of $g(\cdot)$

1. $g(X) = X^n$, $n \in \mathbb{N} \implies \mathbb{E}[g(X)] = \mathbb{E}[X^n]$ is the nth moment of X. In general, moments serve to describe the shape of a distribution. If $n = 0$, then $\mathbb{E}[X^0] = 1$. If $n = 1$, then $\mathbb{E}[X] = \mu_X$ is the mean of X.
2. $g(X) = (X - \mathbb{E}[X])^2 \implies \mathbb{E}[g(X)] = \mathbb{E}[(X - \mathbb{E}[X])^2]$ is the variance of X. Note that

   $$\text{Var}(X) = \sigma_X^2 = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2,$$

   or equivalently

   $$\sigma_X^2 = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2.$$

   Related to this quantity, the standard deviation of X is $\sqrt{\text{Var}(X)} = \sigma_X$.
   The table on the following page summarizes the formulas of the means and variances of the special discrete/continuous distributions introduced earlier, along with their corresponding pmf/pdf formulas.
3. $g(X) = aX + b$, $a, b \in \mathbb{R}$ (i.e., $g(X)$ is a linear function of X). Note that

   $$\begin{aligned} \mu_{aX+b} &= \mathbb{E}[aX + b] = a\mu_X + b, \\ \sigma_{aX+b}^2 &= \text{Var}(aX + b) = a^2\sigma_X^2, \\ \sigma_{aX+b} &= \sqrt{\text{Var}(aX + b)} = |a|\sigma_X. \end{aligned}$$

4. $g(X) = e^{tX}$, $t \in \mathbb{R} \implies \mathbb{E}[g(X)] = \mathbb{E}[e^{tX}]$ is the *moment generating function* (mgf) of X. This quantity is a function of t and is denoted by

   $$\phi_X(t) = \mathbb{E}[e^{tX}].$$

   First of all, $\phi_X(0) = \mathbb{E}[e^{0X}] = \mathbb{E}[1] = 1$. Moreover, making use of the linearity property of the expected value operator, note that

   $$\begin{aligned} \phi_X(t) &= \mathbb{E}\left[\sum_{n=0}^{\infty} \frac{(tX)^n}{n!}\right] \\ &= \mathbb{E}\left[\frac{t^0X^0}{0!} + \frac{t^1X^1}{1!} + \frac{t^2X^2}{2!} + \cdots + \frac{t^nX^n}{n!} + \cdots\right] \\ &= \mathbb{E}[X^0]\frac{t^0}{0!} + \mathbb{E}[X^1]\frac{t^1}{1!} + \mathbb{E}[X^2]\frac{t^2}{2!} + \cdots + \mathbb{E}[X^n]\frac{t^n}{n!} + \cdots, \end{aligned}$$

implying that the nth moment of X is simply the coefficient of $t^n/n!$ in the above series expansion.

Remarks.

(1) Given the mgf of X, we can extract its nth moment via

$$\mathbb{E}[X^n] = \phi_X^{(n)}(0) = \frac{d^n}{dt^n}\phi_X(t)\bigg|_{t=0} = \lim_{t\to 0}\frac{d^n}{dt^n}\phi_X(t), \quad n \in \mathbb{N}.$$

Note that the 0th derivative of a function is simply the function itself.

(2) A mgf uniquely characterizes the probability distribution of a rv (i.e., there exists a one-to-one correspondence between the mgf and the pmf/pdf of a rv). In other words, if two rvs X and Y have the same mgf, then they must have the same probability distribution (which we denote by $X \sim Y$). Thus, by finding the mgf of a rv, one has indeed determined its probability distribution.

Example 1.3.

Suppose that $X \sim \text{BIN}(n, p)$. Find the mgf of X and use it to find $\mathbb{E}[X]$ and $\text{Var}(X)$.

Solution. Recall the binomial series formula $(a+b)^m = \sum_{x=0}^{m}\binom{m}{x}a^x b^{m-x}$, $a, b \in \mathbb{R}$, $m \in \mathbb{N}$. Now, we want $\phi_X(t) = \mathbb{E}[e^{tX}]$.

$$\begin{aligned}
\phi_X(t) = \mathbb{E}[e^{tX}] &= \sum_{x=0}^{n} e^{tx}\binom{n}{x}p^x(1-p)^{n-x} \\
&= \sum_{x=0}^{n}\binom{n}{x}(pe^t)^x(1-p)^{n-x} \\
&= (pe^t + 1 - p)^n \in \mathbb{R} \quad \text{unique to BIN.} \\
\phi_X'(t) &= n(pe^t + 1 - p)^{n-1}pe^t \\
\phi_X''(t) &= n(n-1)(pe^t + 1 - p)^{n-2}(pe^t)^2 + npe^t(pe^t + 1 - p)^{n-1} \\
\mathbb{E}[X] &= \phi_X'(0) = np \\
\mathbb{E}[X^2] &= \phi_X''(0) = n(n-1)p^2 + np
\end{aligned}$$

$$\begin{aligned}
\text{Var}(X) &= \phi_X''(0) - \phi_X'(0)^2 \\
&= n(n-1)p^2 + np - n^2p^2 \\
&= np(1-p).
\end{aligned}$$

1.10 Joint Distributions

The following results are presented for the bivariate case mostly, but these ideas extend naturally to an arbitrary number of rvs.

Definition 1.6. The joint cdf of X and Y is

$$F(a,b) = P(X \leq a, Y \leq b) = P(\{X \leq a\} \cap \{Y \leq b\}), \quad a, b \in \mathbb{R}.$$

Remark. If the joint cdf is known, then we can recover their marginal counterparts as follows:

$$F_X(a) = P(X \leq a) = F(a, \infty) = \lim_{b \to \infty} F(a,b),$$
$$F_Y(b) = P(Y \leq b) = F(\infty, b) = \lim_{a \to \infty} F(a,b).$$

1.11 Jointly Discrete Case

Joint pmf.

$$p(x,y) = P(X = x, Y = y)$$

Marginals.

$$p_X(x) = P(X = x) = \sum_y p(x,y)$$
$$p_Y(y) = P(Y = y) = \sum_x p(x,y)$$

1.12 Multinomial Distribution

Consider an experiment which is repeated $n \in \mathbb{Z}^+$ times, with one of $k \geq 2$ distinct outcomes possible each time. Let $p_1, p_2, \ldots, p_k$ denote the probabilities of the k types of outcomes (with $\sum_{i=1}^k p_i = 1$). If X_i, $i = 1, 2, \ldots, k$, counts the number of type-i outcomes to occur, then $(X_1, X_2, \ldots, X_k)$ is Multinomial (i.e., $(X_1, \ldots, X_k) \sim \text{MN}(n, p_1, \ldots, p_k)$) with joint pmf

$$p(x_1, x_2, \ldots, x_k) = \frac{n!}{x_1! x_2! \cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}, \quad x_i = 0, 1, \ldots, n \ \forall i, \ \sum_{i=1}^k x_i = n.$$

Remark. A $\text{MN}(n, p_1, 1 - p_1)$ distribution simplifies to become the $\text{BIN}(n, p_1)$ distribution.

1.13 Jointly Continuous Case

Joint pdf.

The joint pdf $f(x, y)$ is a non-negative real-valued function which enables one to calculate probabilities of the form

$$P(X \in A, Y \in B) = \int_B \int_A f(x, y)\, dx\, dy = \int_A \int_B f(x, y)\, dy\, dx,$$

where A and B are sets of real numbers (e.g., intervals). As a result,

$$F(a, b) = \int_{-\infty}^{b} \int_{-\infty}^{a} f(x, y)\, dx\, dy = \int_{-\infty}^{a} \int_{-\infty}^{b} f(x, y)\, dy\, dx.$$

Marginals.

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$
$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

Important Relationship.

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

Transformations.

Let (X, Y) be jointly continuous with joint pdf $f(x, y)$ and region of support $S(X, Y)$. Suppose that the rvs V and W are given by $V = b_1(X, Y)$ and $W = b_2(X, Y)$, where the functions $v = b_1(x, y)$ and $w = b_2(x, y)$ define a one-to-one transformation that maps the set $S(X, Y)$ onto the set $S(V, W)$. If x and y are expressed in terms of v and w (i.e., $x = h_1(v, w)$ and $y = h_2(v, w)$), then

$$g(v, w) = \begin{cases} f(h_1(v, w), h_2(v, w))|J|, & \text{if } (v, w) \in S(V, W), \\ 0, & \text{elsewhere,} \end{cases}$$

where J is the Jacobian of the transformation given by

$$J = \frac{\partial x}{\partial v}\frac{\partial y}{\partial w} - \frac{\partial x}{\partial w}\frac{\partial y}{\partial v}.$$

Expectation: If $g(\cdot, \cdot)$ denotes an arbitrary real-valued function, then

$$\mathbb{E}[g(X, Y)] = \begin{cases} \sum_x \sum_y g(x, y)p(x, y), & \text{if } X \text{ and } Y \text{ are jointly discrete,} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y)f(x, y)\, dx\, dy, & \text{if } X \text{ and } Y \text{ are jointly continuous.} \end{cases}$$

Remark. The order of summation/integration is irrelevant and can be interchanged.

1.14 Special choices of $g(\cdot, \cdot)$

1. $g(X,Y) = (X - \mathbb{E}[X])(Y - \mathbb{E}[Y]) \implies \mathbb{E}[g(X,Y)] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$ is the covariance of X and Y. Note that

$$\operatorname{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

   and $\operatorname{Cov}(X,X) = \operatorname{Var}(X)$.
2. $g(X,Y) = aX + bY$, $a, b \in \mathbb{R}$ (i.e., $g(X,Y)$ is a linear combination of X and Y). Note that

$$\begin{aligned}\mathbb{E}[aX + bY] &= a\mathbb{E}[X] + b\mathbb{E}[Y],\\ \operatorname{Var}(aX + bY) &= a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X,Y).\end{aligned}$$

3. $g(X,Y) = e^{sX+tY}$, $s, t \in \mathbb{R} \implies \mathbb{E}[g(X,Y)] = \mathbb{E}[e^{sX+tY}]$ is the joint mgf of X and Y. A joint mgf (denoted by $\phi(s,t)$) also uniquely characterizes a joint probability distribution and can be used to calculate joint moments of X and Y via the formula

$$\mathbb{E}[X^m Y^n] = \phi^{(m,n)}(0,0) = \frac{\partial^{m+n}}{\partial s^m \partial t^n}\phi(s,t)\bigg|_{s=0,t=0} = \lim_{s\to 0, t\to 0} \frac{\partial^{m+n}}{\partial s^m \partial t^n}\phi(s,t), \quad m, n \in \mathbb{N}.$$

1.15 Independence of Random Variables

Definition 1.7. X and Y are independent rvs if

$$F(a,b) = P(X \le a, Y \le b) = P(X \le a)P(Y \le b) = F_X(a)F_Y(b) \quad \forall a, b \in \mathbb{R}.$$

Equivalently, independence exists iff $p(x,y) = p_X(x)p_Y(y)$ (in the jointly discrete case) or $f(x,y) = f_X(x)f_Y(y)$ (in the jointly continuous case) $\forall x, y \in \mathbb{R}$.

Proposition 1.3. *For arbitrary real-valued functions $g(\cdot)$ and $h(\cdot)$, if X and Y are independent, then*

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)].$$

Remark. As a consequence of this property, $\operatorname{Cov}(X,Y) = 0$ if X and Y are independent, implying that $\operatorname{Var}(aX + bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y)$. However, if $\operatorname{Cov}(X,Y) = 0$, we cannot conclude that X and Y are independent (we can only say that X and Y are uncorrelated).

Example 1.4.

Suppose that X and Y have joint pmf (and corresponding marginals) of the form

$p(x,y)$	$y=0$	$y=1$	$p_X(x)$
$x=0$	0.2	0	0.2
$x=1$	0	0.6	0.6
$x=2$	0.2	0	0.2
$p_Y(y)$	0.4	0.6	1

Show that $\mathrm{Cov}(X,Y)=0$, but X and Y are not independent.

Solution.

$$\begin{aligned}
\mathrm{Cov}(X,Y) &= \mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y] \\
\mathbb{E}[XY] &= \sum_x\sum_y xy\,p(x,y) \\
&= 0\cdot 0\cdot 0.2+0\cdot 1\cdot 0+1\cdot 0\cdot 0+1\cdot 1\cdot 0.6+2\cdot 0\cdot 0.2+2\cdot 1\cdot 0 \\
\mathbb{E}[X] &= \sum_x xp_X(x)=0\cdot 0.2+1\cdot 0.6+2\cdot 0.2=1 \\
\mathbb{E}[Y] &= \sum_y yp_Y(y)=0\cdot 0.4+1\cdot 0.6=0.6 \\
\mathrm{Cov}(X,Y) &= \mathbb{E}[XY]-\mathbb{E}[X]\mathbb{E}[Y]=0.6-1\cdot 0.6=0 \\
p(2,0) &= 0.2\neq p_X(2)p_Y(0)=0.08 \\
\therefore\ \text{Although } \mathrm{Cov}(X,Y) &= 0,\ p(2,0)\neq p_X(2)p_Y(0).
\end{aligned}$$

Hence $X \not\perp\!\!\!\perp Y$.

Theorem 1.1. *If $X_1, X_2, \ldots, X_n$ are independent rvs where $\phi_{X_i}(t)$ is the mgf of X_i, $i=1,2,\ldots,n$, then $T=\sum_{i=1}^n X_i$ has mgf*

$$\phi_T(t)=\prod_{i=1}^n \phi_{X_i}(t).$$

Proof. By definition,

$$\phi_T(t)=\mathbb{E}[e^{t\sum_{i=1}^n X_i}]=\mathbb{E}[e^{tX_1+\cdots+tX_n}]=\mathbb{E}[e^{tX_1}\cdots e^{tX_n}].$$

Note that $e^{t\sum_{i=1}^n X_i}=\prod_{i=1}^n e^{tX_i}$, so we have

$$\phi_T(t)=\mathbb{E}\left[\prod_{i=1}^n e^{tX_i}\right].$$

Since $X_1, X_2, \ldots, X_n$ and the rvs $e^{tX_1}, \ldots, e^{tX_n}$ are also independent,

$$\mathbb{E}\left[\prod_{i=1}^{n} e^{tX_i}\right] = \prod_{i=1}^{n} \mathbb{E}[e^{tX_i}] = \prod_{i=1}^{n} \phi_{X_i}(t).$$

□

Remark.

$$\mathbb{E}[e^{tX_1} \cdots e^{tX_n}] = \mathbb{E}[e^{tX_1}] \cdots \mathbb{E}[e^{tX_n}], \quad X_1 \perp X_2 \perp \cdots \perp X_n.$$

Remarks.

(1) Simply put, Theorem 1.1 states that the mgf of a sum of independent rvs is just the product of their individual mgfs.
(2) As a special case of the above result, note that $\phi_T(t) = \phi_{X_1}(t)^n$ if $X_1, X_2, \ldots, X_n$ is an independent and identically distributed (iid) sequence of rvs.

Example 1.5.

Let $X_1, X_2, \ldots, X_m$ be an independent sequence of rvs where $X_i \sim \text{BIN}(n_i, p)$, $i = 1, 2, \ldots, m$. Find the distribution of $T = \sum_{i=1}^{m} X_i$.

Solution. By theorem, mgf of T is the product of the mgfs of X_i's.

For each $X_i \sim \text{BIN}(n_i, p)$, the mgf is defined as:

$$\phi_{X_i}(t) = (pe^t + 1 - p)^{n_i}.$$

Thus,

$$\phi_T(t) = \prod_{i=1}^{m} \phi_{X_i}(t) = \prod_{i=1}^{m} (pe^t + 1 - p)^{n_i} = (pe^t + 1 - p)^{\sum_{i=1}^{m} n_i},$$

i.e. exactly the mgf of a $\text{BIN}(\sum_{i=1}^{m} n_i, p)$.

Remark. As a special case of the above example, if $X_1, X_2, \ldots, X_m$ are iid $\text{BERN}(p)$ rvs, then $T = \sum_{i=1}^{m} X_i \sim \text{BIN}(m, p)$.

1.16 Modes of Convergence

If X_n, $n \in \mathbb{Z}^+$, and X are rvs, then

1. $X_n \to X$ *in distribution* iff

$$\lim_{n\to\infty} P(X_n \leq x) = P(X \leq x) \quad \forall x \in \mathbb{R} \text{ at which } P(X \leq x) \text{ is continuous.}$$

2. $X_n \to X$ *in probability*, iff $\forall \varepsilon > 0$,

$$\lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0.$$

3. $X_n \to X$ *almost surely (a.s.)* iff

$$P\Big(\lim_{n\to\infty} X_n = X\Big) = 1.$$

Remarks.

(1) In probability theory, an event is said to happen a.s. if it happens with probability 1.
(2) The following implications hold true in general:

$$X_n \to X \text{ a.s. } \implies X_n \to X \text{ in probability } \implies X_n \to X \text{ in distribution.}$$

Theorem 1.2 (Strong Law of Large Numbers (SLLN))**.** *If $\{X_i\}_{i=1}^{\infty}$ is an iid sequence of rvs with common mean μ and $\mathbb{E}[|X_1|] < \infty$, then*

$$\bar{X}_n = \frac{X_1 + X_2 + \cdots + X_n}{n} \to \mu \quad \textit{a.s. as } n \to \infty.$$

Remark. The SLLN is one of the most important results in probability and statistics, indicating that the sample mean will, with probability 1, converge to the true mean of the underlying distribution as the sample size approaches infinity. In other words, if the same experiment or study is repeated independently a large number of times, the average of the results of the trials must be close to the mean. The result gets closer to the mean as the number of trials is increased.

2 Conditional Distributions and Conditional Expectation

Overview. Conditioning produces distributions and expectations that depend on observed information. We develop the discrete, continuous, and mixed constructions, then derive the tower property, variance decomposition, and probability identities from first principles.

2.1 A Unified View of Conditioning

Definition 2.1 (Conditional expectation given a random variable)**.** If X is integrable and Y is a random variable, then $\mathbb{E}[X \mid Y]$ denotes $\mathbb{E}[X \mid \sigma(Y)]$: an integrable, $\sigma(Y)$-measurable random variable satisfying

$$\mathbb{E}[X\mathbf{1}_{\{Y\in B\}}] = \mathbb{E}[\mathbb{E}[X \mid Y]\mathbf{1}_{\{Y\in B\}}], \qquad B \in \mathcal{B}(\mathbb{R}).$$

It exists and is unique up to almost-sure equality. If $\mathbb{E}[X \mid Y] = g(Y)$ for a measurable g, the notation $\mathbb{E}[X \mid Y = y] = g(y)$ denotes a version of the conditional mean.

For discrete Y with $\mathbb{P}(Y = y) > 0$, the conditional law is defined by ordinary event conditioning. For continuous Y, the event $\{Y = y\}$ generally has probability zero; conditional densities, when they exist, provide a version of the regular conditional law instead.

Proposition 2.1 (Pull-out and tower identities)**.** *If X is integrable, $g(Y)$ is bounded and measurable, and the expressions are defined, then*

$$\mathbb{E}[g(Y)X \mid Y] = g(Y)\mathbb{E}[X \mid Y], \qquad \mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$$

Proof. The first claimed right-hand side is $\sigma(Y)$-measurable. For each $B \in \mathcal{B}(\mathbb{R})$, the defining identity for $\mathbb{E}[X \mid Y]$, first with simple g and then by bounded approximation, gives

$$\mathbb{E}[g(Y)X\mathbf{1}_{\{Y\in B\}}] = \mathbb{E}[g(Y)\mathbb{E}[X \mid Y]\mathbf{1}_{\{Y\in B\}}].$$

Uniqueness gives the pull-out identity. Taking $B = \mathbb{R}$ in the defining identity establishes the tower identity. □

2.2 Jointly Discrete Case

Definition 2.2 (Conditional pmf)**.** If X_1 and X_2 are both discrete rvs with joint pmf $p(x_1, x_2)$ and marginal pmfs $p_1(x_1)$ and $p_2(x_2)$, respectively, then the *conditional distribution* of X_1 given $X_2 = x_2$, denoted by $X_1|(X_2 = x_2)$, is defined via its *conditional pmf*

$$p_{1|2}(x_1|x_2) = P(X_1 = x_1|X_2 = x_2) = \frac{P(X_1 = x_1, X_2 = x_2)}{P(X_2 = x_2)} = \frac{p(x_1, x_2)}{p_2(x_2)},$$

provided that $p_2(x_2) > 0$. Similarly, the conditional distribution of $X_2|(X_1 = x_1)$ is defined via its conditional pmf

$$p_{2|1}(x_2|x_1) = P(X_2 = x_2|X_1 = x_1) = \frac{p(x_1, x_2)}{p_1(x_1)},$$

provided that $p_1(x_1) > 0$.

Remarks.

(1) If X_1 and X_2 are independent, then $p(x_1, x_2) = p_1(x_1)p_2(x_2)\ \forall x_1, x_2 \in \mathbb{R}$, and so

$$p_{1|2}(x_1|x_2) = p_1(x_1) \quad \text{and} \quad p_{2|1}(x_2|x_1) = p_2(x_2).$$

(2) These ideas extend beyond the simple bivariate case in a natural way. For example, suppose that X_1, X_2, and X_3 are discrete rvs. We can define the conditional distribution of (X_1, X_2) given $X_3 = x_3$ via its conditional pmf as follows:

$$p_{12|3}(x_1, x_2|x_3) = \frac{P(X_1 = x_1, X_2 = x_2, X_3 = x_3)}{P(X_3 = x_3)} = \frac{p(x_1, x_2, x_3)}{p_3(x_3)},$$

provided that $p_3(x_3) > 0$. Alternatively, we can define the conditional distribution of X_2 given $(X_1 = x_1, X_3 = x_3)$ via its conditional pmf given by

$$p_{2|13}(x_2|x_1, x_3) = \frac{p(x_1, x_2, x_3)}{p_{13}(x_1, x_3)},$$

provided that $p_{13}(x_1, x_3) > 0$, where $p_{13}(x_1, x_3)$ is the joint pmf of X_1 and X_3.

2.3 Conditional Expectation

Definition 2.3 (Conditional mean)**.** The *conditional mean* of $X_1|(X_2 = x_2)$ is

$$\mathbb{E}[X_1|X_2 = x_2] = \sum_{x_1} x_1\, p_{1|2}(x_1|x_2).$$

More generally, if $u(\cdot, \cdot)$, $h(\cdot)$, and $g(\cdot)$ are arbitrary real-valued functions, then

$$\mathbb{E}[u(X_1, X_2)|X_2 = x_2] = \sum_{x_1} u(x_1, x_2)p_{1|2}(x_1|x_2)$$

and

$$\mathbb{E}[g(X_1)h(X_2)|X_2 = x_2] = h(x_2)\mathbb{E}[g(X_1)|X_2 = x_2].$$

As an immediate consequence, if $a, b \in \mathbb{R}$, then we obtain

$$\mathbb{E}[ag(X_1) + bh(X_1)|X_2 = x_2] = a\mathbb{E}[g(X_1)|X_2 = x_2] + b\mathbb{E}[h(X_1)|X_2 = x_2].$$

Furthermore, we recall that $\mathbb{E}[X_1 + X_2] = \sum_{x_1} \sum_{x_2} (x_1 + x_2) p(x_1, x_2)$, then it correspondingly follows that

$$\begin{aligned}
\mathbb{E}[X_1 + X_2 | X_3 = x_3] &= \sum_{x_1} \sum_{x_2} (x_1 + x_2) p_{12|3}(x_1, x_2 | x_3) \\
&= \sum_{x_1} \sum_{x_2} x_1 \frac{p(x_1, x_2, x_3)}{p_3(x_3)} + \sum_{x_1} \sum_{x_2} x_2 \frac{p(x_1, x_2, x_3)}{p_3(x_3)} \\
&= \sum_{x_1} x_1 p_{13}(x_1 | x_3) + \sum_{x_2} x_2 p_{23}(x_2 | x_3) \\
&= \mathbb{E}[X_1 | X_3 = x_3] + \mathbb{E}[X_2 | X_3 = x_3].
\end{aligned}$$

In other words, the *conditional expected value operator* is also a *linear operator*. In fact, more generally, if $a_i \in \mathbb{R}$, $i = 1, 2, \ldots, n$, then the same essential approach can be used to show that

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i \,\middle|\, Y = y \right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i | Y = y].$$

2.4 Conditional Variance

Definition 2.4 (Conditional variance)**.** If we take $g(X_1) = (X_1 - \mathbb{E}[X_1 | X_2 = x_2])^2$, then

$$\mathbb{E}[g(X_1) | X_2 = x_2] = \mathbb{E}[(X_1 - \mathbb{E}[X_1 | X_2 = x_2])^2 | X_2 = x_2] = \mathrm{Var}(X_1 | X_2 = x_2)$$

is the *conditional variance* of $X_1 | (X_2 = x_2)$.

As with the calculation of variance, the following result provides an alternative (and oftentimes preferred) way to calculate $\mathrm{Var}(X_1 | X_2 = x_2)$.

Theorem 2.1.

$$\mathrm{Var}(X_1 | X_2 = x_2) = \mathbb{E}[X_1^2 | X_2 = x_2] - \mathbb{E}[X_1 | X_2 = x_2]^2.$$

Proof. Starting with the definition of conditional variance,

$$\begin{aligned}
\mathrm{Var}(X_1 | X_2 = x_2) &= \mathbb{E}[(X_1 - \mathbb{E}[X_1 | X_2 = x_2])^2 | X_2 = x_2] \\
&= \mathbb{E}[X_1^2 - 2X_1 \mathbb{E}[X_1 | X_2 = x_2] + \mathbb{E}[X_1 | X_2 = x_2]^2 | X_2 = x_2] \\
&= \mathbb{E}[X_1^2 | X_2 = x_2] - 2\mathbb{E}[X_1 | X_2 = x_2] \mathbb{E}[X_1 | X_2 = x_2] + \mathbb{E}[X_1 | X_2 = x_2]^2 \\
&= \mathbb{E}[X_1^2 | X_2 = x_2] - \mathbb{E}[X_1 | X_2 = x_2]^2.
\end{aligned}$$

□

Example 2.1.

Suppose that X_1 and X_2 are discrete rvs having joint pmf of the form

$$p(x_1, x_2) = \begin{cases} \frac{1}{5}, & \text{if } x_1 = 1 \text{ and } x_2 = 0, \\ \frac{2}{15}, & \text{if } x_1 = 0 \text{ and } x_2 = 1, \\ \frac{1}{15}, & \text{if } x_1 = 1 \text{ and } x_2 = 2, \\ \frac{1}{5}, & \text{if } x_1 = 2 \text{ and } x_2 = 0, \\ \frac{2}{5}, & \text{if } x_1 = 1 \text{ and } x_2 = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the conditional distribution of $X_1|(X_2 = 1)$. Also, calculate $\mathbb{E}[X_1|X_2 = 1]$ and $\text{Var}(X_1|X_2 = 1)$.

Solution. Note that for problems of this nature, it often helps to create a table:

$p(x_1, x_2)$	0	1	2	$p_1(x_1)$
0	0	$\frac{2}{15}$	0	$\frac{2}{15}$
1	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{1}{15}$	$\frac{2}{3}$
2	$\frac{1}{5}$	0	0	$\frac{1}{5}$
$p_2(x_2)$	$\frac{2}{5}$	$\frac{8}{15}$	$\frac{1}{15}$	1

Then,

$$p_{1|2}(0|1) = P(X_1 = 0|X_2 = 1) = \frac{2/15}{8/15} = \frac{1}{4},$$
$$p_{1|2}(1|1) = P(X_1 = 1|X_2 = 1) = \frac{2/5}{8/15} = \frac{3}{4}.$$

Thus, the conditional pmf of $X_1|(X_2 = 1)$ can be represented as follows:

x_1	0	1
$p_{1|2}(x_1|1)$	$\frac{1}{4}$	$\frac{3}{4}$

We observe that $X_1|(X_2 = 1) \sim \text{BERN}(3/4)$, and so

$$\mathbb{E}[X_1|X_2 = 1] = \frac{3}{4} \quad \text{and} \quad \text{Var}(X_1|X_2 = 1) = \frac{3}{4}\left(1 - \frac{3}{4}\right) = \frac{3}{16}.$$

Example 2.2.

For $i = 1, 2$, suppose that $X_i \sim \text{BIN}(n_i, p)$ where X_1 and X_2 are independent. Find the conditional distribution of X_1 given $X_1 + X_2 = m$.

Solution. We want to find the conditional pmf of $X_1|(Y=m)$, where $Y = X_1 + X_2$. Let this conditional pmf be denoted by

$$p_{X_1|Y}(x_1|m) = P(X_1 = x_1|Y = m).$$

Recalling from Example 1.5 that $X_1 + X_2 \sim \text{BIN}(n_1 + n_2, p)$, we obtain

$$\begin{aligned} p_{X_1|Y}(x_1|m) &= \frac{P(X_1 = x_1, X_1 + X_2 = m)}{P(X_1 + X_2 = m)} \\ &= \frac{P(X_1 = x_1, X_2 = m - x_1)}{\binom{n_1+n_2}{m} p^m (1-p)^{n_1+n_2-m}} \\ &= \frac{p(x_1, m - x_1)}{\binom{n_1+n_2}{m} p^m (1-p)^{n_1+n_2-m}} \\ &= \frac{p_1(x_1) p_2(m - x_1)}{\binom{n_1+n_2}{m} p^m (1-p)^{n_1+n_2-m}}, \end{aligned}$$

since X_1 and X_2 are independent rvs.

$$= \frac{\binom{n_1}{x_1} p^{x_1} (1-p)^{n_1-x_1} \binom{n_2}{m-x_1} p^{m-x_1} (1-p)^{n_2-(m-x_1)}}{\binom{n_1+n_2}{m} p^m (1-p)^{n_1+n_2-m}},$$

provided $0 \le x_1 \le n_1$ and $0 \le m - x_1 \le n_2$.

$$= \frac{\binom{n_1}{x_1}\binom{n_2}{m-x_1}}{\binom{n_1+n_2}{m}}, \quad x_1 = \max\{0, m - n_2\}, \ldots, \min\{n_1, m\}.$$

Remark. Looking at the above conditional pmf, we recognize that

$$X_1|(X_1 + X_2 = m) \sim \text{HG}(n_1 + n_2, n_1, m),$$

which should not be all that surprising. Consider the sequence of $n_1 + n_2$ Bernoulli trials represented visually as follows:

$$\underbrace{1,\ 2,\ \ldots,\ n_1}_{\text{first block}}, \qquad \underbrace{1,\ 2,\ \ldots,\ n_2}_{\text{second block}}$$

Of these $n_1 + n_2$ trials in which m of them were known to be successes, we want x_1 successes to have occurred among the first n_1 trials (thereby implying that $m - x_1$ successes are obtained during the final n_2 trials). Since any of these trials were equally likely to be a success (i.e., the same success probability p is assumed), the desired result ends up being obtained hypergeometric probability.

Example 2.3.

Let $X_1, X_2, \ldots, X_m$ be independent rvs where $X_i \sim \text{POI}(\lambda_i)$, $i = 1, 2, \ldots, m$. Define $Y = \sum_{i=1}^m X_i$. Find the conditional distribution of $X_j|(Y = n)$.

Solution.

Consider $X_i \sim \text{POI}(\lambda_i)$. Find the mgf:

$$\phi_{X_i}(t) = \mathbb{E}[e^{tX_i}] = \sum_{x_i=0}^{\infty} e^{tx_i} e^{-\lambda_i} \frac{\lambda_i^{x_i}}{x_i!} = e^{-\lambda_i} \sum_{x_i=0}^{\infty} \frac{(e^t\lambda_i)^{x_i}}{x_i!} = e^{-\lambda_i} e^{e^t\lambda_i} = e^{\lambda_i(e^t-1)}.$$

Now consider mgf of Y:

$$\phi_Y(t) = \prod_{i=1}^m \phi_{X_i}(t) = \prod_{i=1}^m e^{\lambda_i(e^t-1)} = e^{(\sum_{i=1}^m \lambda_i)(e^t-1)}.$$

By uniqueness we get

$$Y = \sum_{i=1}^m X_i \sim \text{POI}\left(\sum_{i=1}^m \lambda_i\right).$$

Now, get the conditional pmf of $X_j|(Y = n)$.

$$p_{X_j|Y}(x_j|n) = \frac{P(X_j = x_j, Y = n)}{P(Y = n)}.$$

$$P(X_j = x_j, Y = n) = P(X_j = x_j, \sum_{i=1}^m X_i = n) = P(X_j = x_j, \sum_{i\neq j} X_i = n - x_j).$$

$$= P(X_j = x_j) P\left(\sum_{i\neq j} X_i = n - x_j\right) = e^{-\lambda_j} \frac{\lambda_j^{x_j}}{x_j!} \cdot e^{-\sum_{i\neq j}\lambda_i} \frac{(\sum_{i\neq j}\lambda_i)^{n-x_j}}{(n-x_j)!}.$$

Hence,

$$p_{X_j|Y}(x_j|n) = \frac{e^{-\lambda_j}\frac{\lambda_j^{x_j}}{x_j!} \cdot e^{-\sum_{i\neq j}\lambda_i}\frac{(\sum_{i\neq j}\lambda_i)^{n-x_j}}{(n-x_j)!}}{e^{-\sum_{i=1}^m\lambda_i}\frac{(\sum_{i=1}^m\lambda_i)^n}{n!}}$$

$$= \binom{n}{x_j} \left(\frac{\lambda_j}{\sum_{i=1}^m \lambda_i}\right)^{x_j} \left(1 - \frac{\lambda_j}{\sum_{i=1}^m \lambda_i}\right)^{n-x_j}.$$

$$\Rightarrow \quad X_j|(Y = n) \sim \text{BIN}\left(n, \frac{\lambda_j}{\sum_{i=1}^m \lambda_i}\right).$$

Example 2.4.

Suppose that $X \sim \text{POI}(\lambda)$ and $Y|(X = x) \sim \text{BIN}(x, p)$. Find the conditional distribution of $X|(Y = y)$.

Solution.

$$\begin{aligned}
P_{X|Y}(x|y) &= \frac{P(X = x, Y = y)}{P(Y = y)}. \\
P(Y = y|X = x)P(X = x) &= P(X = x, Y = y). \\
P(X = x, Y = y) &= e^{-\lambda}\frac{\lambda^x}{x!}\binom{x}{y}p^y(1-p)^{x-y}. \\
P(Y = y) &= \sum_{x=y}^{\infty} e^{-\lambda}\frac{\lambda^x}{x!}\binom{x}{y}p^y(1-p)^{x-y}. \\
&= e^{-\lambda}\frac{p^y}{y!}\sum_{x=y}^{\infty}\frac{\lambda^x(1-p)^{x-y}}{(x-y)!}. \\
&= e^{-\lambda}\frac{p^y}{y!}\lambda^y\sum_{k=0}^{\infty}\frac{(\lambda(1-p))^k}{k!} = e^{-\lambda}\frac{(\lambda p)^y}{y!}e^{\lambda(1-p)} = e^{-\lambda p}\frac{(\lambda p)^y}{y!}.
\end{aligned}$$

Hence we have that

$$p_{X|Y}(x|y) = \frac{e^{-\lambda}\frac{\lambda^x}{x!}\binom{x}{y}p^y(1-p)^{x-y}}{e^{-\lambda p}\frac{(\lambda p)^y}{y!}} = \frac{e^{-\lambda(1-p)}(\lambda(1-p))^{x-y}}{(x-y)!}.$$

$\therefore$ this is a shifted Poisson:

$$X|Y = y \sim \text{POI}(\lambda(1-p)) + y.$$

Remark. For small $(x, y)\,dy$,

$$P(y \le Y \le y + dy) \approx f_Y(y)dy \quad \Rightarrow \quad f_Y(y) = \lim_{dy\to 0}\frac{P(y \le Y \le y + dy)}{dy},$$

and

$$f_{X,Y}(x, y) = \lim_{dx,dy\to 0}\frac{P(x \le X \le x + dx, y \le Y \le y + dy)}{dx\,dy}.$$

Remark. The above conditional pmf is recognized as that of a *shifted Poisson* distribution (y units to the right). Specifically, we have that

$$X|(Y = y) \sim W + y, \quad \text{where } W \sim \text{POI}(\lambda(1-p)).$$

2.5 Jointly Continuous Case

- **Formulation:** In the jointly discrete case, it was natural to define

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}.$$

Strictly speaking, this no longer makes sense in a continuous context since $f(x, y) \neq P(X = x, Y = y)$ and $f_Y(y) \neq P(Y = y)$. However, for small positive values of dy (as the figure below shows),

$$P(y \leq Y \leq y + dy) \approx f_Y(y)dy.$$

Formally,

$$f_Y(y) = \lim_{dy \to 0} \frac{P(y \leq Y \leq y + dy)}{dy}.$$

In a similar fashion,

$$f(x, y) = \lim_{dx \to 0, dy \to 0} \frac{P(x \leq X \leq x + dx, y \leq Y \leq y + dy)}{dx\, dy},$$

which implies that

$$P(x \leq X \leq x + dx, y \leq Y \leq y + dy) \approx f(x, y)\, dx\, dy.$$

For small positive values of dx and dy, consider now

$$\begin{aligned} P(x \leq X \leq x + dx|y \leq Y \leq y + dy) &= \frac{P(x \leq X \leq x + dx, y \leq Y \leq y + dy)}{P(y \leq Y \leq y + dy)} \\ &\approx \frac{f(x, y)dx\, dy}{f_Y(y)dy} = \frac{f(x, y)}{f_Y(y)}dx. \end{aligned}$$

Definition 2.5 (Conditional pdf)**.** As a result, we formally define the *conditional pdf* of X given $Y = y$ (again to be denoted by $X|(Y = y)$) as

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \lim_{dx \to 0, dy \to 0} \frac{P(x \leq X \leq x + dx|y \leq Y \leq y + dy)}{dx}.$$

Remark. In the jointly continuous case, the conditional probability of an event of the form $\{a \leq X \leq b\}$ given $Y = y$ would be calculated as

$$P(a \leq X \leq b|Y = y) = \int_a^b f_{X|Y}(x|y)\, dx = \int_a^b \frac{f(x, y)}{f_Y(y)}\, dx,$$

which we can also express as

$$P(a \leq X \leq b|Y = y) = \frac{\int_a^b f(x, y)\, dx}{\int_{-\infty}^{\infty} f(x, y)\, dx}.$$

In other words, we could view this as a way of assigning probability to an event $\{a \leq X \leq b\}$ over a "slice", $Y = y$, of the *joint region of support* for the pair of rvs X and Y.

Example 2.5.

Suppose that the joint pdf of X and Y is given by

$$f(x,y) = \begin{cases} 5e^{-3x-y}, & \text{if } 0 < 2x < y < \infty, \\ 0, & \text{elsewhere.} \end{cases}$$

Determine the conditional distribution of $Y|(X = x)$ where $0 < x < \infty$.

Solution. Note that $2x < y < \infty$ so $y \in (2x, \infty)$.

$$f_X(x) = \int_{2x}^{\infty} 5e^{-3x-y}\, dy = -5e^{-3x-y}\Big|_{y=2x}^{\infty} = 5e^{-5x}.$$
$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{5e^{-3x-y}}{5e^{-5x}} = e^{2x-y}, \quad y \in (2x, \infty).$$

$\Rightarrow$ shifted exponential:

$$Y|(X = x) \sim \text{EXP}(1) + 2x.$$

Remark. The conditional pdf of $Y|(X = x)$ is recognized as that of a *shifted exponential* distribution ($2x$ units to the right). Specifically, we have that

$$Y|(X = x) \sim W + 2x, \quad \text{where } W \sim \text{EXP}(1).$$

- **Conditional Expectation:** If X and Y are jointly continuous rvs and $g(\cdot)$ is an arbitrary real-valued function, then the conditional expectation of $g(X)$ given $Y = y$ is

  $$\mathbb{E}[g(X)|Y = y] = \int_{-\infty}^{\infty} g(x) f_{X|Y}(x|y)\, dx,$$

  and so the conditional mean of $X|(Y = y)$ is given by

  $$\mathbb{E}[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx.$$

Example 2.6.

Suppose that the joint pdf of X and Y is given by

$$f(x,y) = \begin{cases} \frac{12}{5}x(2 - x - y), & \text{if } 0 < x < 1 \text{ and } 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the conditional distribution of X given $Y = y$ where $0 < y < 1$, and use it to calculate its conditional mean.

Solution. Using our earlier theory, we wish to find the conditional pdf of $X|(Y=y)$ given by

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}.$$

The region of support for the joint distribution of X and Y looks like: For $0 < y < 1$,

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f(x,y)\,dx \\ &= \int_0^1 \frac{12}{5}x(2-x-y)\,dx \\ &= \frac{12}{5}\int_0^1 (2x - x^2 - xy)\,dx \\ &= \frac{12}{5}\left(x^2 - \frac{1}{3}x^3 - \frac{1}{2}x^2 y\right)\Big|_0^1 = \frac{12}{5}\left(1 - \frac{1}{3} - \frac{1}{2}y\right) = \frac{2(4-3y)}{5}. \end{aligned}$$

Thus,

$$f_{X|Y}(x|y) = \frac{\frac{12}{5}x(2-x-y)}{\frac{2(4-3y)}{5}} = \frac{6x(2-x-y)}{4-3y}, \quad 0 < x < 1.$$

The conditional mean of X given $Y = y$ is

$$\begin{aligned} \mathbb{E}[X|Y=y] &= \int_0^1 x \cdot \frac{6x(2-x-y)}{4-3y}\,dx \\ &= \frac{6}{4-3y}\int_0^1 (2x^2 - x^3 - x^2 y)\,dx \\ &= \frac{6}{4-3y}\left(\frac{2}{3} - \frac{1}{4} - \frac{1}{3}y\right) \\ &= \frac{5-4y}{2(4-3y)}. \end{aligned}$$

- **Conditional Variance:** Likewise, as in the jointly discrete case, we can also consider the notion of conditional variance, which retains the same definition as before:

$$\mathrm{Var}(X|Y=y) = \mathbb{E}[(X - \mathbb{E}[X|Y=y])^2|Y=y] = \mathbb{E}[X^2|Y=y] - \mathbb{E}[X|Y=y]^2.$$

A fact that is becoming more and more evident is that conditional expectation inherits many of the properties from regular expectation. Moreover, the same properties concerning conditional expectation that held in the jointly discrete case continue to hold true in the jointly continuous case (as we are effectively replacing summation with integration).

Example 2.6. (continued)

Calculate $\mathrm{Var}(X|Y=y)$.

Solution. Using our earlier derived results, we first obtain

$$\begin{aligned}
\mathbb{E}[X^2|Y=y] &= \int_0^1 x^2 \cdot \frac{6x(2-x-y)}{4-3y}\,dx \\
&= \frac{6}{4-3y}\int_0^1 (2x^3 - x^4 - x^3 y)\,dx \\
&= \frac{6}{4-3y}\left(\frac{1}{2} - \frac{1}{5} - \frac{1}{4}y\right) \\
&= \frac{3(6-5y)}{10(4-3y)}.
\end{aligned}$$

Therefore, this leads to

$$\begin{aligned}
\mathrm{Var}(X|Y=y) &= \mathbb{E}[X^2|Y=y] - \mathbb{E}[X|Y=y]^2 \\
&= \frac{3(6-5y)}{10(4-3y)} - \frac{(5-4y)^2}{4(4-3y)^2} \\
&= \frac{6(6-5y)(4-3y) - 5(5-4y)^2}{20(4-3y)^2} \\
&= \frac{19 + 2y(5y-14)}{20(4-3y)^2}.
\end{aligned}$$

2.6 Mixed Case

We can also consider conditional distributions where the rvs are neither jointly continuous nor jointly discrete. To consider such a situation, suppose X is a continuous rv having pdf $f_X(x)$ and Y is a discrete rv having pmf $p_Y(y)$. If we focus on the conditional distribution of X given $Y = y$, then let us look at the following quantity:

$$\begin{aligned}
\frac{P(x \le X \le x+dx|Y=y)}{dx} &= \frac{P(x \le X \le x+dx, Y=y)}{dx P(Y=y)} \\
&= \frac{P(Y=y|x \le X \le x+dx)P(x \le X \le x+dx)}{dx P(Y=y)} \\
&= \frac{P(Y=y|x \le X \le x+dx)}{P(Y=y)} \cdot \frac{P(x \le X \le x+dx)}{dx},
\end{aligned}$$

where dx is again a small positive value. By letting $dx \to 0$, we can formally define the conditional pdf of $X|(Y=y)$ as follows:

$$\begin{aligned}
f(x|y) &= \lim_{dx \to 0} \frac{P(x \le X \le x+dx|Y=y)}{dx} \\
&= \frac{P(Y=y|X=x)}{P(Y=y)} \cdot f_X(x) \\
&= \frac{p(y|x) f_X(x)}{p_Y(y)},
\end{aligned}$$

where $p(y|x) = P(Y = y|X = x)$ is defined as the conditional pmf of $Y|(X = x)$. Note that since $f(x|y)$ is a pdf, it follows that

$$\int_{-\infty}^{\infty} f(x|y)\,dx = 1 \quad \Rightarrow \quad p_Y(y) = \int_{-\infty}^{\infty} p(y|x) f_X(x)\,dx.$$

In a similar fashion, we can also write

$$p(y|x) = \frac{f(x|y)p_Y(y)}{f_X(x)}.$$

Since $p(y|x)$ is a pmf, we have that

$$\sum_y p(y|x) = 1 \quad \Rightarrow \quad f_X(x) = \sum_y f(x|y)p_Y(y).$$

Example 2.7.

Suppose that $X \sim U(0,1)$ and $Y|(X = x) \sim \text{BERN}(x)$. Find the conditional distribution of $X|(Y = y)$.

Solution.

$$f_X(x) = \frac{1}{1-0} = 1.$$

$$p_Y(y) = \int_0^1 f_X(x)p(y|x)\,dx = \begin{cases} \int_0^1 x\,dx = \frac{1}{2}, & y = 1, \\ \int_0^1 (1-x)\,dx = \frac{1}{2}, & y = 0. \end{cases}$$

$$f_{X|Y}(x|y) = \frac{p(y|x)f_X(x)}{p_Y(y)} = \frac{x^y(1-x)^{1-y}}{1/2} = 2x^y(1-x)^{1-y}.$$

2.7 Computing Expectations by Conditioning

As before, let $g(\cdot)$ be an arbitrary real-valued function. In general, we recognize that

$$\mathbb{E}[g(X)|Y = y] = v(y),$$

where $v(y)$ is some function of y. With this in mind, let us make the following definition:

$$\mathbb{E}[g(X)|Y] = \mathbb{E}[g(X)|Y = y]\big|_{y=Y} = v(Y).$$

Functions of rvs are, once again, rvs themselves. Therefore, it makes sense to consider the expected value of $v(Y)$. In this regard, we would obtain:

$$\mathbb{E}[\mathbb{E}[g(X)|Y]] = \mathbb{E}[v(Y)] = \begin{cases} \sum_y v(y)p_Y(y), & \text{if } Y \text{ is discrete,} \\ \int_{-\infty}^{\infty} v(y)f_Y(y)\,dy, & \text{if } Y \text{ is continuous.} \end{cases}$$
$$= \begin{cases} \sum_y \mathbb{E}[g(X)|Y=y]p_Y(y), & \text{if } Y \text{ is discrete,} \\ \int_{-\infty}^{\infty} \mathbb{E}[g(X)|Y=y]f_Y(y)\,dy, & \text{if } Y \text{ is continuous.} \end{cases}$$

The following important result is regarded as the *law of total expectation*.

Theorem 2.2. *For rvs X and Y,*

$$\mathbb{E}[g(X)] = \mathbb{E}[\mathbb{E}[g(X)|Y]].$$

Proof. WLOG, X, Y are cts joint rvs with $f_{X,Y}(x,y)$.

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)\,dx.$$
$$\mathbb{E}[\mathbb{E}[g(X)|Y]] = \int_{-\infty}^{\infty} \mathbb{E}[g(X)|Y=y]f_Y(y)\,dy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x)f_{X|Y}(x|y)f_Y(y)\,dx\,dy.$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x)f_{X,Y}(x,y)\,dx\,dy = \int_{-\infty}^{\infty} g(x)\left(\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy\right)dx.$$
$$= \int_{-\infty}^{\infty} g(x)f_X(x)\,dx = \mathbb{E}[g(X)].$$

□

Remark. Using a similar method of proof, the result of Theorem 2.2 can naturally be extended as follows:

$$\mathbb{E}[g(X,Y)] = \mathbb{E}[\mathbb{E}[g(X,Y)|Y]].$$

The usefulness of the law of total expectation is well-demonstrated in the following example.

Example 2.8.

Suppose that $X \sim \text{GEO}_t(p)$ with pmf

$$p_X(x) = (1-p)^{x-1}p, \quad x = 1, 2, 3, \ldots$$

Calculate $\mathbb{E}[X]$ and $\text{Var}(X)$ using the law of total expectation.

Solution. $X \sim \text{GEO}_t(p)$ where X is the # of independent Bernoulli trials until the 1st success.

$$\Rightarrow Y = \begin{cases} 0 & 1^{\text{st}} \text{ is fail} \\ 1 & 1^{\text{st}} \text{ success} \end{cases} \Rightarrow Y \sim \text{BERN}(p) \text{ so } P(Y=0) = 1-p \text{ and } P(Y=1) = p.$$

Note: $X|(Y=1)$ is degenerate at 1 i.e. $X|Y=1=1$, and $X|(Y=0)$ is equiv. in distrib. to $1+X \Rightarrow X|Y=0 \sim 1+X$ in distrib.

By law of total expectation:

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]] = \sum_{y=0}^{1} \mathbb{E}[X|Y=y]P(Y=y) = (1-p)\mathbb{E}[X|Y=0] + p\mathbb{E}[X|Y=1].$$
$$= (1-p)\mathbb{E}[X+1] + p = (1-p)(1+\mathbb{E}[X]) + p = 1 + (1-p)\mathbb{E}[X].$$

Solving for $\mathbb{E}[X]$,

$$\mathbb{E}[X](1-(1-p)) = 1 \Rightarrow \mathbb{E}[X] = \frac{1}{p}.$$

$$\mathbb{E}[X^2] = \mathbb{E}[\mathbb{E}[X^2|Y]] = \sum_{y=0}^{1} \mathbb{E}[X^2|Y=y]P(Y=y).$$
$$= (1-p)\mathbb{E}[X^2|Y=0] + p\mathbb{E}[X^2|Y=1] = (1-p)\mathbb{E}[(1+X)^2] + p.$$
$$= (1-p)\mathbb{E}[1+2X+X^2] + p = (1-p)[1+2\mathbb{E}[X]+\mathbb{E}[X^2]] + p.$$
$$= 1 + \frac{2(1-p)}{p} + (1-p)\mathbb{E}[X^2].$$

Solving for $\mathbb{E}[X^2]$,

$$p\mathbb{E}[X^2] = 1 + \frac{2-2p}{p} \Rightarrow \mathbb{E}[X^2] = \frac{2-p}{p^2}.$$
$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{2-p}{p^2} - \left(\frac{1}{p}\right)^2 = \frac{1-p}{p^2}.$$

Remarks.

(1) Note that the obtained mean and variance agree with known results. Moreover, the above procedure relied only on basic manipulations and did not involve any complicated sums or the differentiation of an mgf.

(2) As part of the above solution, we claimed that $X|(Y=0) \sim Z$ where $Z = 1+X$, and this implied that $\mathbb{E}[X^2|Y=0] = \mathbb{E}[(1+X)^2]$. To see why this holds true formally, consider first

$$p_{X|Y}(x|0) = P(X=x|Y=0) = \frac{P(X=x, Y=0)}{P(Y=0)} = \frac{P(X=x, Y=0)}{1-p}.$$

Note that

$$P(X=x, Y=0) = P(1^{\text{st}} \text{ trial is a failure and } x \text{ total trials needed to get } 1^{\text{st}} \text{ success})$$
$$= P(1^{\text{st}} \text{ trial is a failure, next } x-2 \text{ trials are failures, and } x^{\text{th}} \text{ trial is a success})$$
$$= (1-p)(1-p)^{x-2}p \quad \text{due to independence of trials.}$$

Thus,

$$p_{X|Y}(x|0) = \frac{(1-p)(1-p)^{x-2}p}{1-p} = (1-p)^{x-2}p, \quad x = 2, 3, 4, \ldots$$

On the other hand, note that

$$\begin{aligned} p_Z(z) &= P(Z = z) \\ &= P(1 + X = z) \\ &= P(X = z - 1) \\ &= (1-p)^{(z-1)-1}p \\ &= (1-p)^{z-2}p, \quad z = 2, 3, 4, \ldots \end{aligned}$$

Since these two pmfs are identical, it follows that $X|(Y = 0) \sim Z$. As a further consequence, for an arbitrary real-valued function $g(\cdot)$, we must have that

$$\mathbb{E}[g(X)|Y = 0] = \mathbb{E}[g(Z)] = \mathbb{E}[g(1 + X)].$$

In recognizing that $\mathbb{E}[g(X)|Y = y]$ is a function of y, it similarly follows that $\mathrm{Var}(X|Y = y)$ is also a function of y. Therefore, we can make the following definition:

$$\mathrm{Var}(X|Y) = \mathrm{Var}(X|Y = y)\big|_{y=Y}.$$

Since $\mathrm{Var}(X|Y)$ is a function of Y, it is a rv as well, meaning that we could take its expected value. The following result, usually referred to as the *conditional variance formula*, provides a convenient way to calculate variance through the use of conditioning.

Theorem 2.3. *For rvs X and Y,*

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X|Y)] + \mathrm{Var}(\mathbb{E}[X|Y]).$$

Proof. Consider $\mathbb{E}[\mathrm{Var}(X|Y)]$. Recall that

$$\begin{aligned} \mathrm{Var}(X|Y) = \mathrm{Var}(X|Y = y)\big|_{y=Y} &= \mathbb{E}[X^2|Y] - \mathbb{E}[X|Y]^2 = v(Y) \text{ since fxn of } Y. \\ \Rightarrow \mathbb{E}[\mathrm{Var}(X|Y)] &= \mathbb{E}[\mathbb{E}[X^2|Y]] - \mathbb{E}[\mathbb{E}[X|Y]^2]. \end{aligned}$$

Next,

$$\mathrm{Var}(\mathbb{E}[X|Y]) = \mathbb{E}[\mathbb{E}[X|Y]^2] - \mathbb{E}[\mathbb{E}[X|Y]]^2 = \mathbb{E}[\mathbb{E}[X|Y]^2] - \mathbb{E}[X]^2.$$

Finally,

$$\begin{aligned} \mathbb{E}[\mathrm{Var}(X|Y)] + \mathrm{Var}(\mathbb{E}[X|Y]) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\ &= \mathrm{Var}(X) \quad \text{by defn.} \end{aligned}$$

□

Example 2.9.

Suppose that $\{X_i\}_{i=1}^{\infty}$ is an iid sequence of rvs with common mean μ and common variance σ^2. Let N be a discrete, non-negative integer-valued rv that is independent of each X_i. Find the mean and variance of

$$T = \sum_{i=1}^{N} X_i$$

(referred to as a *random sum*).

Solution.

$$\mathbb{E}[T] = \mathbb{E}[\mathbb{E}[T|N]] \text{ for } T = \sum_{i=1}^{N} X_i.$$

$$\mathbb{E}[T|N=n] = \mathbb{E}\left[\sum_{i=1}^{N} X_i \Big| N = n\right] = \sum_{i=1}^{n} \mathbb{E}[X_i|N=n] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n\mu.$$

By independence of $\{X_i\}_{i=1}^{\infty}$ and N,

$$\mathbb{E}[T|N] = n\mu\big|_{n=N} = N\mu.$$
$$\Rightarrow \mathbb{E}[T] = \mathbb{E}[N\mu] = \mu\mathbb{E}[N].$$

Next, we have that

$$\mathrm{Var}(T) = \mathbb{E}[\mathrm{Var}(T|N)] + \mathrm{Var}(\mathbb{E}[T|N]).$$
$$\mathrm{Var}(T) = \mathbb{E}[\mathrm{Var}(T|N)] + \mathrm{Var}(N\mu) = \mathbb{E}[\mathrm{Var}(T|N)] + \mu^2\,\mathrm{Var}(N).$$
$$\mathrm{Var}(T|N=n) = \mathrm{Var}\left(\sum_{i=1}^{N} X_i \Big| N = n\right) = \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) \text{ by independence.}$$
$$\Rightarrow \mathrm{Var}(T|N) = \mathrm{Var}(T|N=n)\big|_{n=N}.$$
$$\Rightarrow \mathrm{Var}(T) = \mathbb{E}[N\sigma^2] + \mu^2\,\mathrm{Var}(N) = \sigma^2\mathbb{E}[N] + \mu^2\,\mathrm{Var}(N).$$

2.8 Computing Probabilities by Conditioning

For any two rvs, recall that

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]] = \begin{cases} \sum_y \mathbb{E}[X|Y=y]p_Y(y), & \text{if } Y \text{ is discrete,} \\ \int_{-\infty}^{\infty} \mathbb{E}[X|Y=y]f_Y(y)\,dy, & \text{if } Y \text{ is continuous.} \end{cases} \tag{2.1}$$

Now, suppose that A represents some event of interest and we wish to determine $P(A)$. Define an indicator rv X such that

$$X = \begin{cases} 0, & \text{if event } A^c \text{ occurs,} \\ 1, & \text{if event } A \text{ occurs.} \end{cases}$$

Clearly, $P(X = 1) = P(A)$ and $P(X = 0) = 1 - P(A)$, so that $X \sim \text{BERN}(P(A))$. Thus, $\mathbb{E}[X] = P(A)$ and

$$\begin{aligned}\mathbb{E}[X|Y = y] &= \sum_x xP(X = x|Y = y) \\ &= 0 \cdot P(X = 0|Y = y) + 1 \cdot P(X = 1|Y = y) \\ &= P(X = 1|Y = y) \\ &= P(A|Y = y).\end{aligned}$$

Therefore, (2.1) becomes

$$P(A) = \begin{cases} \sum_y P(A|Y = y)p_Y(y), & \text{if } Y \text{ is discrete,} \\ \int_{-\infty}^{\infty} P(A|Y = y)f_Y(y)\,dy, & \text{if } Y \text{ is continuous,} \end{cases} \tag{2.2}$$

which are analogues of the law of total probability. In other words, the expectation formula (2.1) can also be used to calculate probabilities of interest as indicated by (2.2).

Example 2.10.

Suppose that X and Y are independent continuous rvs. Find an expression for $P(X < Y)$.

Solution. Define event A as $A = \{X < Y\}$. Apply (2.2) to get:

$$\begin{aligned}P(X < Y) &= P(A) \\ &= \int_{-\infty}^{\infty} P(A|Y = y)f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty} P(X < Y|Y = y)f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty} P(X < y)f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty} F_X(y)f_Y(y)\,dy.\end{aligned}$$

Remark. If, in addition, X and Y are identically distributed, then the pdf $f_Y(y)$ is equal to $f_X(y)$ and the result in Example 2.10 simplifies to become

$$\begin{aligned}P(X < Y) &= \int_{-\infty}^{\infty} F_X(y)f_X(y)\,dy \\ &= \int_0^1 u\,du \quad \text{where } u = F_X(y) \Rightarrow \frac{du}{dy} = f_X(y) \Rightarrow du = f_X(y)dy \\ &= \left(\frac{u^2}{2}\right)\Big|_0^1 \\ &= \frac{1}{2},\end{aligned}$$

as one would expect.

Example 2.11.

Suppose that $X \sim \text{EXP}(\lambda_1)$ and $Y \sim \text{EXP}(\lambda_2)$ are independent exponential rvs. Show that

$$P(X < Y) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Solution. Since X and Y are both exponential rvs, it immediately follows that

$$f_Y(y) = \lambda_2 e^{-\lambda_2 y}, \quad y \geq 0,$$

and

$$\begin{aligned} F_X(y) &= \int_0^y \lambda_1 e^{-\lambda_1 x}\, dx = \lambda_1 \left(-\frac{1}{\lambda_1} e^{-\lambda_1 x} \right) \Big|_{x=0}^{y} \\ &= 1 - e^{-\lambda_1 y}, \quad y \geq 0. \end{aligned}$$

Therefore, the result in Example 2.10 becomes

$$\begin{aligned} P(X < Y) &= \int_0^\infty (1 - e^{-\lambda_1 y}) \lambda_2 e^{-\lambda_2 y}\, dy \\ &= \int_0^\infty \lambda_2 e^{-\lambda_2 y}\, dy - \lambda_2 \int_0^\infty e^{-(\lambda_1 + \lambda_2) y}\, dy \\ &= 1 - \frac{\lambda_2}{\lambda_1 + \lambda_2} \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2}. \end{aligned}$$

Remark. As a matter of interest, this particular result will be featured quite prominently in Chapter 4.

Example 2.12.

Suppose that W, X, and Y are independent continuous rvs on $(0, \infty)$. If $Z = X|(X < Y)$, then show that $(W, X)|(W < X < Y)$ and $(W, Z)|(W < Z)$ are identically distributed.

Solution. Let us first consider the conditional joint cdf of $(W, X)|(W < X < Y)$.

$$\begin{aligned} G(w, x) &= P(W \leq w, X \leq x | W < X < Y). \\ &= \frac{P(W \leq w, X \leq x, W < X < Y)}{P(W < X < Y)} = \frac{P(W \leq w, X \leq x, W < X, X < Y)}{P(W < X, X < Y)}, \quad w, x \geq 0. \end{aligned}$$

$$\begin{aligned} P(W < X, X < Y) &= \int_0^\infty P(W < s, s < Y | X = s) f_X(s)\, ds \\ &= \int_0^\infty P(W < s, Y > s) f_X(s)\, ds \\ &= \int_0^\infty P(W < s) P(Y > s) f_X(s)\, ds. \end{aligned}$$

and

$$\begin{aligned}
P(W \le w, X \le x, W < X, X < Y) &= \int_0^\infty P(W \le w, X \le x, W < X, X < Y | X = s) f_X(s)\, ds \\
&= \int_0^\infty P(W \le w, s \le x, W < s, s < Y) f_X(s)\, ds \\
&= \int_0^x P(W \le \min\{s, w\}, Y > s) f_X(s)\, ds \\
&= \int_0^x P(W \le \min\{s, w\}) P(Y > s) f_X(s)\, ds.
\end{aligned}$$

Next, consider the conditional rv $Z = X|(X < Y)$.

$$\begin{aligned}
P(Z \le z) = P(X \le z | X < Y) &= \frac{P(X \le z, X < Y)}{P(X < Y)} \\
&= \frac{\int_0^\infty P(X \le z, s < Y | X = s) f_X(s)\, ds}{P(X < Y)} \\
&= \frac{\int_0^z P(Y > s) f_X(s)\, ds}{P(X < Y)}.
\end{aligned}$$

$$\Rightarrow h_Z(z) = \frac{d}{dz} P(Z \le z) = \frac{P(Y > z) f_X(z)}{P(X < Y)}, \quad z \ge 0.$$

The conditional joint cdf of $(W, Z)|(W < Z)$ is

$$\begin{aligned}
P(W \le w, Z \le z | W < Z) &= \frac{P(W \le w, Z \le z, W < Z)}{P(W < Z)} \\
&= \frac{\int_0^\infty P(W \le w, Z \le z, W < Z | Z = s) h_Z(s)\, ds}{P(W < Z)} \\
&= \frac{\int_0^z P(W \le \min\{s, w\}) h_Z(s)\, ds}{P(W < Z)} \\
&= \frac{\int_0^z P(W \le \min\{s, w\}) P(Y > s) f_X(s)\, ds}{P(X < Y) P(W < Z)}.
\end{aligned}$$

Moreover,

$$\begin{aligned}
P(W < Z) &= \int_0^\infty P(W < s) h_Z(s)\, ds \\
&= \frac{P(W < X < Y)}{P(X < Y)}.
\end{aligned}$$

Substitution shows that this joint cdf is $G(w, z)$, proving that $(W, X)|(W < X < Y)$ and $(W, Z)|(W < Z)$ are identically distributed.

Remark. It can likewise be shown that if $V = X|(W < X)$, then $(X, Y)|(W < X < Y)$ and $(V, Y)|(V < Y)$ are identically distributed (see Exercise 2.35).

2.9 Some Further Extensions

If you consider our treatment of the conditional expectation $\mathbb{E}[X|Y = y]$, then one detail you should notice is that this kind of expectation behaves exactly the same as regular (i.e., unconditional) expectation *except* that all pmfs/pdfs used now are conditional on the event $Y = y$. In this sense, conditional expectations essentially satisfy all the properties of regular expectation. Thus, for an arbitrary real-valued function $g(\cdot)$, a corresponding analogue of

$$\mathbb{E}[g(X)] = \begin{cases} \sum_w \mathbb{E}[g(X)|W = w]p_W(w), & \text{if } W \text{ is discrete,} \\ \int_{-\infty}^{\infty} \mathbb{E}[g(X)|W = w]f_W(w)\,dw, & \text{if } W \text{ is continuous,} \end{cases}$$

would be

$$\mathbb{E}[g(X)|Y = y] = \begin{cases} \sum_w \mathbb{E}[g(X)|W = w, Y = y]p_{W|Y}(w|y), & \text{if } W \text{ is discrete,} \\ \int_{-\infty}^{\infty} \mathbb{E}[g(X)|W = w, Y = y]f_{W|Y}(w|y)\,dw, & \text{if } W \text{ is continuous.} \end{cases}$$

We remark that the above relation makes sense, since if we assume (without loss of generality) that X and Y are discrete rvs, then we obtain (in the case when W is discrete too)

$$\begin{aligned} \sum_w \mathbb{E}[g(X)|W = w, Y = y]p_{W|Y}(w|y) &= \sum_w \sum_x g(x)p_{X|W,Y}(x|w, y)p_{W|Y}(w|y) \\ &= \sum_x g(x) \sum_w p_{X|W,Y}(x|w, y)\frac{p_{W,Y}(w, y)}{p_Y(y)} \\ &= \sum_x \frac{g(x)}{p_Y(y)} \sum_w p_{X,W,Y}(x, w, y) \\ &= \sum_x g(x)\frac{p_{X,Y}(x, y)}{p_Y(y)} \\ &= \sum_x g(x)p_{X|Y}(x|y) \\ &= \mathbb{E}[g(X)|Y = y]. \end{aligned}$$

In a similar fashion, if one introduces an event of interest A and defines

$$g(X) = \begin{cases} 0, & \text{if event } A^c \text{ occurs,} \\ 1, & \text{if event } A \text{ occurs,} \end{cases}$$

then we obtain

$$P(A|Y = y) = \begin{cases} \sum_w P(A|W = w, Y = y)p_{W|Y}(w|y), & \text{if } W \text{ is discrete,} \\ \int_{-\infty}^{\infty} P(A|W = w, Y = y)f_{W|Y}(w|y)\,dw, & \text{if } W \text{ is continuous.} \end{cases}$$

Furthermore, if we now define

$$\mathbb{E}[g(X)|W, Y] = \mathbb{E}[g(X)|W = w, Y = y]\big|_{w=W,y=Y},$$

then the law of total expectation extends to become

$$\mathbb{E}[g(X)] = \mathbb{E}[\mathbb{E}[g(X)|W, Y]].$$

Example 2.13.

Consider an experiment in which independent trials, each having success probability $p \in (0, 1)$, are performed until k consecutive successes are achieved where $k \in \mathbb{Z}^+$. Determine the expected number of trials needed to achieve k consecutive successes.

Solution. Let N_k represent the # of trials used to get k consecutive successes. We want to determine $\mathbb{E}[N_k]$. For $k = 1$, note $N_1 \sim \text{GEO}(p)$ so $\mathbb{E}[N_1] = \frac{1}{p}$.

For $k \geq 2$, let E_r be the expected additional number of trials needed when the current run contains r consecutive successes, where $r = 0, 1, \ldots, k$. Clearly, $E_k = 0$. Conditioning on the outcome of the next trial gives, for $r = 0, 1, \ldots, k-1$,

$$E_r = 1 + pE_{r+1} + (1-p)E_0.$$

Working backward from $E_k = 0$ yields

$$\begin{aligned}
E_{k-1} &= 1 + (1-p)E_0, \\
E_{k-2} &= 1 + p + p(1-p)E_0 + (1-p)E_0, \\
&\vdots \\
E_0 &= (1 + p + \cdots + p^{k-1}) + (1-p)(1 + p + \cdots + p^{k-1})E_0.
\end{aligned}$$

Since $(1-p)(1 + p + \cdots + p^{k-1}) = 1 - p^k$, it follows that

$$\begin{aligned}
p^k E_0 &= 1 + p + \cdots + p^{k-1}, \\
\mathbb{E}[N_k] = E_0 &= \frac{1 - p^k}{p^k(1-p)} = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k}.
\end{aligned}$$

3 Discrete-time Markov Chains

Overview. Discrete-time Markov chains are stochastic systems in which the next-step law depends on the present state rather than the entire past. This chapter develops transition matrices, communicating classes, recurrence, limiting and invariant distributions, branching processes, gambler's ruin, and absorbing-chain methods.

3.1 Notation and the Transition Semigroup

Let S be a finite or countable state space, $X_n \in S$, and define the natural filtration $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$. For a time-homogeneous chain with transition matrix $P = (p_{ij})$, set

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i), \qquad p_{ij}^{(n)} = \mathbb{P}(X_n = j \mid X_0 = i).$$

Thus $P^0 = I$ and $P^n = (p_{ij}^{(n)})_{i,j \in S}$. Row distributions evolve by right multiplication: $\mu_n = \mu_0 P^n$.

Proposition 3.1 (Markov operator and Chapman–Kolmogorov identity)**.** *For every bounded function $f : S \to \mathbb{R}$, define*

$$(Pf)(i) = \sum_{j \in S} p_{ij} f(j).$$

Then $(P^n f)(i) = \mathbb{E}[f(X_n) \mid X_0 = i]$ and, for $m, n \geq 0$,

$$p_{ij}^{(n+m)} = \sum_{k \in S} p_{ik}^{(n)} p_{kj}^{(m)}.$$

Proof. For the first step, condition on X_1 to obtain $(Pf)(i) = \mathbb{E}[f(X_1) \mid X_0 = i]$. Iteration and the Markov property give

$$\begin{aligned} \mathbb{E}[f(X_{n+1}) \mid X_0 = i] &= \mathbb{E}[\mathbb{E}[f(X_{n+1}) \mid X_n] \mid X_0 = i] \\ &= \mathbb{E}[(Pf)(X_n) \mid X_0 = i] = (P^n P f)(i). \end{aligned}$$

Choosing $f = \mathbf{1}_{\{j\}}$ and conditioning on $X_n = k$ yields

$$\begin{aligned} p_{ij}^{(n+m)} &= \sum_{k \in S} \mathbb{P}(X_{n+m} = j, X_n = k \mid X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{n+m} = j \mid X_n = k) \mathbb{P}(X_n = k \mid X_0 = i) \\ &= \sum_{k \in S} p_{ik}^{(n)} p_{kj}^{(m)}. \end{aligned}$$

□

3.2 Definitions and Basic Concepts

Definition 3.1. $\{X(t), t \in T\}$ is called a *stochastic process* if $X(t)$ is a rv (or possibly a random vector) for any given $t \in T$. T is referred to as the *index set* and is often interpreted in the context of time. As such, $X(t)$ is often called the *state* of the process at time t. We note that:

$$\text{Index set } T \begin{cases} \text{can be a continuum of values such as } T = \{t : t \geq 0\}, \\ \text{can be a set of discrete points such as } T = \{t_0, t_1, t_2, \ldots\}. \end{cases}$$

Since there is a one-to-one correspondence between the sets $T = \{t_0, t_1, t_2, \ldots\}$ and $\mathbb{N} = \{0, 1, 2, \ldots\}$, we will use $T = \mathbb{N}$ as the general index set for a discrete-time stochastic process (unless otherwise stated). In other words, $\{X(n), n \in \mathbb{N}\}$ or $\{X_n, n \in \mathbb{N}\}$ will represent a general discrete-time stochastic process. Some examples of discrete-time stochastic processes $\{X_n, n \in \mathbb{N}\}$ might include:

1. X_n represents the outcome of the nth toss of a die,
2. X_n represents the price of a stock at the end of day n trading,
3. X_n represents the maximum temperature in Waterloo during the nth month,
4. X_n represents the number of goals scored in game n by the varsity hockey team,
5. X_n represents the number of STAT 333 students in class for the nth lecture.

Definition 3.2. A stochastic process $\{X_n, n \in \mathbb{N}\}$ is said to be a *discrete-time Markov chain* (DTMC) if the following two conditions hold true:

1. For $n \in \mathbb{N}$, X_n is a discrete rv (i.e., the state space S of X_n is of discrete type).
2. For $n \in \mathbb{N}$ and all states $x_0, x_1, \ldots, x_{n+1} \in S$, the Markov property must hold:

$$P(X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1} = x_{n-1}, \ldots, X_1 = x_1, X_0 = x_0) = P(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

In mathematical terms, this property states that the conditional distribution of any *future state* X_{n+1} given the *past states* $X_0, X_1, \ldots, X_{n-1}$ and the *present state* X_n is independent of the past states. In a more informal way, the Markov property tells us, for a random process, that if we know the value taken by the process at a given time, we will not get any additional information about the future behaviour of the process by gathering more knowledge about the past.

Remarks.

1. Unless otherwise stated, the *state space* S of a DTMC $\{X_n, n \in \mathbb{N}\}$ will be assumed to be $\mathbb{N}$.
2. In general, the *sequence of rvs* $\{X_n\}_{n=0}^{\infty}$ are neither independent nor identically distributed.
3. The Markov property does not require "full" information on the past to ensure independence. For example, consider the following conditional probability:

For compactness, write

$$H = \{X_{n-2} = x_{n-2}, \ldots, X_1 = x_1, X_0 = x_0\}.$$

Then

$$P(X_{n+1} = x_{n+1} \mid X_n = x_n, H) = \frac{P(X_{n+1} = x_{n+1}, X_n = x_n, H)}{P(X_n = x_n, H)},$$

which is "missing" the information for X_{n-1}.

However, note that

$$\begin{aligned}
&P(X_{n+1} = x_{n+1}, X_n = x_n, H) \\
&= \sum_{x_{n-1}=0}^{\infty} P(X_{n+1} = x_{n+1}, X_n = x_n, X_{n-1} = x_{n-1}, H) \\
&= \sum_{x_{n-1}=0}^{\infty} P(X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1} = x_{n-1}, H) \\
&\qquad \times P(X_n = x_n, X_{n-1} = x_{n-1}, H) \\
&= P(X_{n+1} = x_{n+1} \mid X_n = x_n) \sum_{x_{n-1}=0}^{\infty} P(X_n = x_n, X_{n-1} = x_{n-1}, H) \\
&\qquad\qquad \text{due to the Markov property} \\
&= P(X_{n+1} = x_{n+1} \mid X_n = x_n) P(X_n = x_n, H).
\end{aligned}$$

Substituting this expression into the numerator of the above equation yields

$$P(X_{n+1} = x_{n+1} \mid X_n = x_n, H) = P(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

It is straightforward to extend the above result to any number of past time points from $0, 1, \ldots, n-1$ with "missing" information. This is the essence of the Markov property.

Definition 3.3. For any pair of states i and j, the *transition probability* from state i at time n to state j at time $n+1$ is given by

$$P_{n,i,j} = P(X_{n+1} = j \mid X_n = i), \quad n \in \mathbb{N}.$$

Let P_n be the associated matrix containing all these transition probabilities, referred to as the *one-step transition probability matrix* (TPM) from time n to time $n+1$. It looks like

$$P_n = [P_{n,i,j}] = \begin{array}{c} \\ 0 \\ 1 \\ \vdots \\ i \\ \vdots \end{array} \begin{array}{c} \begin{array}{ccccccc} 0 & 1 & 2 & \cdots & j & \cdots \end{array} \\ \begin{bmatrix} P_{n,0,0} & P_{n,0,1} & P_{n,0,2} & \cdots & P_{n,0,j} & \cdots \\ P_{n,1,0} & P_{n,1,1} & P_{n,1,2} & \cdots & P_{n,1,j} & \cdots \\ \vdots & \vdots & \vdots & & \vdots & \\ P_{n,i,0} & P_{n,i,1} & P_{n,i,2} & \cdots & P_{n,i,j} & \cdots \\ \vdots & \vdots & \vdots & & \vdots & \end{bmatrix} \end{array},$$

where, for convenience, the states of the DTMC are labelled along the margins of the matrix. For each pair of states i and j, if $P_{n,i,j} = P_{i,j}\ \forall n \in \mathbb{N}$, then we say that the DTMC is *stationary* or *homogeneous*. In this situation, our one-step TPM becomes:

$$P = [P_{i,j}] = \begin{array}{c} \\ 0 \\ 1 \\ \vdots \\ i \\ \vdots \end{array} \begin{array}{c} \begin{array}{ccccccc} 0 & 1 & 2 & \cdots & j & \cdots \end{array} \\ \begin{bmatrix} P_{0,0} & P_{0,1} & P_{0,2} & \cdots & P_{0,j} & \cdots \\ P_{1,0} & P_{1,1} & P_{1,2} & \cdots & P_{1,j} & \cdots \\ \vdots & \vdots & \vdots & & \vdots & \\ P_{i,0} & P_{i,1} & P_{i,2} & \cdots & P_{i,j} & \cdots \\ \vdots & \vdots & \vdots & & \vdots & \end{bmatrix} \end{array}.$$

Remark. In STAT 333, we only consider stationary DTMCs. Moreover, from the construction of the TPM P, it is clear that $P_{i,j} \geq 0\ \forall i, j \in \mathbb{N}$ and $\sum_{j=0}^{\infty} P_{i,j} = 1\ \forall i \in \mathbb{N}$ (i.e., each row sum of P must be 1). Such a matrix whose elements are non-negative and whose row sums are equal to 1 is said to be *stochastic*.

Example 3.1.

On a given day, the weather is either clear, overcast, or raining. If the weather is clear today, then it will be clear, overcast, or raining tomorrow with respective probabilities 0.6, 0.3, and 0.1. If the weather is overcast today, then it will be clear, overcast, or raining tomorrow with respective probabilities 0.2, 0.5, and 0.3. If the weather is raining today, then it will be clear, overcast, or raining tomorrow with respective probabilities 0.4, 0.2, and 0.4. Construct the underlying DTMC and determine its TPM.

Solution. Note that the weather tomorrow only depends on the weather today, implying that the Markov property holds true. Thus, letting X_n denote the state of the weather on the nth day, $\{X_n, n \in \mathbb{N}\}$ is a three-state DTMC. If we let state 0 correspond to clear, state 1 correspond to overcast, and state 2 correspond to raining, then the state space of the DTMC is $S = \{0, 1, 2\}$ and its TPM is given by

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0.4 & 0.2 & 0.4 \end{bmatrix} \end{array}.$$

Definition 3.4. For any pair of states i and j, the n-step transition probability is given by

$$P_{i,j}^{(n)} = P(X_{m+n} = j \mid X_m = i), \quad m, n \in \mathbb{N}.$$

Due to the stationary assumption, this quantity is actually independent of m (which is why we do not include m in its notation). Thus, $P_{i,j}^{(n)} = P(X_n = j \mid X_0 = i)$, $n \in \mathbb{N}$. Furthermore, it is evident that

$$P_{i,j}^{(0)} = P(X_0 = j \mid X_0 = i) = \begin{cases} 0, & \text{if } i \neq j, \\ 1, & \text{if } i = j. \end{cases}$$

In a similar manner, let $P^{(n)} = [P_{i,j}^{(n)}]$ represent the associated n-step TPM. Clearly, when $n = 1$, $P^{(1)} = P$. When $n = 0$, $P^{(0)} = I$, where I represents the identity matrix. Just as with the one-step TPM, it follows that the row sums of $P^{(n)}$ must equal 1 as well.

Proposition 3.2 (Chapman–Kolmogorov Equations)**.** *For $0 \leq m \leq n$,*

$$P_{i,j}^{(n)} = \sum_{k=0}^{\infty} P_{i,k}^{(m)} P_{k,j}^{(n-m)}, \qquad i, j \in \mathbb{N}.$$

Equivalently, $P^{(n)} = P^{(m)} P^{(n-m)}$.

For $n \in \mathbb{Z}^+$, let us consider

$$\begin{aligned} P_{i,j}^{(n)} &= P(X_n = j \mid X_0 = i) \\ &= \sum_{k=0}^{\infty} P(X_n = j \mid X_{n-1} = k, X_0 = i) P(X_{n-1} = k \mid X_0 = i) \\ &= \sum_{k=0}^{\infty} P_{i,k}^{(n-1)} P(X_n = j \mid X_{n-1} = k, X_0 = i) \\ &= \sum_{k=0}^{\infty} P_{i,k}^{(n-1)} P(X_n = j \mid X_{n-1} = k) \quad \text{due to the Markov property} \\ &= \sum_{k=0}^{\infty} P_{i,k}^{(n-1)} P_{k,j}. \end{aligned} \tag{3.1}$$

Recall: If $A = [a_{i,j}]$, $B = [b_{i,j}]$, and $C = AB$ where $C = [c_{i,j}]$, then $c_{i,j} = \sum_k a_{i,k} b_{k,j}$.

As a result, note that (3.1) implies that $P^{(n)} = P^{(n-1)}P$, $n \in \mathbb{Z}^+$. More generally, $P_{i,j}^{(n)}$ can be expressed as

$$P_{i,j}^{(n)} = \sum_{k=0}^{\infty} P_{i,k}^{(m)} P_{k,j}^{(n-m)} \quad \forall i, j \in \mathbb{N} \text{ and } 0 \leq m \leq n,$$

which are referred to as the Chapman-Kolmogorov equations for a DTMC. In matrix form, this translates to $P^{(n)} = P^{(m)}P^{(n-m)}$, $0 \leq m \leq n$. Coming back to $P^{(n)} = P^{(n-1)}P$, $n \in \mathbb{Z}^+$, let us look at a few values of n:

$$\begin{aligned}
&\text{Take } n = 2 \Rightarrow P^{(2)} = P^{(1)}P = PP = P^2, \\
&\text{Take } n = 3 \Rightarrow P^{(3)} = P^{(2)}P = P^2P = P^3, \\
&\text{Take } n = 4 \Rightarrow P^{(4)} = P^{(3)}P = P^3P = P^4.
\end{aligned}$$

Clearly, we see that

$$P^{(n)} = P^n,$$

and so the n-step TPM is simply the one-step TPM multiplied by itself n times.

Initial Conditions.

For $n \in \mathbb{N}$, let us now introduce a particular row vector, which we will denote as either

$$\alpha_n = (\alpha_{n,0}, \alpha_{n,1}, \ldots, \alpha_{n,k}, \ldots),$$

or

$$\alpha_n = \begin{bmatrix} \alpha_{n,0} & \alpha_{n,1} & \cdots & \alpha_{n,k} & \cdots \end{bmatrix},$$

where $\alpha_{n,k} = P(X_n = k)$ $\forall k \in \mathbb{N}$. In other words, $\alpha_{n,k}$ represents the marginal pmf of X_n, $n \in \mathbb{N}$. As a consequence, it follows that

$$\sum_{k=0}^{\infty} \alpha_{n,k} = 1 \quad \forall n \in \mathbb{N}.$$

If we focus on the case when $n = 0$, then α_0 is referred to as the *initial probability row vector* of the DTMC, or simply the initial conditions of the DTMC. For $n \in \mathbb{Z}^+$, note that

$$\begin{aligned}
\alpha_{n,k} &= P(X_n = k) \\
&= \sum_{i=0}^{\infty} P(X_n = k \mid X_m = i) P(X_m = i) \quad \text{where } 0 \leq m \leq n \\
&= \sum_{i=0}^{\infty} \alpha_{m,i} P(X_{n-m} = k \mid X_0 = i) \quad \text{due to the stationary assumption} \\
&= \sum_{i=0}^{\infty} \alpha_{m,i} P_{i,k}^{(n-m)}.
\end{aligned}$$

In matrix form, the above relation implies that

$$\alpha_n = \alpha_m P^{(n-m)} = \alpha_m P^{n-m}, \quad 0 \leq m \leq n,$$

which subsequently leads to

$$\alpha_n = \alpha_0 P^{(n)} = \alpha_0 P^n, \quad n \in \mathbb{N}.$$

Remark. Having knowledge of the initial conditions and the one-step transition probabilities, we can readily calculate various probabilities of possible interest such as

$$\begin{aligned}
&P(X_0 = x_0, X_1 = x_1, \ldots, X_n = x_n) \\
&= P(X_0 = x_0) \prod_{r=1}^{n} P(X_r = x_r \mid X_0 = x_0, \ldots, X_{r-1} = x_{r-1}) \\
&= P(X_0 = x_0) \prod_{r=1}^{n} P(X_r = x_r \mid X_{r-1} = x_{r-1}) \\
&= \alpha_{0,x_0} \prod_{r=1}^{n} P_{x_{r-1},x_r}.
\end{aligned}$$

where the second last equality follows due to the Markov property, or

$$\begin{aligned}
&P(X_{n+1} = x_{n+1}, \ldots, X_{n+m} = x_{n+m} \mid X_n = x_n) \\
&= \prod_{r=1}^{m} P(X_{n+r} = x_{n+r} \mid X_{n+r-1} = x_{n+r-1}) \\
&= \prod_{r=1}^{m} P_{x_{n+r-1},x_{n+r}}.
\end{aligned}$$

The key observation here is that the DTMC is completely characterized by its one-step TPM P and the initial conditions α_0.

Example 3.2.

A particle moves along the states $\{0, 1, 2\}$ according to a DTMC whose TPM is given by

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0 & 0.6 & 0.4 \\ 0.5 & 0 & 0.5 \end{bmatrix} \end{array}.$$

Let X_n denote the position of the particle after the nth move. Suppose that the particle is equally likely to start in any of the three positions.

(a) Calculate $P(X_3 = 1 \mid X_0 = 0)$.

Solution. Because the chain is homogeneous, the three-step transition matrix is $P^3 = P^2P$. Direct matrix multiplication gives

$$P^2 = \begin{bmatrix} 0.54 & 0.26 & 0.20 \\ 0.20 & 0.36 & 0.44 \\ 0.60 & 0.10 & 0.30 \end{bmatrix},$$

$$P^3 = \begin{bmatrix} 0.478 & 0.264 & 0.258 \\ 0.360 & 0.256 & 0.384 \\ 0.570 & 0.180 & 0.250 \end{bmatrix}.$$

In particular, the required entry can be verified directly:

$$\begin{aligned} \mathbb{P}(X_3 = 1 \mid X_0 = 0) &= (P^3)_{0,1} \\ &= 0.54(0.2) + 0.26(0.6) + 0.20(0) \\ &= 0.264. \end{aligned}$$

(b) Calculate $P(X_4 = 2)$.

Solution. Note that $\alpha_4 = [\alpha_{4,0}\ \alpha_{4,1}\ \alpha_{4,2}]$.

$$\begin{aligned} &\Rightarrow\ \alpha_4 = \alpha_0 P^{(4)} \\ &= \alpha_0 P^{(3)} P \\ &= \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 0.478 & 0.264 & 0.258 \\ 0.36 & 0.256 & 0.384 \\ 0.57 & 0.18 & 0.25 \end{bmatrix} \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0 & 0.6 & 0.4 \\ 0.5 & 0 & 0.5 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 0.4636 & 0.254 & 0.2824 \\ 0.444 & 0.2256 & 0.3304 \\ 0.524 & 0.222 & 0.254 \end{bmatrix} \\ &= \begin{bmatrix} 0.4772 & 0.233867 & 0.288933 \end{bmatrix}. \end{aligned}$$

$$\therefore \quad P(X_4 = 2) = \alpha_{4,2} = 0.288933.$$

(c) Calculate $P(X_6 = 0, X_4 = 2)$.

Solution. $P(X = x, Y = y) = P(X = x \mid Y = y)P(Y = y)$.

$$\begin{aligned} P(X_6 = 0, X_4 = 2) &= P(X_4 = 2)P(X_6 = 0 \mid X_4 = 2) \\ &= \alpha_{4,2} P^{(2)}_{2,0} = 0.288933 \times 0.6 \\ &= 0.1733598 \approx 0.173. \end{aligned}$$

(d) Calculate $P(X_9 = 0, X_7 = 2 \mid X_5 = 1, X_2 = 0)$.

Solution. $P(A \cap B \mid C) = P(B \mid C)P(A \mid B \cap C)$.

By the multiplication rule, followed by the Markov property,

$$\begin{aligned}
&\mathbb{P}(X_9 = 0, X_7 = 2 \mid X_5 = 1, X_2 = 0) \\
&\quad = \mathbb{P}(X_7 = 2 \mid X_5 = 1)\, \mathbb{P}(X_9 = 0 \mid X_7 = 2) \\
&\quad = p^{(2)}_{1,2} p^{(2)}_{2,0} \\
&\quad = 0.44(0.6) = 0.264.
\end{aligned}$$

With these basic results in place, we next consider the classification of states in a DTMC.

Definition 3.5. State j is accessible from state i (denoted by $i \to j$) if $\exists n \in \mathbb{N}$ such that $P^{(n)}_{i,j} > 0$. If states i and j are accessible from each other, then states i and j *communicate* (denoted by $i \leftrightarrow j$). In other words, $i \leftrightarrow j$ iff $\exists m, n \in \mathbb{N}$ such that $P^{(n)}_{i,j} > 0$ and $P^{(m)}_{j,i} > 0$.

In terms of accessibility, note that the size of the components of P do not matter. All that matters is which are positive and which are 0. In particular, if state j is not accessible from state i, then $P^{(n)}_{i,j} = 0 \ \forall n \in \mathbb{N}$ and

Write $T_j = \inf\{n \geq 0 : X_n = j\}$ for the first hitting time of state j. By Boole's inequality and the absence of an accessible path,

$$\begin{aligned}
\mathbb{P}(T_j < \infty \mid X_0 = i) &\leq \sum_{n=0}^{\infty} \mathbb{P}(X_n = j \mid X_0 = i) \\
&= \sum_{n=0}^{\infty} p^{(n)}_{ij} = 0.
\end{aligned}$$

implying that $P(\text{DTMC ever visits state } j \mid X_0 = i) = 0$.

3.2.1 Equivalence Relation

The concept of communication forms what is known as an equivalence relation, satisfying the following criteria:

(i) **Reflexivity:** $i \leftrightarrow i$.
Clearly true since $P^{(0)}_{i,i} = 1 > 0$.

(ii) **Symmetry:** $i \leftrightarrow j \Rightarrow j \leftrightarrow i$.
This is obviously true by definition.

(iii) **Transitivity:** $i \leftrightarrow j$ and $j \leftrightarrow k \Rightarrow i \leftrightarrow k$.

To see this holds formally, we know that $\exists n \in \mathbb{N}$ such that $P^{(n)}_{i,j} > 0$. Also, $\exists m \in \mathbb{N}$ such that $P^{(m)}_{j,k} > 0$. Using the Chapman-Kolmogorov equations, we have that

$$P^{(n+m)}_{i,k} = \sum_{\ell=0}^{\infty} P^{(n)}_{i,\ell} P^{(m)}_{\ell,k} \geq P^{(n)}_{i,j} P^{(m)}_{j,k} > 0.$$

Therefore, $P_{i,k}^{(n+m)} > 0$, implying that $i \to k$. Using precisely the same logic, it is straightforward to show that $k \to i$. Thus, by definition, $i \leftrightarrow k$.

The fact that communication is an equivalence relation allows us to partition all the states of a DTMC into various communication classes, so that within each class, all states communicate. However, if states i and j belong to different classes, then $i \leftrightarrow j$ is not true (i.e., at most one of $i \to j$ or $j \to i$ can be true).

Definition 3.6. A DTMC that has only one communication class is said to be *irreducible*. On the other hand, a DTMC is called *reducible* if there are two or more communication classes.

Example 3.2. (continued)

What are the communication classes of the DTMC?

Solution.

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0.7 & 0.2 & 0.1 \\ 0 & 0.6 & 0.4 \\ 0.5 & 0 & 0.5 \end{bmatrix} \end{array}$$

It is clear from the diagram that all states are accessible from each other. $\Rightarrow$ all states communicate.

Thus, $\{0, 1, 2\}$ is the only communication class.

$\therefore$ this is an irreducible DTMC.

Example 3.3.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0.5 & 0 & 0.5 & 0 \end{bmatrix} \end{array}.$$

What are the communication classes of this DTMC?

Solution.

All states are accessible from each other.

$\Rightarrow$ all states communicate.

$\therefore \{0, 1, 2, 3\}$ is the only communication class.

$\therefore$ this DTMC is irreducible.

Example 3.4.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & 0 & 0 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 1 & 0 \\ \frac{3}{4} & \frac{1}{4} & 0 & 0 \end{bmatrix} \end{array}.$$

What are the communication classes of this DTMC?

Solution.

Communication classes are $\{0, 1, 3\}$ and $\{2\}$.

$\therefore$ this DTMC is reducible.

Definition 3.7. The *period* of state i is given by

$$d(i) = \gcd\{n \in \mathbb{Z}^+ : P_{i,i}^{(n)} > 0\},$$

where $\gcd\{\}$ denotes the greatest common divisor of a set of positive integers.

Remark. If $d(i) = 1$, then state i is said to be *aperiodic*. In fact, a DTMC is said to be *aperiodic* if $d(i) = 1 \ \forall i \in \mathbb{N}$. Furthermore, if $P_{i,i}^{(n)} = 0 \ \forall n \in \mathbb{Z}^+$, then we set $d(i) = \infty$.

Example 3.5.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \frac{1}{3} & 0 & 0 & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 1 & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \end{bmatrix} \end{array}.$$

Determine the communication classes of this DTMC and the period of each state.

Solution.

Communication classes: $\{0,3\},\{1\},\{2\}$.

$$\begin{aligned} d(0) &= \gcd\{n \in \mathbb{Z}^+ : P_{0,0}^{(n)} > 0\} \\ P_{0,0}^{(1)} &= \frac{1}{3} > 0 \\ P_{0,0}^{(2)} \geq P_{0,0}P_{0,0} &= \left(\frac{1}{3}\right)^2 > 0 \\ P_{0,0}^{(3)} \geq P_{0,0}P_{0,0}P_{0,0} &= \left(\frac{1}{3}\right)^3 > 0 \\ &\vdots \\ \therefore\ d(0) &= \gcd\{1,2,\ldots\} = 1 \end{aligned}$$

In fact, every single term on the main diagonal of P is positive.

$\Rightarrow$ the same argument holds true for every state.

$$d(0) = d(1) = d(2) = d(3) = 1$$

$\therefore$ the DTMC is aperiodic but NOT irreducible.

Definition 3.8. The *period* of state i is given by

$$d(i) = \gcd\{n \in \mathbb{Z}^+ : P_{i,i}^{(n)} > 0\},$$

where gcd{} denotes the greatest common divisor of a set of positive integers.

Remark. If $d(i) = 1$, then state i is said to be *aperiodic*. In fact, a DTMC is said to be *aperiodic* if $d(i) = 1\ \forall i \in \mathbb{N}$. Furthermore, if $P_{i,i}^{(n)} = 0\ \forall n \in \mathbb{Z}^+$, then we set $d(i) = \infty$.

Example 3.5.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \frac{1}{3} & 0 & 0 & \frac{2}{3} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 1 & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \end{bmatrix} \end{array}.$$

Determine the communication classes of this DTMC and the period of each state.

Solution.

Communication classes: $\{0,3\},\{1\},\{2\}$.

$$\begin{aligned} d(0) &= \gcd\{n \in \mathbb{Z}^+ : P_{0,0}^{(n)} > 0\} \\ P_{0,0}^{(1)} &= \frac{1}{3} > 0 \\ P_{0,0}^{(2)} \geq P_{0,0}P_{0,0} &= \left(\frac{1}{3}\right)^2 > 0 \\ P_{0,0}^{(3)} \geq P_{0,0}P_{0,0}P_{0,0} &= \left(\frac{1}{3}\right)^3 > 0 \\ &\vdots \\ \therefore\ d(0) &= \gcd\{1,2,\ldots\} = 1 \end{aligned}$$

In fact, every single term on the main diagonal of P is positive.

$\Rightarrow$ the same argument holds true for every state.

$$d(0) = d(1) = d(2) = d(3) = 1$$

$\therefore$ the DTMC is aperiodic but NOT irreducible.

Example 3.3. (continued)

Recall that there is only one communication class for this DTMC, namely $\{0,1,2,3\}$. Determine the period of each state.

Solution.

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0.5 & 0 & 0.5 & 0 \end{bmatrix}$$

$$\begin{aligned} d(0) &= \gcd\{n \in \mathbb{Z}^+ : P_{0,0}^{(n)} > 0\} = \gcd\{2,4,6,8,\ldots\} = 2 \\ d(1) &= \gcd\{n \in \mathbb{Z}^+ : P_{1,1}^{(n)} > 0\} = \gcd\{2,4,6,8,\ldots\} = 2 \\ d(2) &= \gcd\{n \in \mathbb{Z}^+ : P_{2,2}^{(n)} > 0\} = \gcd\{2,4,6,8,\ldots\} = 2 \\ d(3) &= \gcd\{n \in \mathbb{Z}^+ : P_{3,3}^{(n)} > 0\} = \gcd\{2,4,6,8,\ldots\} = 2 \end{aligned}$$

The above examples illustrate some of the kinds of periodic behaviour that can be exhibited by DTMCs. However, we do observe that among the states within a given communication class,

it seems as though the periodic behaviour is consistent. This is not a coincidence, as the next theorem indicates.

Theorem 3.1. *If $i \leftrightarrow j$, then $d(i) = d(j)$.*

Proof. Clearly this is true when $i = j$. So, assume $i \neq j$.

Since $i \leftrightarrow j$, then by definition we know $P_{i,j}^{(n)} > 0$ for some $n \in \mathbb{Z}^+$ and $P_{j,i}^{(m)} > 0$ for some $m \in \mathbb{Z}^+$.

Also, since state i is accessible from j and vice versa, then $\exists s \in \mathbb{Z}^+$ such that $P_{j,j}^{(s)} > 0$.

Then we have

$$P_{i,i}^{(n+m)} \geq P_{i,j}^{(n)} P_{j,i}^{(m)} > 0$$

and

$$P_{i,i}^{(n+s+m)} \geq P_{i,j}^{(n)} P_{j,j}^{(s)} P_{j,i}^{(m)} > 0.$$

Thus $d(i)$ divides both $n + m$ and $n + s + m$.

$$\therefore \; d(i) \text{ divides } (n + s + m) - (n + m) = s.$$

Since this holds for any s which satisfies $P_{j,j}^{(s)} > 0$, then it must be the case that $d(i)$ divides $d(j)$.

Using the same line of logic, then $d(j)$ divides $d(i)$.

$$\therefore \; d(i) = d(j).$$

□

Solution. State transition diagram. The DTMC is irreducible with communication class $\{0, 1, 2\}$.

$$\begin{aligned} P_{0,0}^{(1)} &= 0 \\ P_{0,0}^{(2)} \geq P_{0,1}P_{1,0} &= \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} > 0 \\ P_{0,0}^{(3)} \geq P_{0,1}P_{1,2}P_{2,0} &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8} > 0 \\ &\Rightarrow \; d(0) = \gcd\{n \in \mathbb{Z}^+ : P_{0,0}^{(n)} > 0\} = \gcd\{2, 3, \ldots\} = 1 \end{aligned}$$

Since the Markov chain DTMC is irreducible by Thm. 3.1,

$$\Rightarrow \; d(2) = d(1) = d(0) = 1.$$

$\therefore$ this DTMC is aperiodic.

Remark. As the above example demonstrates, it is still possible to observe aperiodic behaviour even though the main diagonal components of P are all zero. More generally, if $d(i) = k$, then this does not necessarily imply that $P_{i,i}^{(k)} > 0$. Instead, it implies that if the DTMC is in state i at time 0, then it is impossible to observe the DTMC in state i at time $n \in \mathbb{Z}^+$ if n is not a multiple of k (i.e., $P_{i,i}^{(n)} = 0$ for such n).

3.3 Transience and Recurrence

We now wish to take a closer look at the likelihood of a DTMC beginning in some state of returning to that particular state. To that end, let us consider the probability that, starting from state i, the first visit of the DTMC to state j occurs at time $n \in \mathbb{Z}^+$, to be denoted by

$$f_{i,j}^{(n)} = P(X_n = j, X_{n-1} \neq j, \ldots, X_2 \neq j, X_1 \neq j \mid X_0 = i), \quad \forall i, j \in \mathbb{N}.$$

Clearly, we see that $f_{i,j}^{(1)} = P_{i,j}$. For $n \geq 2$, however, the determination of $f_{i,j}^{(n)}$ becomes more complicated, and so we wish to construct a procedure which will enable us to compute $f_{i,j}^{(n)}$ for such n. To do so, we consider the quantity $P_{i,j}^{(n)}$ and condition on the time that the first visit to state j is made. Specifically, for $n \in \mathbb{Z}^+$, we obtain

$$\begin{aligned}
P_{i,j}^{(n)} &= P(X_n = j \mid X_0 = i) \\
&= \sum_{k=1}^{n} P(X_n = j, \text{ first visit to state } j \text{ occurs at time } k \mid X_0 = i) \\
&= \sum_{k=1}^{n} P(X_n = j,\ X_k = j,\ X_{k-1} \neq j, \ldots, X_2 \neq j,\ X_1 \neq j \mid X_0 = i) \\
&= \sum_{k=1}^{n} P(X_k = j,\ X_{k-1} \neq j, \ldots, X_2 \neq j,\ X_1 \neq j \mid X_0 = i)\, P(X_n = j \mid X_k = j) \\
&= \sum_{k=1}^{n} f_{i,j}^{(k)} P_{j,j}^{(n-k)}, \qquad (3.2)
\end{aligned}$$

where we applied the Markov property in the second last equality. From (3.2), we can write

$$P_{i,j}^{(n)} = f_{i,j}^{(n)} P_{j,j}^{(0)} + \sum_{k=1}^{n-1} f_{i,j}^{(k)} P_{j,j}^{(n-k)} = f_{i,j}^{(n)} + \sum_{k=1}^{n-1} f_{i,j}^{(k)} P_{j,j}^{(n-k)},$$

implying that

$$f_{i,j}^{(n)} = P_{i,j}^{(n)} - \sum_{k=1}^{n-1} f_{i,j}^{(k)} P_{j,j}^{(n-k)}, \quad n \in \mathbb{Z}^+. \qquad (3.3)$$

For $n \geq 2$, note that (3.3) yields a recursive means to compute $f_{i,j}^{(n)}$. If we now define the related quantity

$$f_{i,j} = P(\text{DTMC ever makes a future visit to state } j \mid X_0 = i),$$

then it follows that

$$\begin{aligned} f_{i,j} &= \sum_{k=1}^{\infty} P(\text{DTMC visits state } j \text{ for } 1^{\text{st}} \text{ time at time } k \mid X_0 = i) \\ &= \sum_{k=1}^{\infty} f_{i,j}^{(k)} \leq 1. \end{aligned}$$

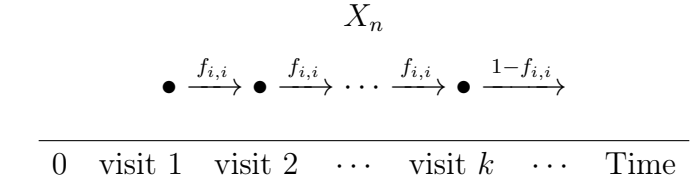
This leads to the following important concept in the study of Markov chains.

Definition 3.9. State i is said to be *transient* if $f_{i,i} < 1$. On the other hand, state i is said to be *recurrent* if $f_{i,i} = 1$.

In what follows, we proceed to look at alternative ways of characterizing the notions of transience and recurrence. As such, let us first define M_i to be a rv which counts the number of (future) times the DTMC visits state i (disregarding the possibility of starting in state i at time 0). If we assume that $f_{i,i} < 1$, then the Markov property and stationary assumption jointly imply that

$$P(M_i = k \mid X_0 = i) = \left(\prod_{n=1}^{k} f_{i,i} \right) (1 - f_{i,i}) = f_{i,i}^k (1 - f_{i,i}), \quad k = 0, 1, 2, \ldots, \tag{3.4}$$

as the DTMC will return to state i k times with probability $f_{i,i}$, and then never return with probability $1 - f_{i,i}$, as visualized below:



We recognize (3.4) as the pmf of a GEO$(1 - f_{i,i})$ rv, thereby implying that

$$\mathbb{E}[M_i \mid X_0 = i] = \frac{1 - (1 - f_{i,i})}{1 - f_{i,i}} = \frac{f_{i,i}}{1 - f_{i,i}} < \infty \quad \text{since } f_{i,i} < 1.$$

However, if $f_{i,i} = 1$, then $P(M_i = \infty \mid X_0 = i) = 1$, immediately implying that $\mathbb{E}[M_i \mid X_0 = i] = \infty$. Therefore, an equivalent way of viewing transience/recurrence is as follows:

$$\mathbb{E}[M_i \mid X_0 = i] = \begin{cases} < \infty, & \text{if state } i \text{ is transient,} \\ \infty, & \text{if state } i \text{ is recurrent.} \end{cases}$$

Following further on the notion of M_i, define a sequence of indicator rvs $\{A_n\}_{n=1}^{\infty}$ such that

$$A_n = \begin{cases} 0, & \text{if } X_n \neq i, \\ 1, & \text{if } X_n = i. \end{cases}$$

With this definition, note that $M_i = \sum_{n=1}^{\infty} A_n$. Now,

$$\begin{aligned}
\mathbb{E}[M_i \mid X_0 = i] &= \mathbb{E}\left[\sum_{n=1}^{\infty} A_n \,\middle|\, X_0 = i\right] \\
&= \sum_{n=1}^{\infty} \mathbb{E}[A_n \mid X_0 = i] \\
&= \sum_{n=1}^{\infty} \{0 \cdot P(A_n = 0 \mid X_0 = i) + 1 \cdot P(A_n = 1 \mid X_0 = i)\} \\
&= \sum_{n=1}^{\infty} P(X_n = i \mid X_0 = i) \\
&= \sum_{n=1}^{\infty} P_{i,i}^{(n)}.
\end{aligned}$$

Therefore, yet another equivalent way of characterizing transience/recurrence is as follows:

$$\sum_{n=1}^{\infty} P_{i,i}^{(n)} = \begin{cases} < \infty, & \text{iff state } i \text{ is transient,} \\ = \infty, & \text{iff state } i \text{ is recurrent.} \end{cases}$$

Remark. A simple way of viewing these concepts is as follows: a recurrent state will be visited infinitely often, whereas a transient state will only be visited finitely often.

As was the case concerning the periodicity of states within the same communication class, the next theorem indicates that transience/recurrence is also a class property.

Theorem 3.2. *If $i \leftrightarrow j$ and state i is recurrent, then state j is recurrent.*

Proof. Since $i \leftrightarrow j$ then $\exists n, m \in \mathbb{N}$ s.t. $P_{i,j}^{(n)} > 0$ and $P_{j,i}^{(m)} > 0$.

Also since state i is recurrent we know

$$\sum_{n=1}^{\infty} P_{i,i}^{(n)} = \infty.$$

For each $s \in \mathbb{Z}^+$, the Chapman–Kolmogorov equations give

$$P_{j,j}^{(m+s+n)} \geq P_{j,i}^{(m)} P_{i,i}^{(s)} P_{i,j}^{(n)}.$$

Consider,

$$\begin{aligned}\sum_{k=1}^{\infty} P_{j,j}^{(k)} &\geq \sum_{s=1}^{\infty} P_{j,j}^{(m+s+n)} \\ &\geq P_{j,i}^{(m)} P_{i,j}^{(n)} \sum_{s=1}^{\infty} P_{i,i}^{(s)} = \infty.\end{aligned}$$

$$\therefore \text{ since } \sum_{k=1}^{\infty} P_{j,j}^{(k)} = \infty, \text{ then state } j \text{ is also recurrent.}$$

□

Remark. An obvious by-product of this theorem is that if $i \leftrightarrow j$ and state i is transient, then state j must also be transient.

The following theorem serves as a companion result to Theorem 3.2.

Theorem 3.3. *If $i \leftrightarrow j$ and state i is recurrent, then*

$$f_{i,j} = P(\textit{DTMC ever makes a future visit to state } j \mid X_0 = i) = 1.$$

Proof. Clearly, the result is true if $i = j$. Therefore, suppose that $i \neq j$. Since $i \leftrightarrow j$, there exists $n_i \in \mathbb{Z}^+$ such that

$$p_0 = P_{i,j}^{(n_i)} > 0.$$

Because state i is recurrent, the DTMC returns to state i infinitely often with probability one. Each time it visits state i, the Markov property implies that the probability of visiting state j within the next n_i time units is at least p_0. Consequently, after r such opportunities, the probability of avoiding state j is at most $(1 - p_0)^r$. Since

$$\lim_{r \to \infty} (1 - p_0)^r = 0,$$

the DTMC eventually visits state j with probability one. Thus, $f_{i,j} = 1$.

□

Remark. Based on the above result, we know that, starting from any state of a recurrent class, a DTMC will visit each state of that class infinitely many times.

At this stage, a natural question to ask is "What do the results that we have accumulated thus far tell us about the behaviour of states within the same communication class?" The answer would be:

1. these states communicate with each other,
2. these states all have the same period,
3. these states are either all recurrent or all transient.

In fact, in an irreducible DTMC, there is only one communication class and so all the states are either recurrent or transient. When the assumption that the DTMC has a finite number of states is included, we obtain the following important result.

Theorem 3.4. *A finite-state DTMC has at least one recurrent state.*

Proof. We wish to prove the existence of at least one recurrent state in a finite-state DTMC, or equivalently, that not all states can be transient. Suppose that $\{0, 1, \ldots, N\}$ represents the states of the DTMC where $N < \infty$. To prove that not all states can be transient, suppose they are and try to get a contradiction.

Now, for each $i = 0, 1, \ldots, N$, if state i is assumed to be transient, then we know that after a finite amount of time (denoted by T_i), state i will never be visited again. As a result, after a finite amount of time $T = \max\{T_0, T_1, \ldots, T_N\}$ has gone by, none of these states will be visited ever again. However, the DTMC must be in some state after time T, but we have exhausted all states for the DTMC to be in. This is our contradiction. Thus, not all states can be transient in a finite-state DTMC.

□

Remarks.

1. Looking at the above result, it is useful to think of it in the following way. There must be at least one recurrent state. After all, the DTMC has to spend its time somewhere, and if it visits each of its finitely many states finitely many times, then where else could it possibly go?
2. As an immediate consequence of Theorem 3.4, an irreducible, finite-state DTMC must be recurrent (i.e., all states of the DTMC are recurrent).

Example 3.3. (continued)

Recall that we considered the irreducible DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0.5 & 0 & 0.5 & 0 \end{bmatrix} \end{array}.$$

Determine whether each state is transient or recurrent.

Solution. Earlier we established that the DTMC is irreducible since this is already a finite-state DTMC as well. Each of the states $0, 1, 2, 3$ are recurrent.

Another interesting property concerning recurrence can also be deduced.

Theorem 3.5. *If state i is recurrent and state i does not communicate with state j, then*

$$P_{i,j}^{(k)} = 0 \quad \forall k \in \mathbb{Z}^+.$$

Proof. Suppose, to the contrary, that $P_{i,j}^{(k)} > 0$ for some $k \in \mathbb{Z}^+$. Because state i is recurrent, a DTMC starting in state i must return to state i with probability one. On the event $\{X_k = j\}$, which has positive probability, the process must therefore be able to reach state i from state j; otherwise, it could never return to state i. Hence $j \to i$. Since $P_{i,j}^{(k)} > 0$ also gives $i \to j$, it follows that $i \leftrightarrow j$, contradicting the assumption. Therefore, $P_{i,j}^{(k)} = 0$ for every $k \in \mathbb{Z}^+$.

□

Example 3.4. (continued)

Recall our earlier DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & 0 & 0 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 1 & 0 \\ \frac{3}{4} & \frac{1}{4} & 0 & 0 \end{bmatrix} \end{array}.$$

Determine whether each state is transient or recurrent.

Solution. $\{0, 1, 3\}$ and $\{2\}$ are the communication classes.

$$\sum_{n=1}^{\infty} P_{2,2}^{(n)} = \sum_{n=1}^{\infty} 1 = \infty \Rightarrow \text{ state 2 is recurrent.}$$

We suspect state 1 to be transient because there is positive probability of never returning to state 1 if going to state 2 occurs. Assume state 1 is recurrent. Then state 1 does NOT communicate with state 2. Then $P_{1,2}^{(n)} = 0$. However from the TPM, $P_{1,2} = \frac{1}{8} \neq 0$, contradiction!

$\therefore$ state 1 must be transient.

Remark. As the above example illustrates, the contrapositive of Theorem 3.5 also provides a test for transience, in that if $\exists k \in \mathbb{Z}^+$ such that $P_{i,j}^{(k)} > 0$ and states i and j do not communicate, then state i must be transient. Moreover, this result implies that once a process enters a recurrent class of states, it can never leave that class. For this reason, a recurrent class is often referred to as a closed class.

Example 3.8.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} \frac{1}{4} & 0 & \frac{3}{4} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & 0 & 1 & 0 \\ 0 & \frac{2}{5} & 0 & \frac{3}{5} \end{bmatrix} \end{array}.$$

Determine whether each state is transient or recurrent.

Solution. $\{0\}$, $\{1,3\}$ and $\{2\}$ are communication classes.

$$\sum_{n=1}^{\infty} P_{0,0}^{(n)} = \sum_{n=1}^{\infty} \left(\frac{1}{4}\right)^n = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{1}{3} < \infty \quad \therefore \text{ state 0 is transient.}$$
$$\sum_{n=1}^{\infty} P_{2,2}^{(n)} = \sum_{n=1}^{\infty} 1 = \infty \quad \therefore \text{ state 2 is recurrent.}$$

$\{1,3\} : f_{1,1}^{(1)} = P_{1,1} = \frac{1}{3}$.

$$f_{1,1}^{(n)} = \left(\frac{2}{3}\right)\left(\frac{2}{5}\right)\left(\frac{3}{5}\right)^{n-2} \quad (n \geq 2)$$
$$\begin{aligned} f_{1,1} &= f_{1,1}^{(1)} + \sum_{n=2}^{\infty} f_{1,1}^{(n)} \\ &= \frac{1}{3} + \sum_{n=2}^{\infty} \left(\frac{2}{3}\right)\left(\frac{2}{5}\right)\left(\frac{3}{5}\right)^{n-2} \\ &= \frac{1}{3} + \frac{2}{3}\cdot\frac{2}{5}\sum_{n=2}^{\infty}\left(\frac{3}{5}\right)^{n-2} \\ &= \frac{1}{3} + \frac{2}{3}\cdot\frac{2}{5}\cdot\frac{1}{1-\frac{3}{5}} = 1 \quad \therefore \{1,3\} \text{ is recurrent.} \end{aligned}$$

Remark. Instead of showing that $f_{1,1} = 1$ in the above example, we could have used an even simpler argument to conclude that $\{1,3\}$ is a recurrent class. In particular, after establishing that $\{0\}$ and $\{2\}$ are transient classes, this DTMC has a finite number of states, and so $\{1,3\}$ must be recurrent due to Theorem 3.4.

Example 3.9.

Consider a DTMC $\{X_n, n \in \mathbb{N}\}$ whose state space S is the set of all integers (i.e., $S = \mathbb{Z}$). Furthermore, suppose that the TPM for this DTMC satisfies

$$P_{i,i-1} = 1 - p \quad \text{and} \quad P_{i,i+1} = p \quad \forall i \in \mathbb{Z} \text{ where } 0 < p < 1.$$

In other words, from any state, either a jump up by one unit or a jump down by one unit takes place in the next transition. As such, X_n is expressed as

$$X_n = \sum_{k=1}^{n} Y_k,$$

where $\{Y_k\}_{k\geq 1}$ is an independent sequence of rvs with

$$Y_k = 1 \text{ w.p. } p \quad \text{and} \quad Y_k = -1 \text{ w.p. } 1-p, \quad k \in \mathbb{Z}^+.$$

This DTMC is well-studied in the literature and is the basis for many applications in a variety of areas (particularly in finance). It is often referred to as the *Random Walk* or *Drunkard's Walk*. Characterize the behaviour of this DTMC in terms of its communication classes, periodicity, and transience/recurrence.

Solution. Since $0 < p < 1$, all states communicate. Thus this DTMC is irreducible.

Starting from state 0, note that 0 cannot possibly be visited in an odd number of transitions; i.e.,

$$P_{0,0}^{(1)} = P_{0,0}^{(3)} = P_{0,0}^{(5)} = \cdots = 0.$$

But it is possible in even # of transitions, i.e. $P_{0,0}^{(2n)} > 0$, $\forall n \in \mathbb{Z}^+$. Thus

$$d(0) = \gcd\{n \in \mathbb{Z}^+ : P_{0,0}^{(n)} > 0\} = \gcd\{2, 4, 6, \ldots\} = 2.$$

To determine if 0 is transient or recurrent, consider:

$$\sum_{n=1}^{\infty} P_{0,0}^{(n)} = \sum_{n=1}^{\infty} P_{0,0}^{(2n)} = \sum_{n=1}^{\infty} \binom{2n}{n} p^n (1-p)^n.$$

Use the Ratio Test: Let $a_n = \binom{2n}{n} p^n (1-p)^n$ where if $L < 1$ converges and $L > 1$ diverges.

$$\begin{aligned} L &= \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{\binom{2n+2}{n+1} p^{n+1}(1-p)^{n+1}}{\binom{2n}{n} p^n (1-p)^n} \\ &= \lim_{n\to\infty} \frac{(2n+2)(2n+1)}{(n+1)(n+1)} p(1-p) = 4p(1-p). \end{aligned}$$

If $p \neq \frac{1}{2} \Rightarrow L < 1 \Rightarrow \sum_{n=1}^{\infty} \binom{2n}{n} p^n (1-p)^n < \infty \therefore$ state 0 is transient. Alternatively, define

$$s_{i,j} = P(\text{DTMC ever makes a future visit to state } j \mid X_0 = i).$$

Recall $f_{i,j} = P(\text{DTMC ever makes a future visit to state } j \mid X_0 = i)$, so we have

$$\begin{aligned} s_{0,0} &= P(A_0 \mid X_0 = 0) \\ &= P(X_1 = -1 \mid X_0 = 0)P(A_0 \mid X_1 = -1, X_0 = 0) \\ &\quad + P(X_1 = 1 \mid X_0 = 0)P(A_0 \mid X_1 = 1, X_0 = 0) \\ &= qs_{-1,0} + ps_{1,0}, \quad \text{where } q = 1 - p. \end{aligned}$$

By spatial homogeneity and the Markov property, $s_{i,0} = s_{1,0}^{i}$ for $i \geq 1$ and $s_{i,0} = s_{-1,0}^{-i}$ for $i \leq -1$. First-step analysis from states 1 and -1 gives

$$s_{1,0} = q + ps_{1,0}^2, \qquad\qquad s_{-1,0} = p + qs_{-1,0}^2.$$

Taking the smallest non-negative solution of each quadratic yields

$$s_{1,0} = \begin{cases} 1, & p \leq q, \\ q/p, & p > q, \end{cases} \qquad\qquad s_{-1,0} = \begin{cases} p/q, & p < q, \\ 1, & p \geq q. \end{cases}$$

Therefore,

$$\begin{aligned} s_{0,0} &= qs_{-1,0} + ps_{1,0} \\ &= \begin{cases} 2p, & p < \frac{1}{2}, \\ 1, & p = \frac{1}{2}, \\ 2q, & p > \frac{1}{2}, \end{cases} \\ &= 1 - |2p - 1|. \end{aligned}$$

If $p \neq \frac{1}{2}$, then $s_{0,0} < 1$. Thus if $p \neq \frac{1}{2}$, state 0 is transient. If $p = \frac{1}{2}$, then $s_{0,0} = 1$. Thus if $p = \frac{1}{2}$, state 0 is recurrent. Hence,

$$\text{state 0 is recurrent iff } p = \frac{1}{2}.$$

Since the DTMC is irreducible, this result applies to all states.

$$\therefore \text{ the random walk on } \mathbb{Z} \text{ is recurrent iff } p = \frac{1}{2},$$

and transient otherwise.

3.4 Limiting Behaviour of DTMCs

The concepts of periodicity and transience/recurrence play an important role in characterizing the limiting behaviour of a DTMC. To demonstrate their influence, let us consider three examples with varying forms of limiting behaviour.

Example 3.10.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{array}.$$

Determine if $\lim_{n\to\infty} P^{(n)}$ exists.

Solution. There are obviously two communication classes, namely $\{0, 2\}$ and $\{1\}$. Each class is recurrent with periods 2 and 1, respectively. Moreover, for $n \in \mathbb{Z}^+$, note that

$$P^{(2n)} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \quad \text{and} \quad P^{(2n-1)} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{array}.$$

As a result, $\lim_{n\to\infty} P^{(n)}$ does not exist since $P^{(n)}$ alternates between these two matrices. For example, $\lim_{n\to\infty} P^{(n)}_{0,0}$ and $\lim_{n\to\infty} P^{(n)}_{0,2}$ do not exist. However, note that some limits do exist such as $\lim_{n\to\infty} P^{(n)}_{0,1} = 0$ and $\lim_{n\to\infty} P^{(n)}_{1,1} = 1$.

Example 3.11.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{bmatrix} \end{array}.$$

Determine if $\lim_{n\to\infty} P^{(n)}$ exists.

Solution. There is clearly only one communication class, and so the DTMC is irreducible. It is also straightforward to verify that the DTMC is aperiodic and recurrent. As we will soon learn, it can be shown that

$$\lim_{n\to\infty} P^{(n)} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} \frac{1}{10} & \frac{3}{10} & \frac{3}{5} \\ \frac{1}{10} & \frac{3}{10} & \frac{3}{5} \\ \frac{1}{10} & \frac{3}{10} & \frac{3}{5} \end{bmatrix} \end{array}.$$

Note that this limiting matrix has identical rows. This implies that $P_{i,j}^{(n)}$ converges to a value as $n \to \infty$ which is the same for all initial states i. In other words, there is a limiting probability that the process will be in state j as $n \to \infty$, and this probability is independent of the initial state i.

Example 3.12.

Consider the DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ 0 & 0 & 1 \end{bmatrix} \end{array}.$$

Examine $\lim_{n\to\infty} P^{(n)}$, and explain why the limiting probability of being in a state can depend on the initial state of this DTMC.

Solution. Clearly, there are three communication classes: $\{0\}$, $\{1\}$, and $\{2\}$. As all main diagonal elements of P are positive, each state is aperiodic. Moreover, states 0 and 2 are recurrent states. In fact, for obvious reasons, states 0 and 2 are examples of what are known as *absorbing states* (i.e., $P_{0,0} = P_{2,2} = 1$). Finally, since $P_{1,0} = \frac{1}{3} > 0$ and states 0 and 1 do not communicate, we conclude that state 1 is transient by the contrapositive of Theorem 3.5. It can be shown that (see the absorption-probability calculation later in this chapter)

$$\lim_{n\to\infty} P^{(n)} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix} \end{array}.$$

In looking at the limiting form of $P^{(n)}$, we remark that unlike the previous example, it can matter from which state one begins in this DTMC.

In the above example, note that the second column of the limiting matrix contains all zeros. Not surprisingly, this is indicative of transient behaviour, implying that one will never end up in state 1 in the long run. This property can be proven more formally in the next theorem.

Theorem 3.6. *For any state i and transient state j of a DTMC, $\lim_{n\to\infty} P_{i,j}^{(n)} = 0$.*

Proof.

$$f_{i,j}^{(n)} = P(X_n = j, X_{n-1} \neq j, \ldots, X_2 \neq j, X_1 \neq j \mid X_0 = i),$$
$$f_{i,j} = P(\text{DTMC ever makes a future visit to state } j \mid X_0 = i) = \sum_{n=1}^{\infty} f_{i,j}^{(n)}.$$

Use (3.2) as a starting point:

$$\begin{aligned}
P_{i,j}^{(n)} &= \sum_{k=1}^{n} f_{i,j}^{(k)} P_{j,j}^{(n-k)}. \\
\sum_{n=1}^{\infty} P_{i,j}^{(n)} &= \sum_{n=1}^{\infty} \sum_{k=1}^{n} f_{i,j}^{(k)} P_{j,j}^{(n-k)} \\
&= \sum_{k=1}^{\infty} \sum_{n=k}^{\infty} f_{i,j}^{(k)} P_{j,j}^{(n-k)} \\
&= \sum_{k=1}^{\infty} f_{i,j}^{(k)} \sum_{n=k}^{\infty} P_{j,j}^{(n-k)} \\
&= \sum_{k=1}^{\infty} f_{i,j}^{(k)} \left(1 + \sum_{\ell=1}^{\infty} P_{j,j}^{(\ell)}\right) \\
&= f_{i,j} \left(1 + \sum_{\ell=1}^{\infty} P_{j,j}^{(\ell)}\right) \leq 1 + \sum_{\ell=1}^{\infty} P_{j,j}^{(\ell)} < \infty \quad \text{since state } j \text{ is transient.}
\end{aligned}$$

Since $\sum_{n=1}^{\infty} P_{i,j}^{(n)} < \infty$, then by the nth term test of infinite series,

$$\therefore \quad \lim_{n\to\infty} P_{i,j}^{(n)} = 0.$$

□

As the previous three examples show, there is variation in the limiting behaviour of a DTMC. In particular, it is worthwhile to determine a set of conditions which ensure the "nice" limiting behaviour witnessed in Example 3.11. To ascertain when such conditions exist, we need to distinguish between two kinds of recurrence. Let

$$N_i = \min\{n \in \mathbb{Z}^+ : X_n = i\},$$

where state i is assumed to be recurrent. Clearly, the conditional rv $N_i \mid (X_0 = i)$ takes on values in $\mathbb{Z}^+$. Moreover, its conditional pmf is given by

$$P(N_i = n \mid X_0 = i) = f_{i,i}^{(n)}, \quad n = 1, 2, 3, \ldots.$$

We observe that this is indeed a pmf since $\sum_{n=1}^{\infty} f_{i,i}^{(n)} = f_{i,i} = 1$, as state i is recurrent. This leads to the introduction of the following important quantity.

Definition 3.10. If state i is recurrent, then its *mean recurrent time* is given by

$$m_i = \mathbb{E}[N_i \mid X_0 = i] = \sum_{n=1}^{\infty} n f_{i,i}^{(n)}.$$

In words, m_i represents the average time it takes the DTMC to make successive visits to state i. Two notions of recurrence can now be defined based on the value of m_i.

Definition 3.11. Suppose that state i is recurrent. State i is said to be *positive recurrent* if $m_i < \infty$. On the other hand, state i is said to be *null recurrent* if $m_i = \infty$.

Remark. A fair question to ask is whether it is even possible for a discrete probability distribution on $\mathbb{Z}^+$ to have an undefined mean (i.e., a mean of ∞). To show that this is indeed possible, consider a rv X with pmf

$$P(X = x) = \frac{1}{x(x+1)}, \quad x = 1, 2, 3, \ldots.$$

Let us first confirm that this is indeed a pmf:

$$\begin{aligned}
\sum_{x=1}^{\infty} \frac{1}{x(x+1)} &= \lim_{n\to\infty} \sum_{x=1}^{n} \frac{1}{x(x+1)} \\
&= \lim_{n\to\infty} \sum_{x=1}^{n} \left(\frac{1}{x} - \frac{1}{x+1} \right) \\
&= \lim_{n\to\infty} \left\{ \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \right\} \\
&= \lim_{n\to\infty} \left(1 - \frac{1}{n+1}\right) \\
&= 1.
\end{aligned}$$

However,

$$\mathbb{E}[X] = \sum_{x=1}^{\infty} x \frac{1}{x(x+1)} = \sum_{x=1}^{\infty} \frac{1}{x+1} = \infty,$$

since the above harmonic series is known to diverge. In other words, a finite mean does not exist!

3.4.1 Some Facts About Positive and Null Recurrence

1. If $i \leftrightarrow j$ and state i is positive recurrent, then state j is also positive recurrent. This means that positive recurrence is also a class property. An obvious by-product of this result is that null recurrence is also a class property too.

2. In a finite-state DTMC, there can never be any null recurrent states.

Remarks.

1. The above facts are provided without formal justification, as their proofs are rather lengthy and depend on material beyond the scope of STAT 333.
2. Positive recurrent, aperiodic states are referred to as *ergodic states*.

Before stating the main result governing the "nice" limiting behaviour demonstrated in Example 3.11, we introduce a special type of probability distribution.

Definition 3.12. A probability distribution $\{p_i\}_{i=0}^{\infty}$ is called a *stationary distribution* of a DTMC if $\{p_i\}_{i=0}^{\infty}$ satisfies the conditions $\sum_{i=0}^{\infty} p_i = 1$ and

$$p_j = \sum_{i=0}^{\infty} p_i P_{i,j} \quad \forall j \in \mathbb{N}.$$

Remark. If we define the row vector $\mathbf{p} = (p_0, p_1, \ldots, p_j, \ldots)$, then the above conditions can be represented in matrix form as

$$\mathbf{pe} = 1 \quad \text{and} \quad \mathbf{p} = \mathbf{p}P,$$

where $\mathbf{e} = (1, 1, \ldots, 1, \ldots)'$ denotes a column vector of ones (in general, the prime notation will be used to represent column vectors). A logical question to ask is "Why is such a distribution called stationary?" To answer this question, suppose that the initial conditions of the DTMC are given by $\boldsymbol{\alpha}_0 = \mathbf{p}$. As a result, we have that $\alpha_{0,j} = P(X_0 = j) = p_j \ \forall j \in \mathbb{N}$. Now, for any $j \in \mathbb{N}$, note that

$$\alpha_{1,j} = P(X_1 = j) = \sum_{i=0}^{\infty} \alpha_{0,i} P_{i,j} = \sum_{i=0}^{\infty} p_i P_{i,j} = p_j = \alpha_{0,j}.$$

The above equation indicates that X_1 has the same probability distribution as X_0 when $\boldsymbol{\alpha}_0 = \mathbf{p}$. More generally, it is straightforward to show (using mathematical induction) that each X_n, $n \in \mathbb{Z}^+$, is identically distributed to X_0, providing that $\boldsymbol{\alpha}_0 = \mathbf{p}$. In other words, if a DTMC is started according to a stationary distribution, then the probability of being in a given state remains unchanged (i.e., stationary) over time.

Remarks.

1. In some texts, the stationary probability distribution is sometimes called the *invariant probability distribution* or *steady-state probability distribution*.
2. A known fact (which again we do not prove formally) is that a stationary distribution will not exist if all the states of the DTMC are either null recurrent or transient. On the other hand, an irreducible DTMC is positive recurrent iff a stationary distribution exists.
3. Stationary distributions are not necessarily unique. This happens when a DTMC has more than one positive recurrent communication class. For instance, it is not difficult to verify that the DTMC in Example 3.10 has an infinite number of stationary distributions (left as an upcoming exercise).

We are now in position to state the fundamental limiting theorem for DTMCs, generally referred to as the Basic Limit Theorem (BLT).

Theorem 3.7 (Basic Limit Theorem)**.** *For an irreducible, recurrent, and aperiodic DTMC,* $\lim_{n\to\infty} P_{i,j}^{(n)}$ *exists and is independent of state* i*, satisfying*

$$\lim_{n\to\infty} P_{i,j}^{(n)} = \pi_j = \frac{1}{m_j}, \quad \forall j \in \mathbb{N}.$$

If the DTMC also happens to be positive recurrent, then $\{\pi_j\}_{j=0}^{\infty}$ *is the unique, positive solution to the system of linear equations defined by*

$$\begin{cases} \pi_j = \sum_{i=0}^{\infty} \pi_i P_{i,j}, & \forall j \in \mathbb{N}, \\ \sum_{j=0}^{\infty} \pi_j = 1. \end{cases}$$

Remarks.

1. A formal proof of the BLT is beyond the scope of STAT 333. However, it is not difficult to understand why $\{\pi_j\}_{j=0}^{\infty}$ (if they exist) satisfies the above system of linear equations. Specifically, recall the Chapman–Kolmogorov equations with $m = n - 1$, namely

   $$P_{i,j}^{(n)} = \sum_{k=0}^{\infty} P_{i,k}^{(n-1)} P_{k,j}, \quad \forall i, j \in \mathbb{N}.$$

   Taking the limit as $n \to \infty$ of both sides of this equation and assuming that it is permissible to pass the limit through the summation sign, we obtain

   $$\lim_{n\to\infty} P_{i,j}^{(n)} = \lim_{n\to\infty} \sum_{k=0}^{\infty} P_{i,k}^{(n-1)} P_{k,j},$$
   $$\pi_j = \sum_{k=0}^{\infty} \lim_{n\to\infty} P_{i,k}^{(n-1)} P_{k,j} = \sum_{k=0}^{\infty} \pi_k P_{k,j}, \quad \forall j \in \mathbb{N},$$

   which is precisely the above system of linear equations.
2. If we define the row vector of limiting probabilities

   $$\boldsymbol{\pi} = (\pi_0, \pi_1, \ldots, \pi_j, \ldots),$$

   then the above system of linear equations can be written succinctly in matrix form as:

   $$\begin{cases} \boldsymbol{\pi} = \boldsymbol{\pi} P, \\ \boldsymbol{\pi} \mathbf{e}' = 1. \end{cases}$$

   Therefore, if a DTMC is irreducible and ergodic, then the BLT states that the limiting probability distribution is the unique stationary distribution.

3. When a DTMC has a finite number of states (i.e., suppose that the state space is $\{0, 1, \ldots, N\}$ where $N < \infty$), the BLT states that there are $N + 1$ linear equations to consider of the form

$$\pi_j = \sum_{i=0}^{N} \pi_i P_{i,j}, \quad j = 0, 1, \ldots, N. \tag{3.8}$$

Along with the condition $\sum_{j=0}^{N} \pi_j = 1$, this leads to $N + 2$ equations in $N + 1$ unknowns, of which a unique solution must exist. In fact, the first $N + 1$ equations given by (3.8) are linearly dependent (implying that there is a redundancy), and so we can drop any one of the equations given by (3.8) and solve the remaining $N + 1$ equations to obtain a unique solution.

4. If the conditions of the BLT are satisfied and state j happens to be null recurrent, then $\pi_j = 0$, which interestingly is similar to the limiting behaviour of a transient state.

Example 3.11. (continued)

Recall that we previously considered a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{bmatrix} \end{array}.$$

Find the limiting probabilities for this DTMC.

Solution. Clearly, this DTMC is irreducible, aperiodic & recurrent, and positive recurrent by finiteness. The conditions of the BLT are satisfied and $\boldsymbol{\pi} = (\pi_0, \pi_1, \pi_2)$ is known to exist. Find $\boldsymbol{\pi}$ to solve the system of linear eq. via $\boldsymbol{\pi} = \boldsymbol{\pi} P$.

$$(\pi_0, \pi_1, \pi_2) = (\pi_0, \pi_1, \pi_2) \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \\ 0 & \frac{1}{3} & \frac{2}{3} \end{bmatrix} \quad \text{s.t. } \boldsymbol{\pi}\mathbf{e}' = 1 \text{ and } \pi_0 + \pi_1 + \pi_2 = 1$$

$$\begin{cases} \pi_0 = \frac{1}{4}\pi_0 + \frac{1}{4}\pi_1 & (1) \\ \pi_1 = \frac{1}{4}\pi_0 + \frac{1}{4}\pi_1 + \frac{1}{3}\pi_2 & (2) \\ \pi_2 = \frac{1}{2}\pi_0 + \frac{1}{2}\pi_1 + \frac{2}{3}\pi_2 & (3) \end{cases}$$

$\iff$ remove (2) with all variables

$$\pi_0 + \pi_1 + \pi_2 = 1$$

We may disregard any one of the first 3 equations and so we select the eqn for π_1 since it involves the most terms.

$$\begin{aligned}
\pi_0 &= \frac{1}{4}\pi_0 + \frac{1}{4}\pi_1 \;\Rightarrow\; \frac{3}{4}\pi_0 = \frac{1}{4}\pi_1 \;\Rightarrow\; \pi_1 = 3\pi_0 \\
\pi_2 &= \frac{1}{2}\pi_0 + \frac{1}{2}\pi_1 + \frac{2}{3}\pi_2 \;\Rightarrow\; \frac{1}{3}\pi_2 = \frac{1}{2}(\pi_0 + \pi_1) \;\Rightarrow\; \pi_2 = 6\pi_0 \\
1 &= \pi_0 + \pi_1 + \pi_2 = \pi_0 + 3\pi_0 + 6\pi_0 = 10\pi_0 \;\Rightarrow\; \pi_0 = \frac{1}{10} \\
\pi_1 &= 3\pi_0 = \frac{3}{10}, \qquad \pi_2 = 6\pi_0 = \frac{3}{5}.
\end{aligned}$$

$$\lim_{n\to\infty} P^{(n)} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} \frac{1}{10} & \frac{3}{10} & \frac{3}{5} \\ \frac{1}{10} & \frac{3}{10} & \frac{3}{5} \\ \frac{1}{10} & \frac{3}{10} & \frac{3}{5} \end{bmatrix} \end{array} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{bmatrix} \pi_0 & \pi_1 & \pi_2 \\ \pi_0 & \pi_1 & \pi_2 \\ \pi_0 & \pi_1 & \pi_2 \end{bmatrix}.$$

$\therefore$ observe each row of $P^{(n)} \to \boldsymbol{\pi}$ as $n \to \infty$.

Recall that the TPM of a DTMC is *stochastic*, with all row sums of P being equal to 1. However, a TPM is said to be *doubly stochastic* if all column sums of P are also equal to 1 (i.e., $\sum_{i=0}^{\infty} P_{i,j} = 1 \;\forall j \in \mathbb{N}$). The following theorem provides an interesting result concerning the limiting behaviour of a class of such DTMCs.

Theorem 3.8. *Suppose that a finite-state DTMC with state space $S = \{0, 1, \ldots, N-1\}$ is irreducible and aperiodic. If the associated TPM is doubly stochastic, then the limiting probabilities $\{\pi_j\}_{j=0}^{N-1}$ exist and are given by*

$$\pi_j = \frac{1}{N}, \quad j = 0, 1, \ldots, N-1.$$

Proof. If the DTMC is irreducible and aperiodic then we know that it must also be positive recurrent. In other words, the conditions of the BLT apply here. To determine the unique limiting distribution, let us propose that:

$$\pi_j = \frac{1}{N} \text{ for } j = 0, 1, \ldots, N-1 \;\Rightarrow\; \sum_{j=0}^{N-1} \pi_j = \sum_{j=0}^{N-1} \frac{1}{N} = 1.$$

Moreover, for $j = 0, \ldots, N-1$, note that

$$\sum_{k=0}^{N-1} \pi_k P_{k,j} = \sum_{k=0}^{N-1} \frac{1}{N} P_{k,j} = \frac{1}{N} \sum_{k=0}^{N-1} P_{k,j} = \frac{1}{N} \cdot 1 = \frac{1}{N} = \pi_j.$$

Since the linear equations of the BLT are satisfied, this must be the unique limiting distribution of the DTMC.

□

The primary interpretation of the limiting distribution of a DTMC is that after the process has been in operation for a "long" period of time, the probability of finding the process in state j is π_j (assuming the conditions of the BLT are met). In such situations, however, another interpretation exists for π_j. Specifically, π_j also represents the "long-run" mean fraction of time that the process spends in state j. To see that this interpretation is valid, define the sequence of indicator rvs $\{A_k\}_{k=1}^n$ as follows:

$$A_k = \begin{cases} 0, & \text{if } X_k \neq j, \\ 1, & \text{if } X_k = j. \end{cases}$$

The fraction of time the DTMC visits state j during the time interval from 1 to n inclusive is therefore given by

$$\frac{1}{n}\sum_{k=1}^{n} A_k.$$

Looking at the quantity

$$\mathbb{E}\left[\frac{1}{n}\sum_{k=1}^{n} A_k \,\middle|\, X_0 = i\right],$$

which is interpreted as the mean fraction of time spent in state j during the time interval from 1 to n inclusive, given that the process starts in state i, note that

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{n}\sum_{k=1}^{n} A_k \,\middle|\, X_0 = i\right] &= \frac{1}{n}\sum_{k=1}^{n} \mathbb{E}[A_k \mid X_0 = i] \\
&= \frac{1}{n}\sum_{k=1}^{n} \big(0 \cdot P(A_k = 0 \mid X_0 = i) + 1 \cdot P(A_k = 1 \mid X_0 = i)\big) \\
&= \frac{1}{n}\sum_{k=1}^{n} P(X_k = j \mid X_0 = i) \\
&= \frac{1}{n}\sum_{k=1}^{n} P_{i,j}^{(k)}.
\end{aligned}$$

Recall: If $\{a_n\}_{n=1}^{\infty}$ is a real sequence such that $a_n \to a$ as $n \to \infty$, then $\frac{1}{n}\sum_{k=1}^{n} a_k \to a$ as $n \to \infty$.

Thus, if the conditions of the BLT are satisfied, then $P_{i,j}^{(n)} \to \pi_j$ as $n \to \infty$. Therefore, applying the above result with $a_n = P_{i,j}^{(n)}$ and $a = \pi_j$, we obtain

$$\mathbb{E}\left[\frac{1}{n}\sum_{k=1}^{n} A_k \,\middle|\, X_0 = i\right] \to \pi_j \quad \text{as } n \to \infty,$$

implying that the long-run mean fraction of time spent in state j is also equal to π_j.

Remark. If one begins in recurrent state j, we realize that the process spends one unit of time in state j every N_j time units. On average, this amounts to one unit of time in state j every $\mathbb{E}[N_j \mid X_0 = j] = m_j$ time units. If the conditions of the BLT are satisfied, then it makes sense intuitively that $\pi_j = 1/m_j$, as the BLT specifies. For a more formal justification in the positive recurrent case, let $\{N_j^{(n)}\}_{n=1}^{\infty}$ be a sequence of rvs where $N_j^{(n)}$ represents the number of transitions between the $(n-1)^{\text{th}}$ and n^{th} visits into state j, as illustrated in the diagram below. By the Markov property and the stationary assumption of the DTMC, $\{N_j^{(n)}\}_{n=1}^{\infty}$ is actually an iid sequence of rvs with common mean $m_j < \infty$. Therefore, the long-run fraction of time spent in state j can be viewed as

$$\pi_j = \lim_{n\to\infty} \frac{n}{\sum_{i=1}^{n} N_j^{(i)}} = \lim_{n\to\infty} \frac{1}{\frac{1}{n}\sum_{i=1}^{n} N_j^{(i)}} = \frac{1}{m_j},$$

where the last equality follows from the SLLN.

$$\begin{aligned}
\text{DTMC begins in state } j \longrightarrow & \underbrace{N_j^{(1)}}_{\text{1st visit back to state } j} \longrightarrow \underbrace{N_j^{(2)}}_{\text{2nd visit back to state } j} \\
\longrightarrow & \underbrace{N_j^{(3)}}_{\text{3rd visit back to state } j} \longrightarrow \cdots \\
\longrightarrow & \underbrace{N_j^{(n-1)}}_{\text{(n–1)th visit back to state } j} \longrightarrow \underbrace{N_j^{(n)}}_{n^{\text{th}} \text{ visit back to state } j} \longrightarrow \text{Time.}
\end{aligned}$$

3.5 Two Interesting Applications

3.5.1 The Galton–Watson Branching Process

The famous *Galton–Watson Branching Process* was first introduced by Francis Galton in 1889 as a simple mathematical model for the propagation of family names. They were reinvented by Leo Szilard in the late 1930s as models for the proliferation of free neutrons in a nuclear fission reaction. Such mathematical models (and their generalizations) continue to play an important role in both the theory and applications of stochastic processes.

In what follows, we assume that a population of individuals (which may represent people, organisms, free neutrons, etc.) evolves in discrete time. Specifically, we define:

$$\begin{aligned} X_0 &\equiv \text{population of the } 0^{\text{th}} \text{ (original) generation,} \\ X_1 &\equiv \text{population of the } 1^{\text{st}} \text{ generation,} \\ &\vdots \\ X_n &\equiv \text{population of the } n^{\text{th}} \text{ generation,} \\ &\vdots \end{aligned}$$

We assume that each individual in a generation produces a random number (possibly 0) of individuals, called offspring, which go on and become part of the very next generation. In other words, it is always the offspring of a current generation which go on to form the next generation. We further assume that individuals produce offspring independently of all others according to the same probability distribution, namely

$$\alpha_m = P(\text{an individual produces } m \text{ offspring}), \quad m = 0, 1, 2, \ldots$$

In addition, for purposes we see later, we assume that $\alpha_0 \in (0, 1)$ and $\alpha_0 + \alpha_1 < 1$.

For $j \in \mathbb{N}$, let $Z_i^{(j)}$ be the number of offspring produced from individual i in the jth generation. Due to the earlier independence assumptions, $\{Z_i^{(j)}\}_{i=1}^{\infty}$ is an iid sequence of rvs with

$$\alpha_m = P(Z_i^{(j)} = m) \quad \forall i \in \mathbb{N}.$$

Moreover, let $\mu = \mathbb{E}[Z_i^{(j)}]$ and $\sigma^2 = \text{Var}(Z_i^{(j)})$ represent the (common) mean and variance, respectively, of the number of offspring produced by a single individual. Based on the above assumptions, the Galton–Watson process $\{X_n, n \in \mathbb{N}\}$ is actually a DTMC taking values in the state space $S = \mathbb{N}$, since it follows that

$$X_n = \sum_{i=1}^{X_{n-1}} Z_i^{(n-1)}, \tag{3.9}$$

implying that the Markov property and stationarity assumption are both satisfied.

In this DTMC, we remark that $P_{0,0} = 1$, since state 0 is obviously an absorbing state. If we now consider state i, $i \in \mathbb{Z}^+$, then we can easily show that state i is transient. In particular, note that states 0 and i do not communicate. Moreover, $P_{i,0} = \alpha_0^i > 0$ (since $\alpha_0 > 0$). By the contrapositive of Theorem 3.5, state i must therefore be transient. Thus, since state 0 is recurrent and states $1, 2, 3, \ldots$ are transient, the following conclusion can be drawn: *The population will either die out completely or its size will grow indefinitely (to ∞).*

Since (3.9) implies that X_n is expressible as a random sum, we can apply the results of Example 2.9 to obtain

$$\mathbb{E}[X_n] = \mu \mathbb{E}[X_{n-1}]$$

and

$$\mathrm{Var}(X_n) = \sigma^2 \mathbb{E}[X_{n-1}] + \mu^2 \,\mathrm{Var}(X_{n-1}).$$

For convenience, let us henceforth assume that $X_0 = 1$ (with probability 1). As it is understood that $X_0 = 1$, for ease of notation, we will suppress writing the condition "$X_0 = 1$" in all expectations and probabilities which follow. Now, let us consider $\mathbb{E}[X_n]$ for several values of n:

$$\begin{aligned}
&\text{Take } n = 1 \Rightarrow \mathbb{E}[X_1] = \mu \mathbb{E}[X_0] = \mu, \\
&\text{Take } n = 2 \Rightarrow \mathbb{E}[X_2] = \mu \mathbb{E}[X_1] = \mu^2, \\
&\text{Take } n = 3 \Rightarrow \mathbb{E}[X_3] = \mu \mathbb{E}[X_2] = \mu^3.
\end{aligned}$$

Based on the above findings, it is straightforward to deduce that

$$\mathbb{E}[X_n] = \mu^n, \quad n \in \mathbb{N}.$$

In a similar fashion, we have:

$$\begin{aligned}
&\text{Take } n = 1 \Rightarrow \mathrm{Var}(X_1) = \sigma^2 \mathbb{E}[X_0] + \mu^2 \,\mathrm{Var}(X_0) = \sigma^2, \\
&\text{Take } n = 2 \Rightarrow \mathrm{Var}(X_2) = \sigma^2 \mathbb{E}[X_1] + \mu^2 \,\mathrm{Var}(X_1) = \sigma^2 \mu + \sigma^2 \mu^2, \\
&\text{Take } n = 3 \Rightarrow \mathrm{Var}(X_3) = \sigma^2 \mathbb{E}[X_2] + \mu^2 \,\mathrm{Var}(X_2) = \sigma^2 \mu^2 + \sigma^2 \mu^3 + \sigma^2 \mu^4.
\end{aligned}$$

Continuing inductively, we find that

$$\mathrm{Var}(X_n) = \sigma^2 \mu^{n-1} \sum_{i=0}^{n-1} \mu^i, \quad n \in \mathbb{N},$$

which simplifies to give

$$\mathrm{Var}(X_n) = \begin{cases} n\sigma^2, & \text{if } \mu = 1, \\ \sigma^2 \mu^{n-1} \left(\dfrac{1 - \mu^n}{1 - \mu} \right), & \text{if } \mu \neq 1. \end{cases}$$

Under the assumption that $X_0 = 1$, let π_0 denote the limiting probability that the population dies out. In other words,

$$\pi_0 = \lim_{n \to \infty} P(X_n = 0).$$

Let us first consider the situation when $\mu < 1$, which is referred to as the *subcritical case*. Clearly, as $n \to \infty$, $\mathbb{E}[X_n] = \mu^n$ and $\mathrm{Var}(X_n) = \sigma^2 \mu^{n-1} \left(\dfrac{1 - \mu^n}{1 - \mu} \right)$ both converge to 0. Therefore, we would expect that if $\mu < 1$, then $\pi_0 = 1$. To prove this formally, note that

$$\mu^n = \mathbb{E}[X_n] = \sum_{j=1}^{\infty} j \cdot P(X_n = j) \geq \sum_{j=1}^{\infty} 1 \cdot P(X_n = j) = P(X_n \geq 1) = 1 - P(X_n = 0).$$

This implies that $1 - \mu^n \leq P(X_n = 0) \leq 1$. Taking the limit as $n \to \infty$ leads to

$$\lim_{n\to\infty}(1 - \mu^n) \leq \lim_{n\to\infty} P(X_n = 0) \leq \lim_{n\to\infty} 1,$$

or simply, $\pi_0 = 1$.

On the other hand, suppose that $\mu \geq 1$. By conditioning on the number of offspring produced by the single individual present in the population at time 0, we obtain

$$\pi_0 = P(\text{population dies out}) = \sum_{j=0}^{\infty} P(\text{population dies out} \mid X_1 = j)\alpha_j.$$

However, with $X_1 = j$, the population will eventually die out iff each of the j families started by the members of the first generation eventually dies out. As each family is assumed to act independently, and since the probability that any particular family dies out is simply π_0, it follows that

$$P(\text{population dies out} \mid X_1 = j) = \pi_0^j,$$

and our above equation becomes

$$\pi_0 = \sum_{j=0}^{\infty} \pi_0^j \alpha_j. \tag{3.10}$$

Thus, the value of π_0 satisfies (3.10), or equivalently, $z = \pi_0$ satisfies the equation

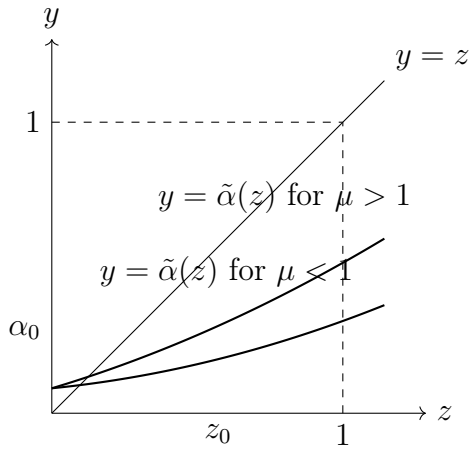
$$z = \tilde{\alpha}(z), \tag{3.11}$$

where $\tilde{\alpha}(z) = \sum_{j=0}^{\infty} \alpha_j z^j$. Note that $\tilde{\alpha}(0) = \alpha_0 > 0$ and $\tilde{\alpha}(1) = \sum_{j=0}^{\infty} \alpha_j = 1$. Consequently, $z = 0$ is not a solution to (3.11), whereas $z = 1$ is a solution to (3.11). The important question which needs to be addressed now is this: Is $z = 1$ the only solution to (3.11) on $[0, 1]$?

To answer this question, we observe the following facts concerning the function $\tilde{\alpha}(z)$:

1. $\tilde{\alpha}(z)$ is clearly a continuous function of z on the interval $[0, 1]$,
2. $\tilde{\alpha}'(z) = \dfrac{d}{dz}\tilde{\alpha}(z) = \sum_{j=1}^{\infty} j z^{j-1}\alpha_j$, and $\tilde{\alpha}'(1) = \sum_{j=1}^{\infty} j\alpha_j = \mu$,
3. $\tilde{\alpha}'(z) > 0$ for $z > 0 \Rightarrow \tilde{\alpha}(z)$ is an increasing function of z on $(0, 1]$,
4. $\tilde{\alpha}''(z) = \dfrac{d^2}{dz^2}\tilde{\alpha}(z) = \sum_{j=2}^{\infty} j(j-1)z^{j-2}\alpha_j$.
5. $\tilde{\alpha}''(z) > 0$ for $z > 0$ since $\alpha_0 + \alpha_1 < 1 \Rightarrow \tilde{\alpha}(z)$ is concave up for $z \in (0, 1]$.

As a result of these facts, the following diagram depicts the behaviour of the function $y = \tilde{\alpha}(z)$ in relation to $y = z$:



In other words, when $\mu = 1$ (i.e., the so-called *critical case*), there is only one root in $[0, 1]$, namely $z = 1$. However, when $\mu > 1$, there is a second root $z = z_0 \in (0, 1)$ which satisfies (3.11). Therefore, this now raises the question: Is $\pi_0 = z_0$ or $\pi_0 = 1$ when $\mu > 1$?

Let $z = z^*$ be any non-negative solution satisfying (3.11). It is straightforward to show by mathematical induction that $z^* \geq P(X_n = 0)$, $n \in \mathbb{N}$ (left as an upcoming exercise). As a result, it follows that

$$\lim_{n\to\infty} z^* \geq \lim_{n\to\infty} P(X_n = 0) \Rightarrow z^* \geq \pi_0,$$

which implies that $z = \pi_0$ is the smallest positive number satisfying (3.11). It is only when $\mu > 1$ (referred to as the *supercritical case*) that π_0 is known to exist in the interval $(0, 1)$.

Example 3.13.

Given the following offspring probabilities, what is the probability that the population dies out in the long run assuming that $X_0 = 1$?

(a) $\alpha_0 = 3/4, \alpha_1 = 1/8, \alpha_2 = 1/8$

Solution.

$$\mu = 0 \cdot \frac{3}{4} + 1 \cdot \frac{1}{8} + 2 \cdot \frac{1}{8} = \frac{3}{8} < 1.$$

Since $\mu < 1$, the population will die out w/ probability 1, i.e.

$$P(X_\infty = 0) = \pi_0 = 1.$$

(b) $\alpha_0 = 1/5, \alpha_1 = 1/10, \alpha_2 = 7/10$

Solution.

$$\mu = 0 \cdot \frac{1}{5} + 1 \cdot \frac{1}{10} + 2 \cdot \frac{7}{10} = \frac{3}{2} > 1.$$

Since $\mu = \frac{3}{2} > 1$, then $\pi_0 \in (0,1)$ is known to exist.

$$z = \tilde{\alpha}(z) = \sum_{j=0}^{\infty} z^j \alpha_j = z^0\alpha_0 + z^1\alpha_1 + z^2\alpha_2.$$

$$z = \frac{1}{5} + \frac{z}{10} + \frac{7z^2}{10} \;\Rightarrow\; 10z = 2 + z + 7z^2 \;\Rightarrow\; 0 = 7z^2 - 8z + 2 = (z-1)(7z-2).$$

$$\frac{2}{7} < 1 \;\Rightarrow\; \pi_0 = \frac{2}{7}.$$

Remarks.

(1) In the case when $X_0 = n$, $n \in \mathbb{Z}^+$, the population will die out iff the families of each of the members of the initial generation die out. As a result, it immediately follows that the extinction probability is simply π_0^n.

(2) For certain choices of the offspring distribution, the Galton-Watson branching process is not very interesting to analyze. For example, when $X_0 = 1$ and $\alpha_r = 1$ for some $r \in \mathbb{N}$, the evolution of the process is purely deterministic (i.e., $P(X_n = r^n) = 1$, $n \in \mathbb{N}$). Another uninteresting case occurs when $\alpha_0\alpha_1 > 0$ and $\alpha_j = 0$ for all $j > 1$. In this situation, the population remains at its initial size $X_0 = 1$ for a random number of generations (according to a geometric distribution), before dying out completely.

3.5.2 The Gambler's Ruin Problem

One of the most powerful ideas in the theory of DTMCs is that many fundamental probabilities and expectations can be computed as the solutions of systems of linear equations. We have already seen one such example of this through the application of the BLT. In what follows, we will continue to illustrate this idea by deriving appropriate linear systems for a number of key probabilities and expectations that arise in certain settings. To motivate this idea, let us consider a famous problem in stochastic processes known as the *Gambler's Ruin Problem*.

Example 3.14.

Consider a gambler who, at each play of a game, has probability $p \in (0,1)$ of winning one unit and probability $q = 1 - p$ of losing one unit. Assume that successive plays of the game are independent. If the gambler initially begins with i units, what is the probability that the gambler's fortune will reach N units ($N < \infty$) before reaching 0 units? This problem is often referred to as the *Gambler's Ruin Problem*, with state 0 representing bankruptcy and state N representing the jackpot.

Solution. X_n is the gambler's fortune after the nth play of the game with $X_0 = i$ with prob. 1. Clearly $\{X_n : n \in \mathbb{N}\}$ is a DTMC with TPM. Note that states 0 and N are treated as absorbing states. This implies that they are both recurrent. States $1, 2, \ldots, N-1$ belong to the same communication class and it is straightforward to verify that it is a transient class. We want

to determine $G(i)$, for $i = 1, \ldots, N-1$, representing the probability starting in i units, the gambler's fortune will eventually reach N units.

$$\lim_{n\to\infty} P^{(n)} = \begin{array}{c|cccccccc} & 0 & 1 & 2 & 3 & \cdots & N-1 & N \\ \hline 0 & 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1-G(1) & 0 & 0 & 0 & \cdots & 0 & G(1) \\ 2 & 1-G(2) & 0 & 0 & 0 & \cdots & 0 & G(2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ N-1 & 1-G(N-1) & 0 & 0 & 0 & \cdots & 0 & G(N-1) \\ N & 0 & 0 & 0 & 0 & \cdots & 0 & 1 \end{array}$$

From above, NOTE that $G(0) = 0$ and $G(N) = 1$. For $i = 1, \ldots, N-1$, let's condition on the outcome of the very first game to get:

$$\begin{aligned} G(i) &= qG(i-1) + pG(i+1). \\ (p+q)G(i) &= qG(i-1) + pG(i+1), \\ p(G(i+1) - G(i)) &= q(G(i) - G(i-1)), \\ G(i+1) - G(i) &= \frac{q}{p}[G(i) - G(i-1)]. \end{aligned}$$

When $p \neq \frac{1}{2}$:

$$\begin{aligned} G(k+1) - G(k) &= \left(\frac{q}{p}\right)^k [G(1) - G(0)], \\ G(k) &= \sum_{j=0}^{k-1} \left(\frac{q}{p}\right)^j [G(1) - G(0)], \\ G(k) &= \frac{1-(q/p)^k}{1-(q/p)} G(1). \end{aligned}$$

Applying finite geometric series when $p \neq q$ and using $k = N$:

$$\begin{aligned} G(N) = 1 &= \frac{1-(q/p)^N}{1-(q/p)} G(1), \\ G(1) &= \frac{1-(q/p)}{1-(q/p)^N}. \end{aligned}$$

Thus,

$$G(k) = \begin{cases} \dfrac{1-(q/p)^k}{1-(q/p)^N}, & p \neq \dfrac{1}{2}, \\ \dfrac{k}{N}, & p = \dfrac{1}{2}. \end{cases}$$

Remarks.

(1) An interesting question to ask is what happens to the gambler's probability of winning the jackpot, given an initial fortune of i units, as N grows "larger" (i.e., $N \to \infty$)? In other words, what happens to the limit of $G(i)$ as $N \to \infty$? Looking at the cases based on the value of p, we see:

(i) When $p = 1/2$, $G(i) = \dfrac{i}{N} \to 0$ as $N \to \infty$,

(ii) When $p < 1/2$, $G(i) = \dfrac{1-(q/p)^i}{1-(q/p)^N} \to 0$ as $N \to \infty$,

(iii) When $p > 1/2$, $G(i) = \dfrac{1-(q/p)^i}{1-(q/p)^N} \to 1-(q/p)^i$ as $N \to \infty$,

since $q/p > (<)\, 1$ when $p < (>)\, 1/2$. Simply put, only when $p > 1/2$ does a positive probability exist that the gambler's fortune will increase indefinitely. Otherwise, the gambler is sure to go broke.

(2) In our study of the *Random Walk* in Example 3.9 featuring a DTMC on the state space $\mathbb{Z}$ with transition probabilities analogous to those in the Gambler's Ruin Problem, we previously showed that

$$f_{0,0} = (1-p) f_{-1,0} + p f_{1,0}.$$

Suppose that $p > 1/2$. First of all, note that For the random walk started at 1, eventual return to 0 is the increasing-boundary limit of the corresponding gambler's-ruin bankruptcy event. Thus

$$\begin{aligned} f_{1,0} &= \lim_{N\to\infty} \mathbb{P}_1(T_0 < T_N) \\ &= 1 - \left(1 - \frac{1-p}{p}\right) \\ &= \frac{1-p}{p}. \end{aligned}$$

In a similar fashion, we also have that Starting at -1 instead, reflection converts the problem into gambler's ruin with upward-step probability $1-p < 1/2$. The limiting bankruptcy probability is therefore

$$\begin{aligned} f_{-1,0} &= \lim_{N\to\infty} \mathbb{P}_1(T_0 < T_N) \\ &= 1 - 0 = 1, \end{aligned}$$

where the probability in this display refers to the reflected walk with upward-step probability $1-p$.

Therefore, we end up ultimately obtaining $f_{0,0} = (1-p)\cdot 1 + p \cdot (1-p)/p = 2(1-p)$, which agrees with our earlier result. The same essential procedure can be adapted to verify that $f_{0,0} = 2p$ if $p < 1/2$.

3.6 Absorbing DTMCs

The *Gambler's Ruin Problem* is actually an example of a more general problem which we turn our attention to next. In particular, consider a DTMC $\{X_n, n \in \mathbb{N}\}$ with a **finite** number of states arranged specifically as follows:

$$0, 1, \ldots, M-1, M, M+1, \ldots, N.$$

transient states absorbing states

The TPM for this DTMC can be expressed as

$$P = \begin{array}{c} \\ \text{transient} \\ \text{absorbing} \end{array} \begin{array}{c} \begin{array}{cc} \text{transient} & \text{absorbing} \end{array} \\ \begin{bmatrix} Q & R \\ 0 & I \end{bmatrix} \end{array},$$

where

$$Q = \begin{array}{c} \\ 0 \\ 1 \\ \vdots \\ M-1 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & \cdots & M-1 \end{array} \\ \begin{bmatrix} Q_{0,0} & Q_{0,1} & \cdots & Q_{0,M-1} \\ Q_{1,0} & Q_{1,1} & \cdots & Q_{1,M-1} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{M-1,0} & Q_{M-1,1} & \cdots & Q_{M-1,M-1} \end{bmatrix} \end{array}, \qquad \text{transient}$$

$$R = \begin{array}{c} \\ 0 \\ 1 \\ \vdots \\ M-1 \end{array} \begin{array}{c} \begin{array}{cccc} M & M+1 & \cdots & N \end{array} \\ \begin{bmatrix} R_{0,M} & R_{0,M+1} & \cdots & R_{0,N} \\ R_{1,M} & R_{1,M+1} & \cdots & R_{1,N} \\ \vdots & \vdots & \ddots & \vdots \\ R_{M-1,M} & R_{M-1,M+1} & \cdots & R_{M-1,N} \end{bmatrix} \end{array}, \qquad \text{absorbing.}$$

0 is a zero matrix of dimension $(N - M + 1) \times M$, and I is an identity matrix of dimension $(N - M + 1) \times (N - M + 1)$. In what follows, let i be a transient state (i.e., $0 \leq i \leq M - 1$) and assume that $X_0 = i$. Let

$$T = \min\{n \in \mathbb{Z}^+ : M \leq X_n \leq N\}$$

be the absorption time rv. For $M \leq k \leq N$ (i.e., k is an absorbing state), consider the absorption probability into state k from state i defined by

$$U_{i,k} = P(X_T = k \mid X_0 = i).$$

Conditioning on the state of the DTMC at time 1, note that

$$\begin{aligned}
U_{i,k} &= P(X_T = k \mid X_0 = i) \\
&= \sum_{j=0}^{N} P(X_T = k \mid X_1 = j, X_0 = i) P(X_1 = j \mid X_0 = i) \\
&= \sum_{j=0}^{N} P(X_T = k \mid X_1 = j, X_0 = i) P_{i,j} \\
&= \sum_{j=0}^{M-1} P_{i,j} P(X_T = k \mid X_1 = j, X_0 = i) + \sum_{j=M}^{N} P(X_T = k \mid X_1 = j, X_0 = i) P_{i,j} \\
&= \sum_{j=0}^{M-1} P_{i,j} P(X_T = k \mid X_1 = j, X_0 = i) + P_{i,k},
\end{aligned}$$

since it follows that for $M \leq j \leq N$, $T \mid (X_1 = j, X_0 = i)$ is degenerate at 1, and so

$$P(X_T = k \mid X_1 = j, X_0 = i) = \delta_{j,k}$$

where

$$\delta_{j,k} = \begin{cases} 0, & \text{if } j \neq k, \\ 1, & \text{if } j = k, \end{cases}$$

denotes the Kronecker delta function. For $0 \leq j \leq M-1$, however, note that

$$P(X_T = k \mid X_1 = j, X_0 = i) = P(X_T = k \mid X_0 = j) = U_{j,k}.$$

To see this, let T_i denote the remaining number of transitions until absorption given that the DTMC is currently in transient state i. Clearly,

$$T \mid (X_0 = i) \sim T_i.$$

Moreover, for transient state j, we have

$$T \mid (X_1 = j, X_0 = i) \sim (1 + T_j) \mid (X_1 = j),$$

due to the Markov property. Therefore,

$$\begin{aligned}
P(X_T = k \mid X_1 = j, X_0 = i) &= P(X_{T_i+1} = k \mid X_1 = j) \\
&= P(X_{T_j} = k \mid X_0 = j) \quad \text{due to the stationary assumption} \\
&= P(X_T = k \mid X_0 = j) \\
&= U_{j,k},
\end{aligned}$$

where the second last equality holds due to T_j and T being equivalent in distribution under the condition $X_0 = j$. As a result, we ultimately end up with

$$U_{i,k} = P_{i,k} + \sum_{j=0}^{M-1} P_{i,j}U_{j,k} = R_{i,k} + \sum_{j=0}^{M-1} Q_{i,j}U_{j,k}, \qquad 0 \leq i \leq M-1,\ M \leq k \leq N. \tag{3.12}$$

In other words, to determine $U_{i,k}$ for a particular pair of values for i and k, the system of M linear equations given by (3.12) must be solved, yielding solutions for $U_{0,k}, U_{1,k}, \ldots, U_{M-1,k}$.

Example 3.12.

(continued) Recall the earlier DTMC we considered having TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ 0 & 0 & 1 \end{bmatrix} \end{array}.$$

We previously claimed that

$$\lim_{n\to\infty} P^{(n)} = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix} \end{array}.$$

Show that the absorption probabilities from transient state 1 into states 0 and 2 are equal to $\lim_{n\to\infty} P_{1,0}^{(n)}$ and $\lim_{n\to\infty} P_{1,2}^{(n)}$, respectively.

Solution. First, relabel the states of this DTMC as follows.

$$\begin{aligned} 0^* &\equiv \text{state 1 in original DTMC} \\ 1^* &\equiv \text{state 0 in original DTMC} \\ 2^* &\equiv \text{state 2 in original DTMC.} \end{aligned}$$

As a result, the new TPM corresponding to states $\{0^*, 1^*, 2^*\}$ looks like

$$P^* = \begin{array}{c} \\ 0^* \\ 1^* \\ 2^* \end{array} \begin{array}{c} \begin{array}{ccc} 0^* & 1^* & 2^* \end{array} \\ \left[\begin{array}{c|cc} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ \hline 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \end{array},$$

so that

$$Q = \begin{bmatrix} \frac{1}{2} \end{bmatrix} \qquad \text{and} \qquad R = \begin{bmatrix} \frac{1}{3} & \frac{1}{6} \end{bmatrix}.$$

Using (3.12) we now find $U_{0^*,1^*}$ and $U_{0^*,2^*}$ to be

$$\begin{aligned} U_{0^*,1^*} &= R_{0^*,1^*} + Q_{0^*,0^*}U_{0^*,1^*} \Rightarrow U_{0^*,1^*} = \frac{1}{3} + \frac{1}{2}U_{0^*,1^*} \\ &\Rightarrow U_{0^*,1^*} = \frac{2}{3} = \lim_{n\to\infty} P_{1,0}^{(n)} \end{aligned}$$

and

$$\begin{aligned} U_{0^*,2^*} &= R_{0^*,2^*} + Q_{0^*,0^*}U_{0^*,2^*} \Rightarrow U_{0^*,2^*} = \frac{1}{6} + \frac{1}{2}U_{0^*,2^*} \\ &\Rightarrow U_{0^*,2^*} = \frac{1}{3} = \lim_{n\to\infty} P_{1,2}^{(n)}. \end{aligned}$$

Remarks.

(1) If we define $U = [U_{i,k}]$ to be the $M \times (N - M + 1)$ matrix of absorption probabilities, then (3.12) can be expressed more succinctly in matrix form as

$$U = R + QU,$$

or equivalently,

$$(I - Q)U = R.$$

A known mathematical fact is that the matrix $I - Q$ is invertible, and so an explicit (matrix) solution for U is given by

$$U = (I - Q)^{-1}R. \tag{3.13}$$

In the previous example, note that

$$\begin{aligned} U = [\, U_{0^*,1^*} \;\; U_{0^*,2^*} \,] &= \left(1 - \frac{1}{2}\right)^{-1} \begin{bmatrix} \frac{1}{3} & \frac{1}{6} \end{bmatrix} \\ = 2 \begin{bmatrix} \frac{1}{3} & \frac{1}{6} \end{bmatrix} &= \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \end{bmatrix}. \end{aligned}$$

(2) As demonstrated in the previous example, the limiting behaviour of the general DTMC we are considering can be characterized as follows:

$$\lim_{n\to\infty} P^{(n)} = \begin{array}{c|cccc|cccc} & 0 & 1 & \cdots & M-1 & M & M+1 & \cdots & N \\ 0 & 0 & 0 & \cdots & 0 & U_{0,M} & U_{0,M+1} & \cdots & U_{0,N} \\ 1 & 0 & 0 & \cdots & 0 & U_{1,M} & U_{1,M+1} & \cdots & U_{1,N} \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ M-1 & 0 & 0 & \cdots & 0 & U_{M-1,M} & U_{M-1,M+1} & \cdots & U_{M-1,N} \\ \hline M & 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ M+1 & 0 & 0 & \cdots & 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ N & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 1 \end{array}.$$

Another way to see this is through the use of Exercise 3.5.1, which states that

$$P^{(n)} = \begin{bmatrix} Q^n & \sum_{i=0}^{n-1} Q^i R \\ 0 & I \end{bmatrix}, \qquad n \in \mathbb{Z}^+.$$

Therefore, it follows that

$$\lim_{n\to\infty} P^{(n)} = \begin{bmatrix} \lim_{n\to\infty} Q^n & \left(\lim_{n\to\infty} \sum_{i=0}^{n-1} Q^i\right) R \\ 0 & I \end{bmatrix}.$$

However, $\lim_{n\to\infty} Q^n = 0$ (as Q has only transient states). Moreover, note that

$$\begin{aligned} (I-Q)\left(\lim_{n\to\infty}\sum_{i=0}^{n-1} Q^i\right) &= \lim_{n\to\infty}\sum_{i=0}^{n-1}(I-Q)Q^i \\ &= \lim_{n\to\infty}\sum_{i=0}^{n-1}(Q^i - Q^{i+1}) \\ &= \lim_{n\to\infty}\left\{(Q^0-Q^1)+(Q^1-Q^2)+\cdots+(Q^{n-1}-Q^n)\right\} \\ &= \lim_{n\to\infty}(I-Q^n) \\ &= I - \lim_{n\to\infty} Q^n \\ &= I. \end{aligned}$$

Therefore, we have that

$$\sum_{i=0}^{\infty} Q^i = \lim_{n\to\infty}\sum_{i=0}^{n-1} Q^i = (I-Q)^{-1}I = (I-Q)^{-1},$$

which yields a formula for an infinite geometric series of matrices. From (3.13), we obtain

$$\lim_{n\to\infty} P^{(n)} = \begin{bmatrix} 0 & U \\ 0 & I \end{bmatrix}.$$

(3) In the *Gambler's Ruin Problem*, if we reorder the states $0, 1, 2, \ldots, N-1, N$ as

$$1, 2, \ldots, N-1, \ N, 0,$$

transient absorbing

then $U_{i,N} = G(i)$ and $U_{i,0} = 1 - G(i)$, $i = 1, 2, \ldots, N-1$.

Example 3.15.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0.4 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.3 & 0.3 & 0.3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}.$$

Suppose that the DTMC begins in state 1. What is the probability that the DTMC ultimately ends up in state 3? How would this probability change if the DTMC begins in state 0 with probability 3/4 and in state 1 with probability 1/4?

Solution. First, we wish to determine $U_{1,3}$:

$$Q = \begin{bmatrix} 0.4 & 0.3 \\ 0.1 & 0.3 \end{bmatrix} \quad \text{and} \quad R = \begin{bmatrix} 0.2 & 0.1 \\ 0.3 & 0.3 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} U_{0,2} & U_{0,3} \\ U_{1,2} & U_{1,3} \end{bmatrix}.$$

Since $U = (I - Q)^{-1}R$, we need to find the inverse.

$$I - Q = \begin{bmatrix} 0.6 & -0.3 \\ -0.1 & 0.7 \end{bmatrix}.$$

Recall: For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

provided $ad - bc \neq 0$.

Applying the formula, we get

$$(I-Q)^{-1} = \begin{bmatrix} \frac{70}{39} & \frac{30}{39} \\ \frac{10}{39} & \frac{60}{39} \end{bmatrix} \Rightarrow U = (I-Q)^{-1}R = \begin{bmatrix} \frac{23}{39} & \frac{16}{39} \\ \frac{20}{39} & \frac{19}{39} \end{bmatrix}.$$

Therefore,

$$U_{1,3} = \frac{19}{39} \approx 0.487.$$

Now under the alternative set of initial conditions, we calculate:

$$\begin{aligned} &P(\text{DTMC ultimately ends in state 3}) \\ &\qquad = P(X_0 = 0)P(\text{ultimately ends in } 3 \mid X_0 = 0) \\ &\qquad\quad + P(X_0 = 1)P(\text{ultimately ends in } 3 \mid X_0 = 1) \\ &\qquad = \frac{3}{4}U_{0,3} + \frac{1}{4}U_{1,3} \\ &\qquad = \frac{3}{4} \cdot \frac{16}{39} + \frac{1}{4} \cdot \frac{19}{39} = \frac{67}{156} \approx 0.429. \end{aligned}$$

An interesting feature of the above methodology is that it can even be used for DTMCs in which the set of absorbing states are replaced by one or more recurrent classes. The following example demonstrates the basic idea.

Example 3.16.

Consider a DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{ccccc} 0 & 1 & 2 & 3 & 4 \end{array} \\ \left[\begin{array}{ccccc} 0.4 & 0.3 & 0.2 & 0.1 & 0 \\ 0.1 & 0.3 & 0.3 & 0.3 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0.7 & 0.3 \\ 0 & 0 & 0 & 0.4 & 0.6 \end{array}\right] \end{array}.$$

Suppose that the DTMC begins in state 1. What is the probability that the DTMC ultimately ends up in state 3?

Solution. We wish to determine $P_{1,3}^{(n)} \Rightarrow \lim_{n\to\infty} P_{1,3}^{(n)}$.

To do so, let's group states 3 & 4 together as a single state, which we will denote 3^*.

As a result of this grouping, our revised TPM has form

$$\begin{array}{c|cc|cc} & 0 & 1 & 2 & 3^* \\ 0 & 0.4 & 0.3 & 0.2 & 0.1 \\ 1 & 0.1 & 0.3 & 0.3 & 0.3 \\ 2 & 0 & 0 & 1 & 0 \\ \hline 3^* & 0 & 0 & 0 & 1 \end{array},$$

which is identical to the TPM from the previous example 3.15.

Using the results of example 3.15 we know that

$$U_{1,3^*} = \frac{19}{39}.$$

Once in state 3^*, the DTMC will remain in the recurrent class $\{3, 4\}$ with associated TPM

$$\begin{array}{c} \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cc} 3 & 4 \end{array} \\ \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \end{array}.$$

For this smaller DTMC, the conditions of the BLT are satisfied, meaning we can determine the limiting probabilities.

π_3 & π_4 by solving the linear system

$$\begin{aligned} (\pi_3 \ \pi_4) = (\pi_3 \ \pi_4) \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \quad \text{s.t.} \quad \pi_3 + \pi_4 = 1. \\ \Rightarrow \pi_3 = \frac{4}{7} \ \& \ \pi_4 = \frac{3}{7}. \\ \therefore \lim_{n\to\infty} P_{1,3}^{(n)} = \pi_3 \cdot U_{1,3^*} = \frac{19}{39} \cdot \frac{4}{7} = 0.278. \end{aligned}$$

Next, for $0 \leq i \leq M-1$, let $v_i = \mathbb{E}[T \mid X_0 = i]$ be the mean absorption time from state i. Once again, conditioning on the state of the DTMC at time 1, we get

$$\begin{aligned} v_i &= \mathbb{E}[T \mid X_0 = i] \\ &= \sum_{j=0}^{M-1} \mathbb{E}[T \mid X_1 = j, X_0 = i] P_{i,j} + \sum_{j=M}^{N} \mathbb{E}[T \mid X_1 = j, X_0 = i] P_{i,j} \\ &= \sum_{j=0}^{M-1} \mathbb{E}[T \mid X_1 = j, X_0 = i] P_{i,j} + \sum_{j=M}^{N} 1 \cdot P_{i,j}, \end{aligned}$$

since $T \mid (X_1 = j, X_0 = i)$ is degenerate at 1 for $M \leq j \leq N$. For $0 \leq j \leq M-1$, our previous arguments lead to

$$T \mid (X_1 = j, X_0 = i) \sim (1 + T_j) \mid (X_1 = j) \sim 1 + T \mid (X_0 = j),$$

which implies that

$$\begin{aligned} v_i &= \sum_{j=0}^{M-1} (1 + \mathbb{E}[T \mid X_0 = j]) P_{i,j} + \sum_{j=M}^{N} 1 \cdot P_{i,j} \\ &= 1 + \sum_{j=0}^{M-1} P_{i,j} v_j \\ &= 1 + \sum_{j=0}^{M-1} Q_{i,j} v_j. \end{aligned} \tag{3.14}$$

If we now define the $M \times 1$ column vector of mean absorption times

$$v' = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{M-1} \end{bmatrix},$$

then (3.14) yields in matrix form

$$v' = e' + Qv'.$$

Therefore, an explicit (matrix) solution for v' is

$$v' = (I - Q)^{-1} e'.$$

Example 3.12.

(continued) Recall the modified TPM

$$P^* = \begin{array}{c|ccc} & 0^* & 1^* & 2^* \\ \hline 0^* & \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 1^* & 0 & 1 & 0 \\ 2^* & 0 & 0 & 1 \end{array}.$$

What is the mean absorption time for this DTMC, given that it begins in state 0^*?

Solution. Using the matrix eq'n for $v' = [v_{0^*}]$ we have

$$v_{0^*} = (I - Q)^{-1} e' = (1 - \tfrac{1}{2})^{-1}(1) = 2.$$

Given the DTMC starts in state 0^* with probab. $\frac{1}{2}$ or becomes absorbed into one of the two absorbing states with probab. $\frac{1}{3}+\frac{1}{6}=\frac{1}{2}$. Then, the # of transitions needed for absorption to occur simply follows a geometric distribn.

$$T \mid (X_0 = 0^*) \sim \text{Geom}\left(\tfrac{1}{2}\right) \Rightarrow \mathbb{E}[T \mid X_0 = 0^*] = \frac{1}{1/2} = 2.$$

Example 3.15.

(continued) If the DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \left[\begin{array}{cc|cc} 0.4 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.3 & 0.3 & 0.3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \end{array}$$

begins in state 1, how long, on average, does it take to end up in either of states 2 or 3?

Solution. Here $v' = [v_0 \; v_1]^\top$ where wish to find v_1.

Note that

$$v' = (I-Q)^{-1}e' = \begin{bmatrix} \frac{70}{39} & \frac{30}{39} \\ \frac{10}{39} & \frac{60}{39} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$
$$\Rightarrow v_1 = \frac{10}{39}\cdot 1 + \frac{60}{39}\cdot 1 = \frac{70}{39} \approx 1.79.$$

As before, assume that $X_0 = i$ where $0 \le i \le M-1$. Let ℓ be a transient state as well, so that $0 \le \ell \le M-1$. Define the following sequence of indicator rvs $\{A_n\}_{n\ge 0}$ such that

$$A_n = \begin{cases} 0, & \text{if } X_n \neq \ell, \\ 1, & \text{if } X_n = \ell. \end{cases}$$

We are interested in computing the quantity

$$W_{i,\ell} = \mathbb{E}\left[\sum_{n=0}^{T-1} A_n \,\middle|\, X_0 = i\right],$$

which represents the mean number of times that state ℓ is visited (including time 0) prior to absorption given that $X_0 = i$. To begin with, note that

$$\begin{aligned}
W_{i,\ell} &= \mathbb{E}\left[\sum_{n=0}^{T-1} A_n \,\middle|\, X_0 = i\right] \\
&= \mathbb{E}\left[A_0 + \sum_{n=1}^{T-1} A_n \,\middle|\, X_0 = i\right] \\
&= \mathbb{E}[A_0 \mid X_0 = i] + \mathbb{E}\left[\sum_{n=1}^{T-1} A_n \,\middle|\, X_0 = i\right] \\
&= 0 \cdot P(A_0 = 0 \mid X_0 = i) + 1 \cdot P(A_0 = 1 \mid X_0 = i) + \mathbb{E}\left[\sum_{n=1}^{T-1} A_n \,\middle|\, X_0 = i\right] \\
&= \delta_{i,\ell} + \mathbb{E}\left[\sum_{n=1}^{T-1} A_n \,\middle|\, X_0 = i\right].
\end{aligned}$$

To find $\mathbb{E}\left[\sum_{n=1}^{T-1} A_n \mid X_0 = i\right]$, we condition on the state of the DTMC at time 1 to obtain

$$\mathbb{E}\left[\sum_{n=1}^{T-1} A_n \,\middle|\, X_0 = i\right] = \sum_{j=0}^{M-1} P_{i,j}\, \mathbb{E}\left[\sum_{n=1}^{T-1} A_n \,\middle|\, X_1 = j\right] + \sum_{j=M}^{N} \mathbb{E}\left[\sum_{n=1}^{T-1} A_n \,\middle|\, X_1 = j, X_0 = i\right] P_{i,j}.$$

When $M \leq j \leq N$, note that

$$\mathbb{E}\left[\sum_{n=1}^{T-1} A_n \,\middle|\, X_1 = j, X_0 = i\right] = \mathbb{E}\left[\sum_{n=1}^{1-1} A_n \,\middle|\, X_1 = j, X_0 = i\right] = 0,$$

since $T \mid (X_1 = j, X_0 = i)$ is degenerate at 1. When $0 \leq j \leq M-1$, recall that

$$T \mid (X_1 = j, X_0 = i) \sim (1 + T_j) \mid (X_1 = j)$$

and

$$T_j \sim T \mid (X_0 = j),$$

where T_j denotes the remaining number of transitions until absorption given that the DTMC is currently in transient state j. Therefore,

$$\begin{aligned}
\sum_{j=0}^{M-1} P_{i,j}\,\mathbb{E}\left[\sum_{n=1}^{T-1} A_n \,\middle|\, X_1 = j\right] &= \sum_{j=0}^{M-1} \mathbb{E}\left[\sum_{n=1}^{1+T_j-1} A_n \,\middle|\, X_1 = j\right] P_{i,j} \\
&= \sum_{j=0}^{M-1} \mathbb{E}\left[\sum_{n=1}^{T_j} A_n \,\middle|\, X_1 = j\right] P_{i,j} \\
&= \sum_{j=0}^{M-1} \mathbb{E}\left[\sum_{m=0}^{T_j-1} A_{m+1} \,\middle|\, X_1 = j\right] P_{i,j} \\
&= \sum_{j=0}^{M-1} P_{i,j}\mathbb{E}\left[\sum_{m=0}^{T-1} A_m \,\middle|\, X_0 = j\right] \quad \text{due to the stationary assumption} \\
&= \sum_{j=0}^{M-1} P_{i,j}\mathbb{E}\left[\sum_{m=0}^{T-1} A_m \,\middle|\, X_0 = j\right] \quad \text{since } T_j \mid (X_0 = j) \sim T \mid (X_0 = j) \\
&= \sum_{j=0}^{M-1} Q_{i,j} W_{j,\ell}.
\end{aligned}$$

Therefore, we ultimately end up with

$$W_{i,\ell} = \delta_{i,\ell} + \sum_{j=0}^{M-1} Q_{i,j} W_{j,\ell}, \quad 0 \le i, \ell \le M-1. \tag{3.15}$$

In matrix notation, define the $M \times M$ matrix $W = [W_{i,\ell}]$, so that (3.15) in matrix form then becomes

$$W = I + QW.$$

Solving for W leads to

$$\begin{aligned}
W - QW &= I \\
(I - Q)W &= I \\
W &= (I - Q)^{-1}.
\end{aligned}$$

Remark. From our earlier result, we recognize that $v' = (I - Q)^{-1}e' = We'$. In other words, by summing the rows of W (i.e., adding up the mean number of times each of the individual transient states is visited), we actually obtain the mean number of transitions to reach absorption.

Finally, let us again consider $0 \le i, \ell \le M-1$ and recall that

$$\begin{aligned}
f_{i,\ell} &= \mathbb{P}(\text{a future visit to state } \ell \mid X_0 = i) \\
&= \mathbb{P}(X_n = \ell \text{ for some } 1 \le n < T \mid X_0 = i).
\end{aligned}$$

Using an elementary conditioning argument (as well as the Markov and stationary properties of the DTMC), note that

$$\begin{aligned}
W_{i,\ell} &= \mathbb{E}\left[\sum_{n=0}^{T-1} A_n \,\middle|\, X_0 = i\right] \\
&= \mathbb{E}\left[\sum_{n=0}^{T-1} A_n \,\middle|\, X_0 = i,\ X_n = \ell \text{ for some } 1 \le n \le T-1\right] \cdot f_{i,\ell} \\
&\quad + \mathbb{E}\left[\sum_{n=0}^{T-1} A_n \,\middle|\, X_0 = i,\ X_n \neq \ell \text{ for all } 1 \le n \le T-1\right] \cdot (1 - f_{i,\ell}) \\
&= (\delta_{i,\ell} + W_{\ell,\ell}) \cdot f_{i,\ell} + \delta_{i,\ell} \cdot (1 - f_{i,\ell}) \\
&= \delta_{i,\ell} + W_{\ell,\ell} f_{i,\ell},
\end{aligned}$$

which immediately leads to

$$f_{i,\ell} = \frac{W_{i,\ell} - \delta_{i,\ell}}{W_{\ell,\ell}}.$$

Example 3.15.

(continued) Recall the DTMC with TPM

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0.4 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.3 & 0.3 & 0.3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}.$$

Given $X_0 = 1$, what is the average number of visits to state 0 prior to absorption? Also, what is the probability that the DTMC ever makes a visit to state 0?

Solution. We wish to find $W_{1,0}$ where

$$W = \begin{array}{c} \\ 0 \\ 1 \end{array} \begin{array}{c} \begin{array}{cc} 0 & 1 \end{array} \\ \begin{bmatrix} W_{0,0} & W_{0,1} \\ W_{1,0} & W_{1,1} \end{bmatrix} \end{array}.$$

From earlier,

$$W = (I - Q)^{-1} = \begin{bmatrix} \frac{70}{39} & \frac{30}{39} \\ \frac{10}{39} & \frac{60}{39} \end{bmatrix}, \quad W_{1,0} = \frac{10}{39} \approx 0.257.$$

$$f_{1,0} = \frac{W_{1,0} - \delta_{1,0}}{W_{0,0}} = \frac{W_{1,0}}{W_{0,0}} = \frac{10/39}{70/39} = \frac{10}{70} = \frac{1}{7} \approx 0.143.$$

Remark. For $\ell = i$, the probability of the DTMC visiting state i in the future (given that $X_0 = i$) is

$$f_{i,i} = \frac{W_{i,i} - \delta_{i,i}}{W_{i,i}} = 1 - \frac{1}{W_{i,i}},$$

where $W_{i,i}$ is the expected number of visits to state i (including time 0) before absorption. From the above relation, $W_{i,i} = (1 - f_{i,i})^{-1}$. However, recall that for transient state i, the random number M_i of future visits to state i, given $X_0 = i$, has conditional pmf given by (3.4), namely

$$P(M_i = k \mid X_0 = i) = f_{i,i}^k (1 - f_{i,i}), \quad k = 0, 1, 2, \ldots,$$

which we recognize as the pmf of a $\text{GEO}_{f_{i,i}}$ rv. Therefore, the expected number of future visits to state i is

$$\mathbb{E}[M_i \mid X_0 = i] = \frac{1 - (1 - f_{i,i})}{1 - f_{i,i}} = \frac{1}{1 - f_{i,i}} - 1 = W_{i,i} - 1,$$

which is in agreement with the definition of $W_{i,i}$, since we count the initial visit to state i occurring at time 0.

4 The Exponential Distribution and the Poisson Process

Overview. Exponential waiting times and Poisson counting processes form the foundation of continuous-time event models. We derive memorylessness, exponential races, interarrival distributions, Poisson counts, conditional arrival times, and nonhomogeneous and compound generalizations.

4.1 Notation and the Connection Between Counts and Waiting Times

Write $N(t)$ for the number of arrivals by time $t \geq 0$, S_n for the time of the nth arrival, and $T_n = S_n - S_{n-1}$, where $S_0 = 0$, for the interarrival times. The basic duality is

$$\{N(t) \geq n\} = \{S_n \leq t\}, \qquad \{N(t) = n\} = \{S_n \leq t < S_{n+1}\}.$$

For a rate-λ homogeneous Poisson process, $N(t) \sim \text{Poisson}(\lambda t)$ while the T_n are independent $\text{Exp}(\lambda)$ random variables. These two characterizations are shown in the chapter; they are not independent modeling assumptions.

Proposition 4.1 (Gamma distribution of the nth arrival)**.** *If $T_1, \ldots, T_n$ are independent* $\text{Exp}(\lambda)$ *variables with $\lambda > 0$, then $S_n = \sum_{i=1}^{n} T_i$ has density*

$$f_{S_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \qquad t > 0.$$

Equivalently, S_n has an Erlang distribution of shape n and rate λ.

Proof. For $n = 1$ the formula is the exponential density. Assuming it holds for n, convolution with the density of T_{n+1} gives, for $t > 0$,

$$\begin{aligned} f_{S_{n+1}}(t) &= \int_0^t f_{S_n}(s) f_{T_{n+1}}(t-s)\,\mathrm{d}s \\ &= \int_0^t \frac{\lambda^n s^{n-1} e^{-\lambda s}}{(n-1)!} \lambda e^{-\lambda(t-s)}\,\mathrm{d}s \\ &= \frac{\lambda^{n+1} e^{-\lambda t}}{(n-1)!} \int_0^t s^{n-1}\,\mathrm{d}s \\ &= \frac{\lambda^{n+1} t^n e^{-\lambda t}}{n!}. \end{aligned}$$

Induction completes the proof. □

4.2 Properties of the Exponential Distribution

If a rv X has an exponential distribution with parameter $\lambda > 0$ (i.e., $X \sim \text{EXP}(\lambda)$ where λ is often referred to as the "rate"), then we have the following basic distributional results in place:

- pdf: $f(x) = \lambda e^{-\lambda x},\ x > 0,$
- cdf: $F(x) = P(X \le x) = \int_0^x \lambda e^{-\lambda y}\, dy = 1 - e^{-\lambda x},\ x \ge 0,$
- tpf: $\bar{F}(x) = P(X > x) = 1 - F(x) = e^{-\lambda x},\ x \ge 0,$
- mgf: $\phi_X(t) = \mathbb{E}[e^{tX}] = \int_0^\infty e^{tx} \cdot \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t} \int_0^\infty (\lambda - t) e^{-(\lambda - t)x} dx = \frac{\lambda}{\lambda - t},\ t < \lambda,$
- mean: $\mathbb{E}[X] = 1/\lambda,$
- variance: $\text{Var}(X) = 1/\lambda^2.$

Proposition 4.2 (Minimum of Independent Exponentials)**.** *Let $\{X_i\}_{i=1}^n$ be a sequence of independent rvs, where*

$$X_i \sim EXP(\lambda_i), \qquad i = 1, 2, \ldots, n.$$

Define $Y = \min\{X_1, X_2, \ldots, X_n\}$ to be their smallest order statistic. Then

$$Y = \min\{X_1, X_2, \ldots, X_n\} \sim EXP\left(\sum_{i=1}^n \lambda_i\right).$$

Proof. Clearly, Y takes on possible values in the state space $S = (0, \infty)$. To determine the distribution of Y, consider its tpf:

$$\begin{aligned}
\bar{F}_Y(y) &= P(Y > y) \\
&= P(\min\{X_1, X_2, \ldots, X_n\} > y) \\
&= P(X_1 > y, X_2 > y, \ldots, X_n > y) \\
&= P(X_1 > y) P(X_2 > y) \cdots P(X_n > y) \quad \text{by independence} \\
&= e^{-\lambda_1 y} \cdot e^{-\lambda_2 y} \cdots e^{-\lambda_n y} \quad \text{provided that } y \ge 0 \\
&= e^{-(\lambda_1 + \lambda_2 + \cdots + \lambda_n) y},\ y \ge 0.
\end{aligned}$$

which we recognize as the tpf of an $\text{EXP}(\sum_{i=1}^n \lambda_i)$ rv. Therefore, the stated result follows. □

Remark. As a special case of this result, if we additionally assume that $X_1, X_2, \ldots, X_n$ are iid $\text{EXP}(\lambda)$ rvs, then

$$Y = \min\{X_1, X_2, \ldots, X_n\} \sim \text{EXP}(n\lambda).$$

Example 4.1.

Let $\{X_i\}_{i=1}^n$ be a sequence of independent rvs where $X_i \sim \text{EXP}(\lambda_i),\ i = 1, 2, \ldots, n$.

(a) For $j \in \{1, 2, \ldots, n\}$, determine $P(X_j = \min\{X_1, X_2, \ldots, X_n\})$.

Solution.

$$\begin{aligned}
P(X_j = \min\{X_1, \ldots, X_n\}) & \\
&= P(X_j < X_i \text{ for every } i \neq j) \\
&= \int_0^\infty P(X_j < X_i \text{ for every } i \neq j \mid X_j = x)\lambda_j e^{-\lambda_j x}\, dx \\
&= \int_0^\infty \prod_{i \neq j} P(X_i > x)\, \lambda_j e^{-\lambda_j x}\, dx \\
&= \int_0^\infty \prod_{i \neq j} e^{-\lambda_i x}\, \lambda_j e^{-\lambda_j x}\, dx \\
&= \lambda_j \int_0^\infty \exp\left\{-\left(\sum_{i=1}^n \lambda_i\right) x\right\} dx \\
&= \frac{\lambda_j}{\sum_{i=1}^n \lambda_i} \int_0^\infty \left(\sum_{i=1}^n \lambda_i\right) \exp\left\{-\left(\sum_{i=1}^n \lambda_i\right) x\right\} dx \\
&= \frac{\lambda_j}{\sum_{i=1}^n \lambda_i} \qquad (4.1) \\
\therefore \quad P(X_j = \min\{X_1, \ldots, X_n\}) &= \frac{\lambda_j}{\sum_{i=1}^n \lambda_i}.
\end{aligned}$$

(b) Show that the conditional rv $X_1 \mid (X_1 < X_2 < \cdots < X_n)$ is identically distributed to the rv $\min\{X_1, X_2, \ldots, X_n\}$.

Solution. Let $Y = X_1 \mid (X_1 < X_2 < \cdots < X_n)$ for $y \geq 0$.

The tpf of Y is given by,

$$\bar{F}_Y(y) = P(X_1 > y \mid X_1 < X_2 < \cdots < X_n) = \frac{P(y < X_1 < \cdots < X_n)}{P(X_1 < \cdots < X_n)}.$$

Note that

$$\begin{aligned}
&P(y < X_1 < \cdots < X_n) \\
&\qquad = \int_y^\infty \int_{x_1}^\infty \int_{x_2}^\infty \cdots \int_{x_{n-2}}^\infty \left\{ \prod_{i=1}^n \lambda_i e^{-\lambda_i x_i} \right\} dx_n \cdots dx_2 dx_1 \\
&\qquad = \left(\prod_{i=1}^n \lambda_i \right) \int_y^\infty e^{-\lambda_1 x_1} \int_{x_1}^\infty e^{-\lambda_2 x_2} \cdots \int_{x_{n-2}}^\infty e^{-\lambda_{n-1} x_{n-1}} \\
&\cdot \int_{x_{n-1}}^\infty \lambda_n e^{-\lambda_n x_n} dx_n \cdots dx_2 dx_1 \\
&\qquad = \frac{\prod_{i=1}^n \lambda_i}{\lambda_{n-1} + \lambda_n} \int_y^\infty e^{-\lambda_1 x_1} \int_{x_1}^\infty e^{-\lambda_2 x_2} \cdots \int_{x_{n-2}}^\infty (\lambda_{n-1} + \lambda_n) \\
&\cdot e^{-(\lambda_{n-1}+\lambda_n) x_{n-1}} dx_{n-1} \cdots dx_2 dx_1 \\
&\qquad = \frac{\prod_{i=1}^n \lambda_i}{\prod_{i=1}^{n-1} \left(\sum_{j=i}^n \lambda_j \right)} \exp\left\{ - \left(\sum_{i=1}^n \lambda_i \right) y \right\}
\end{aligned}$$

By (4.3)

$$\begin{aligned}
P(X_1 < \cdots < X_n) &= P(0 < X_1 < \cdots < X_n) \\
&= \frac{\prod_{i=1}^n \lambda_i}{\prod_{i=1}^{n-1} \left(\sum_{j=i}^n \lambda_j \right)}. \qquad (4.4) \\
\Rightarrow \quad \bar{F}_Y(y) &= \exp\left\{ - \left(\sum_{i=1}^n \lambda_i \right) y \right\}, \quad \text{for } y \geq 0.
\end{aligned}$$

tpf of EXP($\sum_{i=1}^n \lambda_i$) r.v.

since,

$$\min\{X_1, \ldots, X_n\} \sim \text{EXP}\left(\sum_{i=1}^n \lambda_i \right).$$

Recall $\quad P(X_j = \min\{X_1, \ldots, X_n\}) = \dfrac{\lambda_j}{\sum_{i=1}^n \lambda_i}$

Remarks.

(1) In the case when $n = 2$, note that (4.1) simplifies to become

$$\text{By 4.1} \qquad P(X_1 = \min\{X_1, X_2\}) = P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2},$$

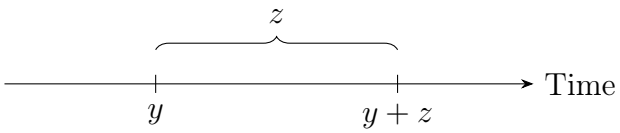
which agrees with the result of Example 2.11.

(2) Interestingly, looking at (4.4), we see that

$$\begin{aligned}\mathbb{P}(X_1 < \cdots < X_n) &= \prod_{i=1}^{n-1} \frac{\lambda_i}{\lambda_i + \lambda_{i+1} + \cdots + \lambda_n} \\ &= \prod_{i=1}^{n-1} \mathbb{P}\left(X_i = \min_{i \le j \le n} X_j \right).\end{aligned}$$

Definition 4.1 (Memoryless Property)**.** A rv X is *memoryless* iff

$$P(X > y + z \mid X > y) = P(X > z) \quad \forall y, z \geq 0.$$



Note that if we express $P(X > y+z \mid X > y)$ as $P(X - y > z \mid X > y)$ and think of X as being the lifetime of some component, then the memoryless property (or sometimes referred to as the *forgetfulness property*) states that the distribution of the remaining lifetime is independent of the time the component has already lasted. In other words, such a probability distribution is independent of its history.

An equivalent way to define the memoryless property is given by the following theorem.

Theorem 4.1. *X is memoryless iff*

$$P(X > y + z) = P(X > y)P(X > z) \quad \forall y, z \geq 0.$$

Proof. *Note that*

$$P(X > y + z \mid X > y) = \frac{P(X > y + z, X > y)}{P(X > y)} = \frac{P(X > y + z)}{P(X > y)}.$$

If X is memoryless then

$$P(X > y + z \mid X > y) = P(X > z) \;\Rightarrow\; P(X > z) = \frac{P(X > y + z)}{P(X > y)}$$

$$\begin{aligned}&\forall y, z \geq 0 \quad \text{then} \\ &\quad P(X > y + z) = P(X > y)P(X > z).\end{aligned}$$

Conversely if

$$P(X > y + z) = P(X > y)P(X > z)$$

then

$$P(X > y + z \mid X > y) = \frac{P(X > y + z)}{P(X > y)} = \frac{P(X > y)P(X > z)}{P(X > y)} = P(X > z).$$
$$\Rightarrow \quad X \text{ is memoryless by defn.}$$

□

This leads to one of the main results concerning the exponential distribution.

Theorem 4.2. *An exponential distribution is memoryless.*

Proof. Suppose $X \sim \text{EXP}(\lambda)$. For $y, z \geq 0$ we have

$$P(X > y + z)$$
$$= e^{-\lambda(y+z)} = e^{-\lambda y}e^{-\lambda z} = P(X > y)P(X > z)$$

by properties of exponents.

□

Example 4.2.

Suppose that a computer has 3 switches which govern the transfer of electronic impulses. These switches operate simultaneously and independently of one another, with lifetimes that are exponentially distributed with mean lifetimes of 10, 5, and 4 years, respectively.

(a) What is the probability that the time until the very first switch breakdown exceeds 6 years?

Solution. $X_i =$ lifetime of switch $i = 1, 2, 3$.

$X_i \sim \text{EXP}(\lambda_i)$ $\lambda_1 = \frac{1}{10}, \lambda_2 = \frac{1}{5}$ and $\lambda_3 = \frac{1}{4}$

Time until first breakdown is the r.v. $T = \min\{X_1, X_2, X_3\}$.

Since the lifetimes are indep.

$T \sim \text{EXP}(\lambda)$ where $\lambda = \frac{1}{10} + \frac{1}{5} + \frac{1}{4} = \frac{11}{20}$

TPF:

$$P(T > 6) = \exp\left\{-\frac{11}{20} \cdot 6\right\} = e^{-3.3} = 0.0369$$
$$= e^{-\sum \lambda_i \cdot x} \quad \text{with } x = 6 \text{ here.}$$

Also

(b) What is the probability that switch 2 outlives switch 1?

Solution. i.e. $P(X_1 < X_2)$

$$P(X_1 < X_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{\frac{1}{10}}{\frac{1}{10} + \frac{1}{5}} = \frac{1}{3}$$

by equation 4.

(c) What is the probability that switch 1 has the longest lifetime, followed next by switch 3 and then switch 2?

Solution.

$$\begin{aligned} &P(X_2 < X_3 < X_1) \\ &\qquad = \frac{\lambda_2}{\lambda_2 + \lambda_3 + \lambda_1} \cdot \frac{\lambda_3}{\lambda_3 + \lambda_1} = \frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{4} + \frac{1}{10}} \cdot \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{10}} \\ &\qquad = \frac{20}{77} \approx 0.260 \end{aligned}$$

$P(X_2 < X_3 < X_1)$

split the 3 way comparison

$\Rightarrow$ Solves by memoryless property. (d) If switch 3 is known to have lasted 2 years, what is the probability it will last at most 3 more years?

Solution.

$$\begin{aligned} &P(X_3 \leq 5 \mid X_3 > 2) \\ &\qquad = 1 - P(X_3 > 5 \mid X_3 > 2) \\ &\qquad = 1 - P(X_3 > 2 + 3 \mid X_3 > 2) \\ &\qquad = 1 - P(X_3 > 3) = 1 - e^{-\lambda_3 \cdot 3} = 1 - e^{-0.75} \approx 0.528 \end{aligned}$$

(e) Considering only switches 1 and 2, what is the expected amount of time until they have both suffered a breakdown?

Solution. Want $\mathbb{E}[\max\{X_1, X_2\}]$.

Note

$$\min\{X_1, X_2\} + \max\{X_1, X_2\} = X_1 + X_2$$

Taking expectations;

$$\begin{aligned} \mathbb{E}[\min\{X_1, X_2\} + \max\{X_1, X_2\}] &= \mathbb{E}[X_1 + X_2] \\ \mathbb{E}[\max\{X_1, X_2\}] &= \mathbb{E}[X_1] + \mathbb{E}[X_2] - \mathbb{E}[\min\{X_1, X_2\}] \\ &= 10 + 5 - \frac{1}{\frac{1}{10} + \frac{1}{5}} = 11.67. \end{aligned}$$

Note:

$$\int_0^\infty e^{-\lambda z} f_Z(z)\, dz = \int_0^\infty P(X > z) f_Z(z)\, dz$$

TPF

$$= \int_0^\infty P(X > z \mid Z = z) f_Z(z)\, dz = P(X > Z)$$

memoryless property.

Remarks.

(1) The exponential distribution is the unique continuous distribution possessing the memoryless property (incidentally, the geometric distribution is the unique discrete distribution which is memoryless, which is not all that surprising in light of Exercise 2.2.3.To prove this statement, suppose that X is a continuous rv satisfying the memoryless property. Let $\bar{F}(x) = P(X > x)$, which is a continuous function of x. By Theorem 4.2, it follows that

$$\bar{F}(y + z) = \bar{F}(y)\bar{F}(z) \quad \forall y, z \geq 0.$$

Note that

$$\bar{F}\left(\frac{2}{n}\right) = \bar{F}\left(\frac{1}{n} + \frac{1}{n}\right) = \bar{F}^2\left(\frac{1}{n}\right).$$

As a result, it immediately follows that $\bar{F}(m/n) = \bar{F}^m(1/n)$. Furthermore,

$$\bar{F}(1) = \bar{F}\left(\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n}\right) = \bar{F}^n\left(\frac{1}{n}\right),$$

or equivalently

$$\bar{F}\left(\frac{1}{n}\right) = (\bar{F}(1))^{1/n}.$$

Thus, $\bar{F}(x) = (\bar{F}(1))^x$ for all rational values of x, and by the continuity of $\bar{F}(x)$, this implies that $\bar{F}(x) = (\bar{F}(1))^x \ \forall x \geq 0$. However, note that we can write

$$\bar{F}(x) = e^{x \ln(\bar{F}(1))} = e^{-\lambda x},$$

where $\lambda = -\ln(\bar{F}(1)) > 0$. In other words,

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x},$$

which shows that X is exponentially distributed.

(2) The memoryless property of the exponential distribution even holds in a broader setting. Specifically, if $X \sim \text{EXP}(\lambda)$, then

$$P(X > Y + Z \mid X > Y) = P(X > Z), \tag{4.5}$$

where Y and Z are independently distributed non-negative valued rvs which are both independent of X. The equality defined by (4.5) is referred to as the generalized memoryless property. To prove the above result holds, note that

$$P(X > Y + Z \mid X > Y) = \frac{P(X > Y + Z, X > Y)}{P(X > Y)}.$$

Without loss of generality, assume that Y and Z are independent continuous rvs, so that

$$\begin{aligned}
&P(X > Y + Z, X > Y) \\
&\qquad = \int_0^\infty P(X > Y + Z, X > Y \mid Y = y) f_Y(y)\, dy \\
&\qquad = \int_0^\infty P(X > y + Z, X > y) f_Y(y)\, dy \\
&\qquad = \int_0^\infty P(X > y + Z) f_Y(y)\, dy \\
&\qquad = \int_0^\infty \left(\int_0^\infty P(X > y + z \mid Z = z) f_Z(z)\, dz \right) f_Y(y)\, dy \\
&\qquad = \int_0^\infty \left(\int_0^\infty P(X > y + z) f_Z(z)\, dz \right) f_Y(y)\, dy \\
&\qquad = \int_0^\infty \left(\int_0^\infty e^{-\lambda(y+z)} f_Z(z)\, dz \right) f_Y(y)\, dy \\
&\qquad = \int_0^\infty \left(\int_0^\infty e^{-\lambda z} f_Z(z)\, dz \right) e^{-\lambda y} f_Y(y)\, dy \\
&\qquad = \int_0^\infty P(X > Z) e^{-\lambda y} f_Y(y)\, dy \\
&\qquad = P(X > Z) \int_0^\infty e^{-\lambda y} f_Y(y)\, dy \\
&\qquad = P(X > Z) P(X > Y),
\end{aligned}$$

since we have for independent continuous rv Y (and similarly for Z)

$$P(X > Y) = \int_0^\infty P(X > y) f_Y(y)\, dy = \int_0^\infty e^{-\lambda y} f_Y(y)\, dy.$$

Thus,

$$P(X > Y + Z \mid X > Y) = \frac{P(X > Y + Z, X > Y)}{P(X > Y)} = \frac{P(X > Z) P(X > Y)}{P(X > Y)} = P(X > Z).$$

(3) The generalized memoryless property implies that $(X-Y) \mid (X>Y) \sim \text{EXP}(\lambda)$ regardless of the distribution of Y. To see this, let Z be an rv with a degenerate distribution at z. In this case, (4.5) becomes

$$P(X > Y + z \mid X > Y) = P(X > z) = e^{-\lambda z},$$

since $X \sim \text{EXP}(\lambda)$. Thus,

$$P(X - Y > z \mid X > Y) = e^{-\lambda z},$$

and so $(X-Y) \mid (X>Y) \sim \text{EXP}(\lambda)$.

Example 4.3.

Let X_1 and X_2 be independent rvs where $X_i \sim \text{EXP}(\lambda_i)$, $i = 1, 2$. Given $X_1 < X_2$, show that X_1 and $X_2 - X_1$ are conditionally independent rvs.

Solution. For $x, y \geq 0$ consider, To verify conditional independence, consider the joint conditional tpf:

$$\begin{aligned}
P(X_1 > x,\ X_2 - X_1 > y \mid X_1 < X_2) &= \frac{P(X_1 > x,\ X_2 - X_1 > y,\ X_1 < X_2)}{P(X_1 < X_2)} \\
&= \frac{\lambda_1 + \lambda_2}{\lambda_1} P(X_1 > x,\ X_2 > X_1 + y,\ X_2 > X_1) \\
&= \frac{\lambda_1 + \lambda_2}{\lambda_1} P(X_1 > x,\ X_2 > X_1 + y) \\
&= \frac{\lambda_1 + \lambda_2}{\lambda_1} \int_0^\infty P(X_1 > x,\ X_2 > w + y \mid X_1 = w) \lambda_1 e^{-\lambda_1 w} dw \\
&= \frac{\lambda_1 + \lambda_2}{\lambda_1} \int_x^\infty P(X_2 > w + y) \lambda_1 e^{-\lambda_1 w} dw \\
&= e^{-\lambda_2 y} \int_x^\infty (\lambda_1 + \lambda_2) e^{-(\lambda_1 + \lambda_2) w} dw \\
&= e^{-\lambda_2 y} e^{-(\lambda_1 + \lambda_2) x}.
\end{aligned}$$

Consider $(X_2 - X_1) \mid (X_1 < X_2) \sim (X_2 - X_1) \mid (X_2 > X_1) \sim \text{EXP}(\lambda_2)$ by generalized memoryless property.

Next, consider the conditional r.v.

$X_1 \mid (X_1 < X_2) \sim \min\{X_1, X_2\} \sim \text{EXP}(\lambda_1 + \lambda_2)$

Thus we have verified that

$$P(X_1 > x,\ X_2 - X_1 > y \mid X_1 < X_2) = P(X_1 > x \mid X_1 < X_2)\, P(X_2 - X_1 > y \mid X_1 < X_2).$$

4.2.1 The Erlang Distribution

Recall that if $X \sim \text{Erlang}(n, \lambda)$ where $n \in \mathbb{Z}^+$ and $\lambda > 0$, then its pdf is of the form

$$f(x) = \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!},\ x > 0.$$

Letting $n = 1$, the above pdf simplifies to become $f(x) = \lambda e^{-\lambda x},\ x > 0$, which is the $\text{EXP}(\lambda)$ pdf. To obtain the corresponding cdf of an $\text{Erlang}(n, \lambda)$ rv, we consider

$$F(x) = P(X \leq x) = \int_0^x \frac{\lambda^n y^{n-1} e^{-\lambda y}}{(n-1)!} dy = \frac{\lambda^n}{(n-1)!} \int_0^x y^{n-1} e^{-\lambda y} dy,\ x \geq 0.$$

Assume that $n \geq 2$ and apply integration by parts (i.e., $\int u\, dv = uv - \int v\, du$) to the above integral. In particular, choose

$$u = y^{n-1} \Rightarrow \frac{du}{dy} = (n-1)y^{n-2} \Rightarrow du = (n-1)y^{n-2} dy,$$

and

$$dv = e^{-\lambda y} dy \Rightarrow \int 1 \cdot dv = \int e^{-\lambda y} dy \Rightarrow v = -\frac{1}{\lambda} e^{-\lambda y},$$

so that

$$\begin{aligned} \int_0^x y^{n-1} e^{-\lambda y} dy &= -\frac{1}{\lambda} y^{n-1} e^{-\lambda y} \Big|_{y=0}^{y=x} + \frac{n-1}{\lambda} \int_0^x y^{n-2} e^{-\lambda y} dy \\ &= -\frac{1}{\lambda} x^{n-1} e^{-\lambda x} + \frac{n-1}{\lambda} \int_0^x y^{n-2} e^{-\lambda y} dy. \end{aligned}$$

Therefore,

$$\begin{aligned} F(x) &= \frac{\lambda^n}{(n-1)!} \left[-\frac{1}{\lambda} x^{n-1} e^{-\lambda x} + \frac{n-1}{\lambda} \int_0^x y^{n-2} e^{-\lambda y} dy \right] \\ &= \frac{\lambda^{n-1}}{(n-2)!} \int_0^x y^{n-2} e^{-\lambda y} dy - \frac{(\lambda x)^{n-1}}{(n-1)!} e^{-\lambda x}. \end{aligned}$$

If we continue to apply integration by parts until the "y-term" in the integrand has a power of zero, then it is possible to show that

$$F(x) = 1 - e^{-\lambda x} \sum_{j=0}^{n-1} \frac{(\lambda x)^j}{j!},\ x \geq 0. \tag{4.7}$$

Remark. Substituting $n = 1$ into (4.7), we immediately obtain

$$F(x) = \left(1 - e^{-\lambda x}\right) \sum_{j=0}^{1-1} \frac{(\lambda x)^j}{j!} = 1 - e^{-\lambda x}, \ x \geq 0,$$

which is clearly the cdf of an EXP(λ) rv.

To determine the mgf of $X \sim \text{Erlang}(n, \lambda)$, we consider

$$\begin{aligned}
\phi_X(t) &= \int_0^\infty e^{tx} \frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!} dx \\
&= \frac{\lambda^n}{(n-1)!} \int_0^\infty x^{n-1} e^{-\tilde{\lambda} x} dx \quad \text{where we define } \tilde{\lambda} = \lambda - t \\
&= \frac{\lambda^n}{\tilde{\lambda}} \int_0^\infty \frac{\tilde{\lambda}^n x^{n-1} e^{-\tilde{\lambda} x}}{(n-1)!} dx \quad \text{provided that } \tilde{\lambda} = \lambda - t > 0 \\
&= \left(\frac{\lambda}{\lambda - t}\right)^n, \ t < \lambda.
\end{aligned}$$

However, note that

$$\phi_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^n = \prod_{i=1}^n \left(\frac{\lambda}{\lambda - t}\right), \ t < \lambda,$$

is the product of n terms, where each term is the mgf of an EXP(λ) rv. Let $\{Y_i\}_{i=1}^n$ be an iid sequence of EXP(λ) rvs, with $\phi_{Y_i}(t) = \frac{\lambda}{\lambda - t}, \ t < \lambda$ for $i = 1, 2, \ldots, n$. Since $\phi_X(t) = \prod_{i=1}^n \phi_{Y_i}(t)$, it follows that an Erlang distribution can be viewed as the distribution of a sum of iid exponential rvs. As a result, the mean and variance of an Erlang(n, λ) rv X are simply obtained as

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^n Y_i\right] = \sum_{i=1}^n \mathbb{E}[Y_i] = \frac{n}{\lambda}$$

and

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^n Y_i\right) = \sum_{i=1}^n \text{Var}(Y_i) = \frac{n}{\lambda^2}.$$

4.3 The Poisson Process

Definition 4.2. A *counting process* $\{N(t), t \geq 0\}$ is a stochastic process in which $N(t)$ represents the number of events that happen (or occur) by time t, where the index t measures time over a continuous range.

Some examples of counting processes $\{N(t), t \geq 0\}$ might include:

(1) $N(t)$ represents the number of automobile accidents at a specified intersection by week t,

(2) $N(t)$ represents the number of births by year t in Canada,

(3) $N(t)$ represents the number of visits to a particular webpage by time t,

(4) $N(t)$ represents the number of customers who enter a store by time t,

(5) $N(t)$ represents the number of accident claims reported to an insurance company by time t.

4.3.1 Basic Properties of Counting Processes

(1) $N(0) = 0$.

(2) $N(t)$ is a non-negative integer $\forall t \geq 0$ (i.e., $N(t) \in \mathbb{N}\ \forall t \geq 0$).

(3) If $s < t$, then $N(s) \leq N(t)$.

(4) $N(t) - N(s)$ counts the number of events to occur in the time interval $(s, t]$ for $s < t$.

We now introduce two important properties associated with counting processes.

Definition 4.3. A counting process $\{N(t), t \geq 0\}$ has *independent increments* if $N(t_1) - N(s_1)$ is independent of $N(t_2) - N(s_2)$ when $(s_1, t_1] \cap (s_2, t_2] = \varnothing$ for all choices of s_1, t_1, s_2, t_2 (i.e., the number of events in non-overlapping time intervals are assumed to be independent of each other).

Definition 4.4. A counting process $\{N(t), t \geq 0\}$ has *stationary increments* if the distribution of the number of events in $(s, s + t]$ (i.e., $N(s + t) - N(s)$) depends only on t, the length of the time interval. In this case, $N(s + t) - N(s)$ has the same probability distribution as $N(0 + t) - N(0) = N(t)$, the number of events occurring in the interval $[0, t]$.

Before introducing the formal definition of a Poisson process, we first introduce a few mathematical tools which are needed.

Definition 4.5. A function $y = f(x)$ is said to be "$o(h)$" (i.e., little-o of h) if

$$\lim_{h \to 0} \frac{f(h)}{h} = 0.$$

Remark. An $o(h)$ function $y = f(x)$ is one in which $f(h)$ approaches 0 faster than h does.

Examples:

(1) $y = f(x) = x$. Note that

$$\lim_{h\to 0} \frac{f(h)}{h} = \lim_{h\to 0} \frac{h}{h} = \lim_{h\to 0} 1 \neq 0.$$

Thus, $y = x$ is not little-o of h.

(2) $y = f(x) = x^2$. Note that

$$\lim_{h\to 0} \frac{f(h)}{h} = \lim_{h\to 0} \frac{h^2}{h} = \lim_{h\to 0} h = 0.$$

Thus, $y = x^2$ is little-o of h. In fact, a function of the form $y = x^r$ is clearly little-o of h provided that $r > 1$.

(3) Suppose that $\{f_i(x)\}_{i=1}^n$ is a sequence of $o(h)$ functions. Consider the linear combination of $o(h)$ functions, namely $y = \sum_{i=1}^n c_i f_i(x)$, and note that

$$\lim_{h\to 0} \frac{\sum_{i=1}^n c_i f_i(h)}{h} = \sum_{i=1}^n c_i \lim_{h\to 0} \frac{f_i(h)}{h} = 0.$$

Thus, a linear combination of $o(h)$ functions is still little-o of h.

Remark. In most cases, this result is even true when $n = \infty$.

Definition 4.6. A counting process $\{N(t), t \geq 0\}$ is said to be a *Poisson process at rate* λ if the following three conditions hold true:

(1) The process has both independent and stationary increments.

(2) For $h > 0$, $P(N(h) = 1) = \lambda h + o(h)$.

(3) For $h > 0$, $P(N(h) \geq 2) = o(h)$.

Remarks.

(1) Condition (2) in the above definition implies that in a "small" interval of time, the probability of a single event occurring is essentially proportional to the length of the interval.

(2) Condition (3) in the above definition implies that two or more events occurring in a "small" interval of time is rare.

(3) Conditions (2) and (3) yield

$$\begin{aligned} P(N(h)=0) &= 1 - P(N(h) > 0) \\ &= 1 - P(N(h)=1) - P(N(h) \geq 2) \\ &= 1 - (\lambda h + o(h)) - o(h) \\ &= 1 - \lambda h - o(h) - o(h) \\ &= 1 - \lambda h + o(h). \end{aligned}$$

Ultimately for a Poisson process $\{N(t), t \geq 0\}$ at rate λ, we would like to know the distribution of the rv $N(s+t) - N(s)$, representing the number of events occurring in the interval $(s, s+t]$, $s, t \geq 0$. Since a Poisson process has stationary increments, this rv has the same probability distribution as $N(t)$. The following theorem specifies the distribution of $N(t)$.

Theorem 4.3. *If $\{N(t), t \geq 0\}$ is a Poisson process at rate λ, then $N(t) \sim POI(\lambda t)$.*

Proof. For $t \geq 0$ and $u \leq 0$, define $\phi_t(u) = \mathbb{E}[e^{uN(t)}]$.

For $h \geq 0$, consider

$$\begin{aligned} \phi_{t+h}(u) &= \mathbb{E}[e^{uN(t+h)}] \\ &= \mathbb{E}[e^{u(N(t+h)-N(t)+N(t))}] \\ &= \mathbb{E}[e^{u(N(t+h)-N(t))}e^{uN(t)}] \\ &= \mathbb{E}[e^{u(N(t+h)-N(t))}]\mathbb{E}[e^{uN(t)}] \quad \text{(indep. increments)} \\ &= \mathbb{E}[e^{uN(h)}]\mathbb{E}[e^{uN(t)}] \quad \text{(stationary increments)} \\ &= \phi_h(u) \cdot \phi_t(u) \end{aligned} \tag{4.8}$$

Note that for $j \geq 2$, we see that

$$\begin{aligned} 0 \leq P(N(h)=j) &\leq P(N(h) \geq 2) \\ &\Rightarrow \lim_{h \to 0} \frac{P(N(h)=j)}{h} = 0 \end{aligned}$$

and so

$$\lim_{h \to 0} \frac{P(N(h)=j)}{h} = 0$$

by the squeeze theorem, implying that $P(N(h) = j)$ is $o(h)$. Using the result from above, we get

$$\begin{aligned}
\phi_h(u) &= \mathbb{E}[e^{uN(h)}] \\
&= \sum_{j=0}^{\infty} e^{uj} P(N(h) = j) \\
&= P(N(h) = 0) + e^u P(N(h) = 1) + \sum_{j \geq 2} e^{uj} P(N(h) = j) \\
&= 1 - \lambda h + o(h) + e^u(\lambda h + o(h)) + o(h) \\
&= 1 - \lambda h + e^u \lambda h + o(h)
\end{aligned}$$

Returning to (4.8) we have

$$\begin{aligned}
\phi_{t+h}(u) &= \phi_t(u)\left[1 - \lambda h + e^u \lambda h + o(h)\right] \\
\phi_{t+h}(u) - \phi_t(u) &= \lambda h(e^u - 1)\phi_t(u) + o(h) \\
\frac{\phi_{t+h}(u) - \phi_t(u)}{h} &= \lambda(e^u - 1)\phi_t(u) + \frac{o(h)}{h}
\end{aligned}$$

Now take the limit!

$$\begin{aligned}
\lim_{h \to 0} \frac{\phi_{t+h}(u) - \phi_t(u)}{h} &= \lambda(e^u - 1)\phi_t(u) \\
&\Rightarrow \frac{d}{dt}\phi_t(u) = \lambda(e^u - 1)\phi_t(u) \\
\frac{d}{dt}\phi_t(u)/\phi_t(u) &= \lambda(e^u - 1) \\
\frac{d}{dt}[\ln \phi_t(u)] &= \lambda(e^u - 1) \\
\int_0^t d[\ln \phi_s(u)] &= \int_0^t \lambda(e^u - 1) ds \\
\ln \phi_t(u) - \ln \phi_0(u) &= \lambda t(e^u - 1) \\
\ln \phi_t(u) &= \lambda t(e^u - 1) \\
\phi_t(u) &= \exp\{\lambda t(e^u - 1)\}
\end{aligned}$$

But $\phi_0(u) = \mathbb{E}[e^{uN(0)}] = \mathbb{E}[e^0] = 1$. We recognize this form, for $u \leq 0$, as the transform of a POI(λt) rv. By the uniqueness property, we conclude that

$$N(t) \sim \text{POI}(\lambda t),$$

as required.

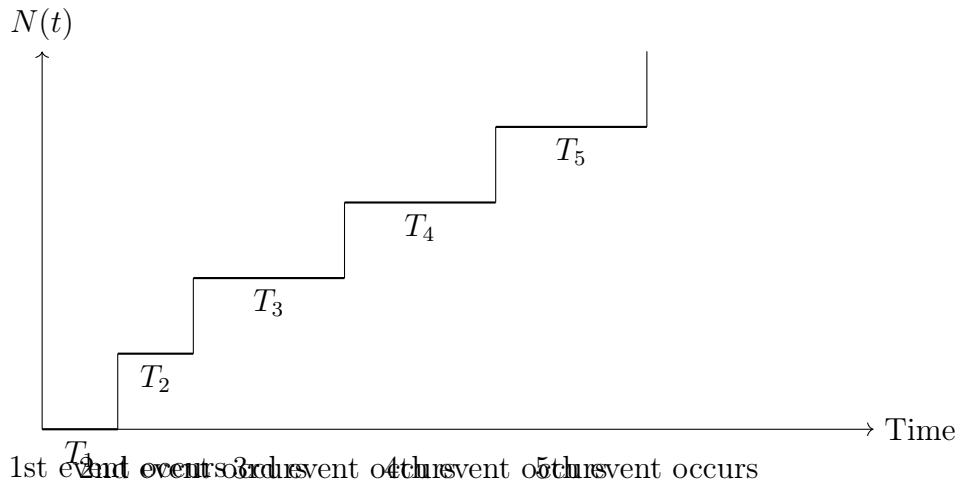
□

Remark. As a direct consequence of Theorem 4.3, for all $s, t \geq 0$, we have

$$P(N(s+t) - N(s) = k) = P(N(t) = k) = \frac{e^{-\lambda t}(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \ldots$$

4.3.2 Interarrival Times

Define T_1 to be the elapsed time (from time 0) until the first event occurs. In general, for $i \geq 2$, let T_i be the elapsed time between the occurrences of the $(i-1)$th event and the ith event. The sequence $\{T_i\}_{i=1}^{\infty}$ is called the interarrival or interevent time sequence. The diagram below depicts the relationship between $N(t)$ and $\{T_i\}_{i=1}^{\infty}$.



A very important result linking a Poisson process to its interarrival time sequence now follows.

Theorem 4.4. *If $\{N(t), t \geq 0\}$ is a Poisson process at rate $\lambda > 0$, then $\{T_i\}_{i=1}^{\infty}$ is a sequence of iid EXP(λ) rvs.*

Proof. We first have, for every $t \geq 0$,

$$\begin{aligned} \mathbb{P}(T_1 > t) &= \mathbb{P}(N(t) = 0) \\ &= e^{-\lambda t}, \end{aligned}$$

so $T_1 \sim \text{Exp}(\lambda)$. To establish the *joint* distribution and independence of all interarrival times, fix $n \geq 1$ and ordered times $0 < s_1 < \cdots < s_n$. Set $s_0 = 0$. The probability of exactly one event in each small interval $(s_i, s_i + h_i]$ and no other events between time 0 and $s_n + h_n$ can be computed from the independent increment property. Dividing by $\prod_{i=1}^{n} h_i$ and letting every $h_i \downarrow 0$ gives the joint density of the first n arrival epochs:

$$\begin{aligned} f_{S_1,\ldots,S_n}(s_1, \ldots, s_n) &= \prod_{i=1}^{n} \left(\lambda e^{-\lambda(s_i - s_{i-1})}\right) \\ &= \lambda^n e^{-\lambda s_n}, \qquad 0 < s_1 < \cdots < s_n. \end{aligned}$$

Now use the transformation $t_i = s_i - s_{i-1}$ for $i = 1, \ldots, n$. Its inverse is $s_i = \sum_{j=1}^{i} t_j$, and the associated triangular Jacobian matrix has determinant 1. Consequently,

$$\begin{aligned} f_{T_1,\ldots,T_n}(t_1, \ldots, t_n) &= \lambda^n \exp\left(-\lambda \sum_{i=1}^{n} t_i\right) \\ &= \prod_{i=1}^{n} \lambda e^{-\lambda t_i}, \qquad t_i > 0. \end{aligned}$$

The factorization of the joint density shows that $T_1, \ldots, T_n$ are independent and each has the $\text{Exp}(\lambda)$ distribution. Since n was arbitrary, the entire interarrival sequence is independent and identically distributed. □

For $n \in \mathbb{Z}^+$, define S_n to be the total elapsed time until the nth event occurs. In other words, S_n denotes the arrival time of the nth event, or the waiting time until the nth event occurs. Clearly, $S_n = \sum_{i=1}^{n} T_i$. If $\{N(t), t \geq 0\}$ is a Poisson process at rate λ, then $\{T_i\}_{i=1}^{\infty}$ is a sequence of iid $\text{EXP}(\lambda)$ rvs by Theorem 4.4, implying that

$$S_n = \sum_{i=1}^{n} T_i \sim \text{Erlang}(n, \lambda).$$

sum of iid $\text{EXP}(\lambda)$ is $\text{Erlang}(n, \lambda)$ S_n is sum of n iid interarrival times.

From our earlier results on the Erlang distribution, we have $\mathbb{E}[S_n] = n/\lambda$, $\text{Var}(S_n) = n/\lambda^2$, and

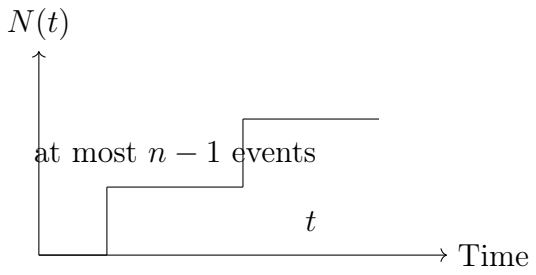
$$\text{TPF of } S_n \qquad P(S_n > t) = e^{-\lambda t} \sum_{j=0}^{n-1} \frac{(\lambda t)^j}{j!}, \quad t \geq 0.$$

where to expect arrival of nth event.

Remarks.

(1) The above formula for the tpf of S_n could have been obtained without reference to the Erlang distribution. In particular, note that

$$\begin{aligned} P(S_n > t) &= P(\text{arrival time of the } n^{\text{th}} \text{ event occurs after time } t) \\ &= P(\text{at most } n-1 \text{ events occur by time } t) \\ &= P(N(t) \leq n-1) \\ &= \sum_{j=0}^{n-1} \frac{e^{-\lambda t}(\lambda t)^j}{j!} \quad \text{since } N(t) \sim \text{POI}(\lambda t) \\ &= e^{-\lambda t} \sum_{j=0}^{n-1} \frac{(\lambda t)^j}{j!}. \end{aligned}$$



(2) If $\{X_i\}_{i=1}^{\infty}$ represents an iid sequence of EXP(λ) rvs and one constructs a counting process $\{N(t), t \geq 0\}$ defined by $N(t) = \max\{n \in \mathbb{N} : \sum_{i=1}^{n} X_i \leq t\}$, then $\{N(t), t \geq 0\}$ is actually a Poisson process at rate λ. In other words, $\{N(t), t \geq 0\}$ has both independent and stationary increments (due to the memoryless property that the sequence of rvs $\{X_i\}_{i=1}^{\infty}$ possesses), in addition to the fact that

$$P(N(t) \leq k) = P\left(\sum_{i=1}^{k+1} X_i > t\right) = e^{-\lambda t} \sum_{j=0}^{k} \frac{(\lambda t)^j}{j!} \quad \text{since } \sum_{i=1}^{k+1} X_i \sim \text{Erlang}(k+1, \lambda),$$

which subsequently leads to

$$\begin{aligned} P(N(t) = k) &= P(N(t) \leq k) - P(N(t) \leq k-1) \\ &= e^{-\lambda t} \sum_{j=0}^{k} \frac{(\lambda t)^j}{j!} - e^{-\lambda t} \sum_{j=0}^{k-1} \frac{(\lambda t)^j}{j!} \\ &= \frac{e^{-\lambda t}(\lambda t)^k}{k!}, \quad k = 0, 1, 2, \ldots \end{aligned}$$

Example 4.4.

At a local insurance company, suppose that fire damage claims come into the company according to a Poisson process at rate 3.8 expected claims per year. $\lambda = 3.8$

(a) What is the probability that exactly 5 claims occur in the time interval $(2.5, 5]$ (measured in years)?

Solution. Let $N(t)$ be the number of claims arriving to the company in the interval $[0, t]$. Since $\{N(t), t \geq 0\}$ is a Poisson process with $\lambda = 3.8$ we want to find

$$P(N(5) - N(2.5) = 5) = \frac{e^{-3.8(5-2.5)}(3.8(5-2.5))^5}{5!} \approx 0.0483.$$

$\sim$ POI(3.8×2.5) by thm POI(λt).

(b) What is the probability that the time between the 2nd and 4th claims is between 2 and 5 months?

Solution. Let T be the time between the 2nd and 4th claims.

As the graph indicates

$$T = T_3 + T_4 \quad \text{where } T_i \sim \text{EXP}(3.8) \text{ for } i = 3, 4$$

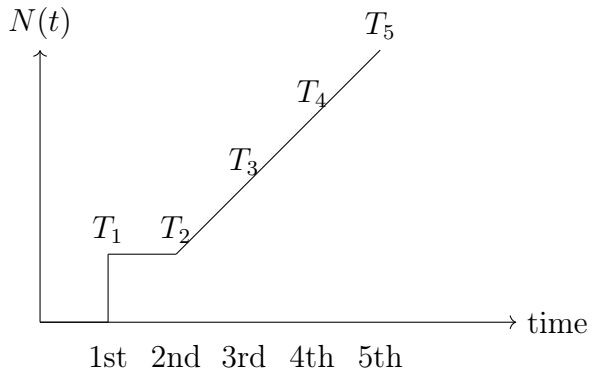
Thus $T \sim \text{Erlang}(2, 3.8)$

Recall:

$$P(T \leq t) = 1 - e^{-3.8t} \sum_{j=0}^{1} \frac{(3.8t)^j}{j!} = 1 - e^{-3.8t}(1 + 3.8t), \quad t \geq 0$$

we want to calculate

$$\begin{aligned} P\left(\frac{2}{12} < T < \frac{5}{12}\right) &= P\left(T > \frac{2}{12}\right) - P\left(T > \frac{5}{12}\right) \\ &= e^{-3.8(\frac{2}{12})}\left(1 + 3.8\frac{2}{12}\right) - e^{-3.8(\frac{5}{12})}\left(1 + 3.8\frac{5}{12}\right) \\ &\approx 0.337. \end{aligned}$$



4.4 Further Properties of the Poisson Process

In this section, we shed further light on a number of interesting and important mathematical properties associated with the Poisson process. To begin with, the binomial distribution also arises in the context of Poisson processes, as the following theorem establishes.

Theorem 4.5. *If $\{N(t), t \geq 0\}$ is a Poisson process at rate λ, then $N(s) \mid (N(t) = n) \sim BIN(n, s/t)$ where $s < t$.*

Proof. We wish to determine the conditional distribution of $N(s) \mid (N(t) = n)$ for $s < t$.

Clearly, $N(s) \mid (N(t) = n)$ takes on values in the set $\{0, 1, \ldots, n\}$. Therefore, for $m = 0, 1, \ldots, n$ note that

$$\begin{aligned}
P(N(s) = m \mid N(t) = n) &= \frac{P(N(s) = m, N(t) = n)}{P(N(t) = n)} \\
&= \frac{P(N(s) = m, N(t) - N(s) + N(s) = n)}{P(N(t) = n)} \\
&= \frac{P(N(s) = m, N(t) - N(s) = n - m)}{P(N(t) = n)} \\
&= \frac{P(N(s) = m)P(N(t) - N(s) = n - m)}{P(N(t) = n)} \quad \text{by independent increments} \\
&= \frac{\frac{e^{-\lambda s}(\lambda s)^m}{m!} \frac{e^{-\lambda(t-s)}(\lambda(t-s))^{n-m}}{(n-m)!}}{\frac{e^{-\lambda t}(\lambda t)^n}{n!}} \\
&= \binom{n}{m} \left(\frac{s}{t}\right)^m \left(1 - \frac{s}{t}\right)^{n-m},
\end{aligned}$$

which we recognize as the pmf of a BIN$(n, s/t)$.

□

4.4.1 Comparison of Event Occurrences

Suppose now that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ are independent Poisson processes at rates λ_1 and λ_2, respectively. Let $S_m^{(1)}$ be the arrival time of the mth event for $\{N_1(t), t \geq 0\}$. Likewise, let $S_n^{(2)}$ be the arrival time of the nth event for $\{N_2(t), t \geq 0\}$. Based on our knowledge of arrival times, we know that $S_m^{(1)} = \sum_{i=1}^m T_i^{(1)}$ where $\{T_i^{(1)}\}_{i=1}^\infty$ is a sequence of iid EXP(λ_1) rvs. Similarly, $S_n^{(2)} = \sum_{j=1}^n T_j^{(2)}$ where $\{T_j^{(2)}\}_{j=1}^\infty$ is a sequence of iid EXP(λ_2) rvs. Moreover, the sequences $\{T_i^{(1)}\}_{i=1}^\infty$ and $\{T_j^{(2)}\}_{j=1}^\infty$ are independent.

We are interested in the probability that the mth event from the first process happens before the nth event of the second process, or equivalently, $P(S_m^{(1)} < S_n^{(2)})$. Before looking at the general case, let us first examine a couple of special cases:

Take $m = n = 1$:

$$P(S_1^{(1)} < S_1^{(2)}) = P(T_1^{(1)} < T_1^{(2)}) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

Take $m = 2, n = 1$:

$$
\begin{aligned}
P(S_2^{(1)} < S_1^{(2)}) &= P(T_1^{(1)} < T_1^{(2)})P(S_2^{(1)} < S_1^{(2)} \mid T_1^{(1)} < T_1^{(2)}) \\
&\quad + P(T_1^{(1)} > T_1^{(2)})P(S_2^{(1)} < S_1^{(2)} \mid T_1^{(1)} > T_1^{(2)}) \\
&= P(T_1^{(1)} < T_1^{(2)})P(T_1^{(1)} + T_2^{(1)} < T_1^{(2)} \mid T_1^{(1)} < T_1^{(2)}) \\
&= P(T_1^{(1)} < T_1^{(2)})P(T_2^{(1)} < T_1^{(2)} - T_1^{(1)} \mid T_1^{(2)} > T_1^{(1)}) \\
&= P(T_1^{(1)} < T_1^{(2)})P(T_2^{(1)} < T_1^{(2)}) \\
&= \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^2.
\end{aligned}
$$

In the general case, we realize, through the continued application of the memoryless property, that $P(S_m^{(1)} < S_n^{(2)})$ is equivalent to the probability of observing m "successes" (where the success probability is $\lambda_1/(\lambda_1+\lambda_2)$) occur before n "failures" (where the failure probability is $\lambda_2/(\lambda_1+\lambda_2)$) in a sequence of independent Bernoulli trials. Specifically, in a sequence of $m+j$ Bernoulli trials (in which m are "successes" and j are "failures"), the $(m+j)^{\text{th}}$ trial must always be a "success" (i.e., the mth one) and the number of "failures" j must be no larger than $n-1$, which ultimately leads to

$$
P(S_m^{(1)} < S_n^{(2)}) = \sum_{j=0}^{n-1} \binom{m+j-1}{m-1} \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^m \left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^j. \tag{4.9}
$$

Example 4.4.

(continued)

(c) If exactly 12 claims have occurred within the first 5 years, how many claims, on average, occurred within the first 3.5 years? How would this change if no claim history of the first 5 years was given?

Solution. We want to calculate $\mathbb{E}[N(3.5) \mid N(5) = 12]$.

Using the binomial result from Theorem 4.5 we have $s = 3.5$, $t = 5$ and $n = 12$ and so

$$
\mathbb{E}[N(3.5) \mid N(5) = 12] = np = 12\left(\frac{3.5}{5}\right) = \frac{42}{5} = 8.4.
$$

On the other hand,

$$
\mathbb{E}[N(3.5)] = \lambda(3.5) = 3.8 \cdot 3.5 = 13.3 \neq 8.4,
$$

so this implies that conditioning on knowledge of $N(5)$ does affect the mean of $N(3.5)$.

(d) At another competing insurance company, suppose that fire damage claims arrive to the company according to a Poisson process with rate 2.9 expected claims per year. What is the

probability that 3 claims arrive to this company before 2 claims arrive to the other (first) company? Assume that the insurance companies operate independently of each other.

Solution. Let $N_1(t)$ denote the # of claims arriving to the first company by time t, whereas $N_2(t)$ denotes the number of claims arriving to this new (second) company.

Using 4.9 we need to calculate

$$\begin{aligned}\mathbb{P}(S_3^{(2)} < S_2^{(1)}) &= 1 - \mathbb{P}(S_2^{(1)} < S_3^{(2)}) \\ &= 1 - \sum_{j=0}^{2} \binom{j+1}{1} \left(\frac{3.8}{6.7}\right)^2 \left(\frac{2.9}{6.7}\right)^j \\ &\approx 0.219.\end{aligned}$$

4.4.2 Splitting and Merging Poisson Processes

The next property we examine concerns the classification (i.e., splitting) of events from a Poisson process into (potentially) several different types. Specifically, for a Poisson process $\{N(t), t \geq 0\}$ at rate λ, suppose that events can be independently classified as being one of k possible types, with probability p_i of being of type i, $i = 1, 2, \ldots, k$, with $\sum_{i=1}^{k} p_i = 1$. As such, let $\{N_i(t), t \geq 0\}$ be the associated counting process for type-i events, $i = 1, 2, \ldots, k$. Clearly, by construction, $\sum_{i=1}^{k} N_i(t) = N(t)$. First of all, for $s, t \geq 0$ and $i = 1, 2, \ldots, k$, note that

$$\begin{aligned}&\mathbb{P}(N_i(s+t) - N_i(s) = m_i) \\ &\quad= \sum_{n=m_i}^{\infty} \binom{n}{m_i} p_i^{m_i} (1-p_i)^{n-m_i} \mathbb{P}(N(s+t) - N(s) = n) \\ &\quad= \sum_{n=m_i}^{\infty} \binom{n}{m_i} p_i^{m_i} (1-p_i)^{n-m_i} \mathbb{P}(N(t) = n) \\ &\quad= \mathbb{P}(N_i(t) = m_i).\end{aligned}$$

which proves that $\{N_i(t), t \geq 0\}$ also has the stationary increments property.

Next, suppose that $(s_1, t_1]$ and $(s_2, t_2]$ are non-overlapping time intervals. For $i = 1, 2, \ldots, k$, note that by the independent increments property of $\{N(t), t \geq 0\}$, the number of events in each of these intervals, $N(t_1) - N(s_1)$ and $N(t_2) - N(s_2)$, are independent. Therefore, in combination with the fact that the classification of each event is an independent process, it must hold that the number of type-i events to occur in these intervals, $N_i(t_1) - N_i(s_1)$ and $N_i(t_2) - N_i(s_2)$, are also independent, implying that $\{N_i(t), t \geq 0\}$ possesses independent increments.

Finally, consider Write $m = \sum_{i=1}^{k} m_i$. Conditioning on the total number of events and using the multinomial distribution gives

$$\begin{aligned}
&\mathbb{P}(N_1(t) = m_1, \ldots, N_k(t) = m_k) \\
&\quad = \mathbb{P}(N_1(t) = m_1, \ldots, N_k(t) = m_k \mid N(t) = m)\mathbb{P}(N(t) = m) \\
&\quad = \frac{m!}{\prod_{i=1}^{k} m_i!} \left(\prod_{i=1}^{k} p_i^{m_i} \right) \frac{e^{-\lambda t}(\lambda t)^m}{m!} \\
&\quad = \prod_{i=1}^{k} \frac{e^{-\lambda p_i t}(\lambda p_i t)^{m_i}}{m_i!} \\
&\quad = \prod_{i=1}^{k} \mathbb{P}(N_i(t) = m_i),
\end{aligned}$$

where

$$P(N_i(t) = m_i) = \frac{e^{-\lambda p_i t}(\lambda p_i t)^{m_i}}{m_i!}$$

is the pmf of a Poisson rv with mean $\lambda p_i t$, thereby proving that $N_1(t), N_2(t), \ldots, N_k(t)$ are independent Poisson rvs. As a result, we have

$$\begin{aligned}
P(N_i(h) = 1) &= \frac{e^{-\lambda p_i h}(\lambda p_i h)^1}{1!} \\
&= \lambda p_i h \sum_{j=0}^{\infty} \frac{(-\lambda p_i h)^j}{j!} \\
&= \lambda p_i h(1 - \lambda p_i h + o(h)) \\
&= \lambda p_i h + o(h),
\end{aligned}$$

and

$$\begin{aligned}
P(N_i(h) \geq 2) &= 1 - P(N_i(h) = 0) - P(N_i(h) = 1) \\
&= 1 - \frac{e^{-\lambda p_i h}(\lambda p_i h)^0}{0!} - \lambda p_i h - o(h) \\
&= 1 - (1 - \lambda p_i h + o(h)) - \lambda p_i h - o(h) \\
&= o(h).
\end{aligned}$$

Hence, by definition, we have shown that $\{N_1(t), t \geq 0\}, \{N_2(t), t \geq 0\}, \ldots, \{N_k(t), t \geq 0\}$ are independent Poisson processes in which $\{N_i(t), t \geq 0\}$ has rate λp_i, $i = 1, 2, \ldots, k$.

Example 4.4.

(continued)

(e) Suppose that fire damage claims can be categorized as being either commercial, business, or residential. At the first insurance company, past history suggests that 15% of the claims are commercial, 25% of them are business, and the remaining 60% are residential. Over the course of the next 4 years, what is the probability that the company experiences fewer than 5 claims in each of these 3 categories?

Solution. Let $N_C(t)$ denote the # of commercial claims by time t. Likewise, let $N_B(t)$ and $N_R(t)$ represent the # of business and residential claims by time t, respectively.

It follows that

$$\begin{aligned} N_C(4) &\sim \text{POI}(3.8 \cdot 0.15 \cdot 4) = \text{POI}(2.28), \\ N_B(4) &\sim \text{POI}(3.8 \cdot 0.25 \cdot 4) = \text{POI}(3.8), \\ N_R(4) &\sim \text{POI}(3.8 \cdot 0.60 \cdot 4) = \text{POI}(9.12). \end{aligned}$$

We want to calculate

$$\begin{aligned} P(N_C(4) < 5, N_B(4) < 5, N_R(4) < 5) &= P(N_C(4) < 5)P(N_B(4) < 5)P(N_R(4) < 5) \\ &= \left(\sum_{i=0}^{4} \frac{e^{-2.28}(2.28)^i}{i!} \right) \left(\sum_{i=0}^{4} \frac{e^{-3.8}(3.8)^i}{i!} \right) \left(\sum_{i=0}^{4} \frac{e^{-9.12}(9.12)^i}{i!} \right) \\ &\approx 0.0313. \end{aligned}$$

Remark. It is also possible to merge independent Poisson processes together. In particular, if $\{N_1(t), t \geq 0\}, \{N_2(t), t \geq 0\}, \ldots, \{N_m(t), t \geq 0\}$ are m independent Poisson processes at respective rates $\lambda_1, \lambda_2, \ldots, \lambda_m$, then it is straightforward to show that $\{N(t), t \geq 0\}$, where $N(t) = \sum_{i=1}^{m} N_i(t)$, is also a Poisson process at rate $\sum_{i=1}^{m} \lambda_i$.

4.4.3 Conditional Distribution of Arrival Times

Theorem 4.5 indicated that the conditional distribution of $N(s) \mid (N(t) = n)$, where $s < t$, is binomial with n trials and success probability s/t. In other words, it is possible to view each event that occurred within $[0, t]$ as being independent of the others, and the probability of any one event landing within the interval $[0, s]$ as being governed by the cdf of a $U(0, t)$ rv evaluated at s. The idea that we can view t as a uniform probability is no coincidence. In fact, the following result confirms this notion.

Theorem 4.6. *Suppose that $\{N(t), t \geq 0\}$ is a Poisson process at rate λ. Given $N(t) = 1$, the conditional distribution of the first arrival time is uniformly distributed on $(0, t)$ (i.e., $S_1 \mid (N(t) = 1) \sim U(0, t)$).*

Proof. In order to identify the conditional distribution of $S_1 \mid (N(t) = 1)$ we consider the cdf of $S_1 \mid (N(t) = 1)$ to be denoted by:

$$G(s) = P(S_1 \leq s \mid N(t) = 1)$$

Note that

$$\begin{aligned}
G(s) &= P(S_1 \leq s \mid N(t) = 1) \\
&= \frac{P(S_1 \leq s, N(t) = 1)}{P(N(t) = 1)} \\
&= \frac{P(1 \text{ event in } [0, s] \cap 0 \text{ events in } (s, t])}{P(N(t) = 1)} \\
&= \frac{P(N(s) = 1, N(t) - N(s) = 0)}{P(N(t) = 1)} \\
&= \frac{P(N(s) = 1)P(N(t) - N(s) = 0)}{P(N(t) = 1)} \quad \text{by independent increments} \\
&= \frac{(\lambda s)e^{-\lambda s}e^{-\lambda(t-s)}}{(\lambda t)e^{-\lambda t}} \\
&= \frac{s}{t}.
\end{aligned}$$

The pdf of $S_1 \mid (N(t) = 1)$ is

$$g(s) = G'(s) = \frac{d}{ds}\left(\frac{s}{t}\right) = \frac{1}{t}, \quad 0 < s < t.$$

$S_1 \mid (N(t) = 1) \sim U(0, t)$.

□

Theorem 4.6 specifies how S_1 behaves distributionally when $N(t) = 1$. A natural question to ask is: *How are the n arrival times $S_1, S_2, \ldots, S_n$ distributed if it is known that exactly n arrivals have occurred by time t?* Before we can address this more general question, we must first familiarize ourselves with some distributional results about order statistics.

In what follows, let $\{Y_i\}_{i=1}^n$ be an iid sequence of rvs having a common continuous distribution on $(0, \infty)$ with cdf $F(y) = P(Y_i \leq y)$ and pdf $f(y) = F'(y)$ for each $i = 1, 2, \ldots, n$.

Definition 4.7. The sequence of rvs $\{Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)}\}$ are called the *order statistics* of $\{Y_1, Y_2, \ldots, Y_n\}$, satisfying:

$$
\begin{aligned}
Y_{(1)} &= \text{1st smallest among } \{Y_1, Y_2, \ldots, Y_n\},\\
Y_{(2)} &= \text{2nd smallest among } \{Y_1, Y_2, \ldots, Y_n\},\\
&\vdots\\
Y_{(n)} &= n^{\text{th}} \text{ smallest among } \{Y_1, Y_2, \ldots, Y_n\}.
\end{aligned}
$$

Remark. By its very definition, we observe that

$$Y_{(1)} < Y_{(2)} < \cdots < Y_{(n)} \quad \text{almost surely,}$$

and moreover,

$$Y_{(1)} = \min\{Y_1, Y_2, \ldots, Y_n\} \quad \text{and} \quad Y_{(n)} = \max\{Y_1, Y_2, \ldots, Y_n\}.$$

We wish to determine the joint distribution of the random vector $(Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)})$. Let the joint cdf of $(Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)})$ be denoted by $G(y_1, y_2, \ldots, y_n) = P(Y_{(1)} \le y_1, Y_{(2)} \le y_2, \ldots, Y_{(n)} \le y_n)$. Also, define

$$g(y_1, y_2, \ldots, y_n) = \frac{\partial^n G(y_1, y_2, \ldots, y_n)}{\partial y_1 \partial y_2 \cdots \partial y_n}$$

to be the corresponding joint pdf of $(Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)})$.

To begin with, consider the case when $n = 2$ and assume that $0 < y_1 < y_2 < \infty$. Note that

$$
\begin{aligned}
P(Y_{(1)} \le y_1, Y_{(2)} \le y_2) &= P(Y_{(1)} \le y_1, y_1 < Y_{(2)} \le y_2) + P(Y_{(1)} \le y_1, Y_{(2)} \le y_1)\\
&= G(y_1, y_2) - G(y_1, y_1),
\end{aligned}
$$

and so

$$
\begin{aligned}
G(y_1, y_2) &= P(Y_{(1)} \le y_1, y_1 < Y_{(2)} \le y_2) + G(y_1, y_1),\\
\frac{\partial^2 G(y_1, y_2)}{\partial y_1 \partial y_2} &= \frac{\partial^2 P(Y_{(1)} \le y_1, y_1 < Y_{(2)} \le y_2)}{\partial y_1 \partial y_2} + \frac{\partial^2 G(y_1, y_1)}{\partial y_1 \partial y_2}.
\end{aligned}
$$

$$\frac{\partial^2 G(y_1, y_1)}{\partial y_1 \partial y_2} = 0,$$

and hence

$$g(y_1, y_2) = \frac{\partial^2 P(Y_{(1)} \le y_1, y_1 < Y_{(2)} \le y_2)}{\partial y_1 \partial y_2}.$$

As a result of this above equality, the joint pdf of $Y_{(1)}$ and $Y_{(2)}$ can be obtained by taking partial derivatives of the quantity $P(Y_{(1)} \leq y_1, y_1 < Y_{(2)} \leq y_2)$ with respect to y_1 and y_2. This fact is true for general n (as can be readily verified), so that

$$g(y_1, \ldots, y_n) = \frac{\partial^n}{\partial y_1 \cdots \partial y_n} \mathbb{P}\big(Y_{(1)} \leq y_1, \ldots, Y_{(n)} \leq y_n\big)\,,$$

for $0 < y_1 < \cdots < y_n < \infty$, wherever the derivative exists.

If we now examine the case when $n = 2$ again with $0 < y_1 < y_2 < \infty$, we see that

$$\begin{aligned}
P(Y_{(1)} \leq y_1, y_1 < Y_{(2)} \leq y_2) &= P(Y_{(1)} \leq y_1, y_1 < Y_{(2)} \leq y_2, \{Y_1 < Y_2\} \cup \{Y_1 > Y_2\}) \\
&= P(Y_{(1)} \leq y_1, y_1 < Y_{(2)} \leq y_2, Y_1 < Y_2) \\
&\quad + P(Y_{(1)} \leq y_1, y_1 < Y_{(2)} \leq y_2, Y_1 > Y_2) \\
&= P(Y_1 \leq y_1, y_1 < Y_2 \leq y_2, Y_1 < Y_2) \\
&\quad + P(Y_2 \leq y_1, y_1 < Y_1 \leq y_2, Y_2 < Y_1) \\
&= 2P(Y_1 \leq y_1, y_1 < Y_2 \leq y_2) \quad \text{since } Y_1, Y_2 \text{ are identically distributed} \\
&= 2P(Y_1 \leq y_1)P(y_1 < Y_2 \leq y_2) \quad \text{since } Y_1, Y_2 \text{ are independent rvs} \\
&= 2F(y_1)[F(y_2) - F(y_1)].
\end{aligned}$$

Thus, we subsequently obtain

$$\begin{aligned}
g(y_1, y_2) &= \frac{\partial^2 P(Y_{(1)} \leq y_1, y_1 < Y_{(2)} \leq y_2)}{\partial y_1 \partial y_2} \\
&= \frac{\partial^2}{\partial y_1 \partial y_2} \{2F(y_1)[F(y_2) - F(y_1)]\} \\
&= 2f(y_1)f(y_2), \quad 0 < y_1 < y_2 < \infty.
\end{aligned}$$

To verify that this is indeed a joint pdf, note that

$$\begin{aligned}
\int_0^\infty \int_0^\infty g(y_1, y_2)\, dy_1 dy_2 &= \int_0^\infty 2f(y_2) \int_0^{y_2} f(y_1)\, dy_1 dy_2 \\
&= \int_0^\infty 2f(y_2)F(y_2)\, dy_2.
\end{aligned}$$

Let

$$u = F(y_2) \implies \frac{du}{dy_2} = f(y_2) \implies du = f(y_2)dy_2,$$

so that

$$\int_0^\infty \int_0^\infty g(y_1, y_2) dy_1 dy_2 = 2\int_0^1 u du = 2\left(\frac{u^2}{2}\right)\Big|_{u=0}^{u=1} = 1.$$

Thus, $g(y_1, y_2) = 2f(y_1)f(y_2)$, $0 < y_1 < y_2 < \infty$, is a legitimate joint pdf. As a result, the marginal pdf of $Y_{(1)}$ is given by

$$g_1(y_1) = \int_{y_1}^{\infty} 2f(y_1)f(y_2)dy_2 = 2f(y_1)[1 - F(y_1)], \quad 0 < y_1 < \infty,$$

whereas the marginal pdf of $Y_{(2)}$ is given by

$$g_2(y_2) = \int_{0}^{y_2} 2f(y_1)f(y_2)dy_1 = 2F(y_2)f(y_2), \quad 0 < y_2 < \infty.$$

Remarks:

(1) The above results can be extended beyond the $n = 2$ case. In fact, it can be shown that the joint pdf of $(Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)})$ is given by

$$g(y_1, y_2, \ldots, y_n) = n! \prod_{i=1}^{n} f(y_i), \quad 0 < y_1 < y_2 < \cdots < y_n < \infty, \tag{4.10}$$

the marginal cdf of $Y_{(i)}$, $i = 1, 2, \ldots, n$, is given by

$$G_i(y_i) = P(Y_{(i)} \leq y_i) = 1 - \sum_{j=0}^{i-1} \binom{n}{j} F(y_i)^j [1 - F(y_i)]^{n-j}, \quad 0 \leq y_i < \infty,$$

and the marginal pdf of $Y_{(i)}$ is given by

$$g_i(y_i) = G_i'(y_i) = \frac{n!}{(n-i)!(i-1)!} F(y_i)^{i-1} f(y_i)[1 - F(y_i)]^{n-i}, \quad 0 < y_i < \infty. \tag{4.11}$$

(2) If $\{Y_i\}_{i=1}^n$ happens to be a sequence of iid $U(0, t)$ rvs, then (4.10) and (4.11) simplify to become

$$g(y_1, y_2, \ldots, y_n) = \frac{n!}{t^n}, \quad 0 < y_1 < y_2 < \cdots < y_n < t, \tag{4.12}$$

and

$$g_i(y_i) = \frac{n! y_i^{i-1} (t - y_i)^{n-i}}{(n-i)!(i-1)! t^n}, \quad 0 < y_i < t.$$

With these results in place, we are now in position to state another important result concerning the Poisson process.

Theorem 4.7. *Let $\{N(t), t \geq 0\}$ be a Poisson process at rate λ. Given $N(t) = n$, the conditional joint distribution of the n arrival times is identical to the joint distribution of the n order statistics from the $U(0, t)$ distribution. In other words,*

$$(S_1, S_2, \ldots, S_n) | (N(t) = n) \sim (Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)}),$$

where $\{Y_i\}_{i=1}^n$ is an iid sequence of $U(0, t)$ rvs.

Proof. Fix $0 < s_1 < \cdots < s_n < t$ and put $s_0 = 0$. Consider the event A that exactly one arrival occurs in each interval $(s_{i-1}, s_i]$ for $1 \leq i \leq n$ and that no other arrivals occur before time t. Equivalently,

$$A = \{S_1 \leq s_1,\ s_1 < S_2 \leq s_2, \ldots,\ s_{n-1} < S_n \leq s_n,\ N(t) = n\}.$$

The independence of increments and the Poisson count probabilities imply

$$\begin{aligned}\mathbb{P}(A \mid N(t) = n) &= \frac{\prod_{i=1}^{n}\left[e^{-\lambda(s_i - s_{i-1})}\lambda(s_i - s_{i-1})\right]\ e^{-\lambda(t-s_n)}}{e^{-\lambda t}(\lambda t)^n/n!} \\ &= \frac{n!}{t^n}\prod_{i=1}^{n}(s_i - s_{i-1}).\end{aligned}$$

Taking one derivative with respect to each upper endpoint (or, equivalently, recognizing the joint law of uniform order statistics) yields the conditional joint density

$$f_{S_1,\ldots,S_n|N(t)=n}(s_1, \ldots, s_n) = \frac{n!}{t^n}, \qquad 0 < s_1 < \cdots < s_n < t.$$

Note that the form of this joint pdf agrees with (4.12)

$$\implies (S_1, S_2, \ldots, S_n)|(N(t) = n) \sim (Y_{(1)}, Y_{(2)}, \ldots, Y_{(n)}).$$

□

Remark: What this result essentially implies is that under the condition that n events have occurred by time t in a Poisson process, the unordered event locations are distributed independently and uniformly over the interval $[0, t]$; the ordered arrival times are their order statistics.

Example 4.5.

Cars arrive to a toll bridge according to a Poisson process at rate λ, where each car pays a toll of \$1 upon arrival. Calculate the mean and variance of the total amount collected by time t, *discounted back to time 0* where $\alpha > 0$ is the discount rate per unit time.

Solution.

Let $N(t)$ count the number of cars arriving to the toll bridge by time t where $N(t) \sim POI(\lambda t)$. If S_i denotes the arrival time of the ith car $i \in \mathbb{Z}^+$, then the discounted value (i.e. back to time 0) of \$1 by ith arrival is given by $1 \cdot e^{-\alpha S_i}$ as demonstrated by the following diagram. If we let T represent the discounted total amount collected by time t. Then,

$$T = \sum_{i=1}^{N(t)} e^{-\alpha S_i} \qquad \text{(Goal: Determine } E[T] \text{ and } Var(T)\text{)}.$$

First of all,

$$E[T] = E[E[T \mid N(t)]].$$

Consider,

$$\begin{aligned} E[T \mid N(t) = n] &= E\left[\sum_{i=1}^{n} e^{-\alpha S_i} \,\middle|\, N(t) = n\right] \\ &= E\left[\sum_{i=1}^{n} e^{-\alpha Y_{(i)}}\right] \\ &= E\left[\sum_{i=1}^{n} e^{-\alpha Y_i}\right], \end{aligned}$$

where $Y_i \sim U(0, t)$.

$$\begin{aligned} E[T \mid N(t) = n] &= \sum_{i=1}^{n} E[e^{-\alpha Y_i}] \\ &= nE[e^{-\alpha Y_1}]. \end{aligned}$$

Note:

$$\begin{aligned} E[e^{-\alpha Y_1}] &= \int_0^t e^{-\alpha y_1} \cdot \frac{1}{t}\, dy_1 \\ &= \frac{1}{t}\left(\frac{-e^{-\alpha y_1}}{\alpha}\right)\Big|_{y_1=0}^{y_1=t} \\ &= \frac{1 - e^{-\alpha t}}{\alpha t}. \end{aligned}$$

Thus,

$$E[T \mid N(t) = n] = n\left(\frac{1 - e^{-\alpha t}}{\alpha t}\right),$$

which implies

$$\begin{aligned} E[T \mid N(t)] &= E[T \mid N(t) = n]\Big|_{n=N(t)} \\ &= N(t)\left(\frac{1 - e^{-\alpha t}}{\alpha t}\right). \end{aligned}$$

Hence,

$$\begin{aligned} E[T] &= E\left[N(t)\left(\frac{1 - e^{-\alpha t}}{\alpha t}\right)\right] \\ &= \frac{1 - e^{-\alpha t}}{\alpha t} E[N(t)] = \frac{\lambda}{\alpha}(1 - e^{-\alpha t}). \end{aligned}$$

To find the variance of T, use

$$\begin{aligned} Var(T) &= Var(E[T \mid N(t)]) + E[Var(T \mid N(t))] \\ &= Var\left\{ N(t) \left(\frac{1 - e^{-\alpha t}}{\alpha t} \right) \right\} + E[Var(T \mid N(t))] \\ &= \frac{(1 - e^{-\alpha t})^2}{\alpha^2 t^2} Var(N(t)) + E[Var(T \mid N(t))]. \end{aligned}$$

So,

$$\begin{aligned} Var(T \mid N(t) = n) &= Var\left(\sum_{i=1}^{N(t)} e^{-\alpha S_i} \,\middle|\, N(t) = n \right) \\ &= Var\left(\sum_{i=1}^{n} e^{-\alpha S_i} \,\middle|\, N(t) = n \right) \\ &= Var\left(\sum_{i=1}^{n} e^{-\alpha Y_{(i)}} \right) \\ &= \sum_{i=1}^{n} Var(e^{-\alpha Y_i}) \\ &= n Var(e^{-\alpha Y_1}) \\ &= n \left\{ E[(e^{-\alpha Y_1})^2] - E[e^{-\alpha Y_1}]^2 \right\} \\ &= n \left\{ E[e^{-2\alpha Y_1}] - \left(\frac{1 - e^{-\alpha t}}{\alpha t} \right)^2 \right\}. \end{aligned}$$

$$Var(e^{-\alpha Y_1}) = \frac{1 - e^{-2\alpha t}}{2\alpha t} - \left(\frac{1 - e^{-\alpha t}}{\alpha t} \right)^2.$$

Thus,

$$\begin{aligned} Var(T \mid N(t)) &= Var(T \mid N(t) = n)\Big|_{n=N(t)} \\ &= N(t) \left\{ \frac{1 - e^{-2\alpha t}}{2\alpha t} - \left(\frac{1 - e^{-\alpha t}}{\alpha t} \right)^2 \right\}. \end{aligned}$$

Then,

$$\begin{aligned} Var(T) &= \frac{(1-e^{-\alpha t})^2}{\alpha^2 t^2} Var(N(t)) \\ &\quad + E\left[N(t)\left\{\frac{1-e^{-2\alpha t}}{2\alpha t} - \left(\frac{1-e^{-\alpha t}}{\alpha t}\right)^2\right\}\right] \\ &= \frac{(1-e^{-\alpha t})^2}{\alpha^2 t^2}\lambda t + \left\{\frac{1-e^{-2\alpha t}}{2\alpha t} - \left(\frac{1-e^{-\alpha t}}{\alpha t}\right)^2\right\}\lambda t \\ &= \frac{\lambda}{2\alpha}(1-e^{-2\alpha t}). \end{aligned}$$

Example 4.6.

Satellites are launched at times according to a Poisson process at rate 3 per year. During the past year, it was observed that only two satellites were launched. What is the joint probability that the first of these two satellites was launched in the first 5 months of the year and the second satellite was launched prior to the last 2 months of the year?

Solution. Let $\{N(t), t \geq 0\}$ be the Poisson process at rate $\lambda = 3$. We are interested in:

$$P\left(S_1 \leq \frac{5}{12}, S_2 \leq \frac{10}{12} \middle| N(1) = 2\right)$$

To do so, we use Theorem 4.7, specifically (4.12) to obtain the conditional joint pdf of $(S_1, S_2)|(N(1) = 2)$ as:
$g(s_1, s_2) = \frac{2!}{1^2} = 2$ for $0 < s_1 < s_2 < 1$.

$$\begin{aligned} &P\left(S_1 \leq \frac{5}{12}, S_2 \leq \frac{10}{12} \middle| N(1) = 2\right) \\ &= \int_0^{\frac{5}{12}} \int_{s_1}^{\frac{10}{12}} 2\, ds_2\, ds_1 \\ &= 2\int_0^{\frac{5}{12}} \left(\frac{10}{12} - s_1\right) ds_1 \\ &= 2\left(\frac{10s_1}{12} - \frac{s_1^2}{2}\right)\Big|_{s_1=0}^{s_1=\frac{5}{12}} \\ &= \frac{25}{48} \approx 0.521. \end{aligned}$$

4.5 Two Important Generalizations

4.5.1 The Non-homogeneous Poisson Process

Oftentimes, we find the Poisson process difficult to apply in applications of real-life phenomena, largely due to the fact that the Poisson process assumes a constant arrival rate of λ for all time. In what follows, we consider a more general type of process in which the arrival rate is allowed to vary as a function of time.

Definition 4.8. The counting process $\{N(t), t \geq 0\}$ is a *non-homogeneous* (or *non-stationary*) Poisson process with rate function $\lambda(t)$ if the following three conditions hold true:

1. $\{N(t), t \geq 0\}$ has independent increments.
2. $P(N(t+h) - N(t) = 1) = h\lambda(t) + o(h)$.
3. $P(N(t+h) - N(t) \geq 2) = o(h)$.

Many applications that generate random points in time are modelled more realistically with non-homogeneous processes. For instance, the rate of customers entering a supermarket is not the same during the entire day. Similarly, the average arrival rate of vehicles on a highway fluctuates between its maximum during rush hours and its minimum during low traffic times. The mathematical cost of this generalization, however, is that we lose the stationary increments property.

For $s_1, s_2 \geq 0$, the following theorem specifies the distribution of $N(s_1 + s_2) - N(s_1)$.

Theorem 4.8. *If $\{N(t), t \geq 0\}$ is a non-homogeneous Poisson process with rate function $\lambda(t)$, then $N(s_1 + s_2) - N(s_1) \sim POI(m(s_1 + s_2) - m(s_1))$ where the mean value function $m(t)$ is given by*

$$m(t) = \int_0^t \lambda(\tau) d\tau, \quad t \geq 0.$$

Proof. Let $\phi_u(s_1, s_2) = E[e^{u\{N(s_1+s_2)-N(s_1)\}}]$ denote the mgf of $N(s_1 + s_2) - N(s_1)$, where u serves as the argument of the mgf. For $h \geq 0$, we first note that

$$\begin{aligned}
\phi_u(s_1, s_2 + h) &= E[e^{u\{N(s_1+s_2+h)-N(s_1)\}}] \\
&= E[e^{u\{N(s_1+s_2+h)-N(s_1+s_2)+N(s_1+s_2)-N(s_1)\}}] \\
&= E[e^{u\{N(s_1+s_2+h)-N(s_1+s_2)\}} e^{u\{N(s_1+s_2)-N(s_1)\}}] \\
&= E[e^{u\{N(s_1+s_2+h)-N(s_1+s_2)\}}] E[e^{u\{N(s_1+s_2)-N(s_1)\}}] \text{ due to independent increments} \\
&= \phi_u(s_1 + s_2, h)\phi_u(s_1, s_2). \qquad (4.13)
\end{aligned}$$

Applying a similar approach to that used in the proof of Theorem 4.3, it can be shown that (4.13) ultimately gives rise to the first-order differential equation (see Exercise 4.4.1)

$$\begin{aligned}
\frac{d}{ds_2}\phi_u(s_1,s_2) &= \lambda(s_1+s_2)(e^u-1)\phi_u(s_1,s_2)\\
\frac{d}{dt}\phi_u(s_1,t) &= \lambda(s_1+t)(e^u-1)\phi_u(s_1,t) \text{ if we let } s_2=t\\
\frac{\frac{d}{dt}\phi_u(s_1,t)}{\phi_u(s_1,t)} &= \lambda(s_1+t)(e^u-1)\\
\frac{d}{dt}(\ln\phi_u(s_1,t)) &= \lambda(s_1+t)(e^u-1)\\
d(\ln\phi_u(s_1,t)) &= (e^u-1)\lambda(s_1+t)dt\\
\int_0^{s_2} d(\ln\phi_u(s_1,t)) &= (e^u-1)\int_0^{s_2}\lambda(s_1+t)dt\\
\ln\phi_u(s_1,t)\Big|_{t=0}^{t=s_2} &= (e^u-1)\int_0^{s_2}\lambda(s_1+t)dt\\
\ln\phi_u(s_1,s_2)-\ln\phi_u(s_1,0) &= (e^u-1)\int_{s_1}^{s_1+s_2}\lambda(\tau)d\tau.
\end{aligned}$$

However, $\phi_u(s_1,0) = E[e^{u\{N(s_1)-N(s_1)\}}] = E[e^0] = 1$, and so

$$\begin{aligned}
\ln\phi_u(s_1,s_2) &= (e^u-1)\int_{s_1}^{s_1+s_2}\lambda(\tau)\,d\tau,\\
\phi_u(s_1,s_2) &= \exp\left\{\left(\int_{s_1}^{s_1+s_2}\lambda(\tau)\,d\tau\right)(e^u-1)\right\}, \quad u\in\mathbb{R}.
\end{aligned}$$

Since

$$\begin{aligned}
\int_{s_1}^{s_1+s_2}\lambda(\tau)\,d\tau &= \int_0^{s_1+s_2}\lambda(\tau)\,d\tau - \int_0^{s_1}\lambda(\tau)\,d\tau\\
&= m(s_1+s_2)-m(s_1),
\end{aligned}$$

we recognize $\phi_u(s_1,s_2)$ as the mgf of a $POI(m(s_1+s_2)-m(s_1))$ rv. By the mgf uniqueness property, $N(s_1+s_2)-N(s_1)\sim POI(m(s_1+s_2)-m(s_1))$. □

Remarks:

(1) As a direct consequence of Theorem 4.8, for all $s,t\geq 0$, we have

$$P(N(s+t)-N(s)=n) = e^{-[m(s+t)-m(s)]}\frac{[m(s+t)-m(s)]^n}{n!}, \quad n=0,1,2,\ldots$$

(2) If the rate function $\lambda(\tau)=\lambda\ \forall\tau\geq 0$, then note that $\int_s^{s+t}\lambda(\tau)d\tau = \int_s^{s+t}\lambda d\tau = \lambda(s+t-s) = \lambda t$ and $P(N(s+t)-N(s)=n) = \frac{e^{-\lambda t}(\lambda t)^n}{n!}$, $n=0,1,2,\ldots$ This is as expected, since $\{N(t), t\geq 0\}$ simplifies to become the standard (i.e., stationary) Poisson process.

Example 4.7.

Requests for technical support within the statistics department occur according to a non-homogeneous Poisson process $\{N(t), t \geq 0\}$ having rate function

$$\lambda(t) = \begin{cases} 5t/2, & \text{if } 0 \leq t < 1, \\ t^2/2 + 2, & \text{if } 1 \leq t < 4, \\ (9-t)(t-2), & \text{if } 4 \leq t \leq 9, \end{cases}$$

where t is measured in hours from the start of the workday. What is the probability that four requests occur in the first two hours of the workday and ten more occur in the final two hours of the workday?

Solution. we wish to calculate

$$\begin{aligned} &P(N(2) = 4, N(9) - N(7) = 10) \\ &\qquad = P(N(2) = 4)P(N(9) - N(7) = 10) \quad \text{by indep. increments.} \end{aligned}$$

We need

$$\begin{aligned} m(2) - m(0) &= \int_0^2 \lambda(t)\, dt \\ &= \int_0^1 \frac{5}{2} t\, dt + \int_1^2 \left(\frac{t^2}{2} + 2\right) dt \\ &= \left(\frac{5}{4}t^2\right)\Big|_{t=0}^{t=1} + \left(\frac{t^3}{6} + 2t\right)\Big|_{t=1}^{t=2} \\ &= \frac{5}{4} + \left(\frac{8}{6} + 4 - \frac{1}{6} - 2\right) = \frac{53}{12}, \\ m(9) - m(7) &= \int_7^9 \lambda(t)\, dt \\ &= \int_7^9 (9-t)(t-2)\, dt \\ &= \int_7^9 (-18 + 11t - t^2)\, dt \\ &= \left(-18t + \frac{11}{2}t^2 - \frac{t^3}{3}\right)\Big|_{t=7}^{t=9} \\ &= \frac{38}{3} \end{aligned}$$

As a result, we obtain:

$$\begin{aligned} &P(N(2) = 4, N(9) - N(7) = 10) \\ &\qquad = \frac{e^{-53/12}(53/12)^4}{4!} \cdot \frac{e^{-38/3}(38/3)^{10}}{10!} \\ &\qquad \approx 0.0177. \end{aligned}$$

4.5.2 The Compound Poisson Process

Another restriction we might wish to relax concerns the assumption that arrivals in a Poisson process occur strictly one after the other. For instance, vehicles crossing the Canada-USA border would usually have more than just one passenger on board. Individuals arriving to a sporting event often arrive in groups of two or more. There are also applications where each arrival might generate a random monetary amount. For example, in an insurance company where claims occur according to a Poisson process, the claim sizes themselves are (random) amounts of money, and one would typically be interested in the total amount of money paid out by the insurance company by time t. In any of the above scenarios, each arrival in a Poisson process comes with an associated real-valued rv that represents the *value* of the arrival in a sense. This gives rise to the following definition.

Definition 4.9. Let $\{Y_i\}_{i=1}^{\infty}$ be an iid sequence of rvs. Let $\{N(t), t \geq 0\}$ be a Poisson process at rate λ, independent of each Y_i, $i = 1, 2, 3, \ldots$. If

$$X(t) = \sum_{i=1}^{N(t)} Y_i,$$

then the process $\{X(t), t \geq 0\}$ is a **compound Poisson process**.

Remarks.

(1) The above definition is another way of generalizing the Poisson process, since if we choose the distribution of each Y_i to be degenerate at 1 (i.e., $Y_i = 1$ with probability 1 $\forall i \in \mathbb{Z}^+$), then $\{X(t), t \geq 0\}$ and $\{N(t), t \geq 0\}$ are identical processes.

(2) The compound Poisson process $\{X(t), t \geq 0\}$ inherits the independent and stationary increments properties from the Poisson process $\{N(t), t \geq 0\}$. To see this formally, let $(s_1, t_1]$ and $(s_2, t_2]$ be time intervals such that $t_1 \leq s_2$ (resulting in $(s_1, t_1] \cap (s_2, t_2] = \varnothing$).

Making use of the above assumptions concerning $\{Y_i\}_{i=1}^{\infty}$ and $\{N(t), t \geq 0\}$, note that

$$
\begin{aligned}
&P(X(t_1)-X(s_1)\le a_1, X(t_2)-X(s_2)\le a_2)\\
&\quad= P\left(\sum_{i=1}^{N(t_1)} Y_i - \sum_{i=1}^{N(s_1)} Y_i \le a_1, \sum_{i=1}^{N(t_2)} Y_i - \sum_{i=1}^{N(s_2)} Y_i \le a_2\right)\\
&\quad= P\left(\sum_{i=N(s_1)+1}^{N(t_1)} Y_i \le a_1, \sum_{i=N(s_2)+1}^{N(t_2)} Y_i \le a_2\right)\\
&\quad= \sum_{m_1=0}^{\infty}\sum_{n_1=0}^{\infty}\sum_{m_2=0}^{\infty}\sum_{n_2=0}^{\infty} P\left(\sum_{i=N(s_1)+1}^{N(t_1)} Y_i \le a_1, \sum_{i=N(s_2)+1}^{N(t_2)} Y_i \le a_2 \,\middle|\, \begin{matrix} N(s_1)=m_1, N(t_1)-N(s_1)=n_1,\\ N(s_2)-N(t_1)=m_2, N(t_2)-N(s_2)=n_2\end{matrix}\right)\\
&\qquad\times P(N(s_1)=m_1, N(t_1)-N(s_1)=n_1, N(s_2)-N(t_1)=m_2, N(t_2)-N(s_2)=n_2)\\
&\quad= \sum_{m_1=0}^{\infty}\sum_{n_1=0}^{\infty}\sum_{m_2=0}^{\infty}\sum_{n_2=0}^{\infty} P\left(\sum_{i=m_1+1}^{m_1+n_1} Y_i \le a_1, \sum_{i=m_1+n_1+m_2+1}^{m_1+n_1+m_2+n_2} Y_i \le a_2\right)\\
&\qquad\times P(N(s_1)=m_1)P(N(t_1)-N(s_1)=n_1)P(N(s_2)-N(t_1)=m_2)P(N(t_2)-N(s_2)=n_2)\\
&\quad= \sum_{m_1=0}^{\infty}\sum_{n_1=0}^{\infty}\sum_{m_2=0}^{\infty}\sum_{n_2=0}^{\infty} P\left(\sum_{i=m_1+1}^{m_1+n_1} Y_i \le a_1\right) P\left(\sum_{i=m_1+n_1+m_2+1}^{m_1+n_1+m_2+n_2} Y_i \le a_2\right)\\
&\qquad\times P(N(s_1)=m_1)P(N(t_1)-N(s_1)=n_1)P(N(s_2)-N(t_1)=m_2)P(N(t_2)-N(s_2)=n_2)\\
&\quad= \left\{\sum_{n_1=0}^{\infty} P\left(\sum_{i=1}^{n_1} Y_i \le a_1\right) P(N(t_1-s_1)=n_1)\right\}\left\{\sum_{n_2=0}^{\infty} P\left(\sum_{i=1}^{n_2} Y_i \le a_2\right) P(N(t_2-s_2)=n_2)\right\}\\
&\quad= P\left(\sum_{i=1}^{N(t_1-s_1)} Y_i \le a_1\right) P\left(\sum_{i=1}^{N(t_2-s_2)} Y_i \le a_2\right)\\
&\quad= P(X(t_1-s_1)\le a_1)P(X(t_2-s_2)\le a_2).
\end{aligned}
$$

(3) For $t > 0$, determining the probability distribution of $X(t)$ is, generally speaking, a challenging mathematical problem. On the other hand, the mean and variance of $X(t)$ are readily accessible. In particular, by making use of the results for the mean and variance of a random sum (see Example 2.9), we immediately obtain

$$E[X(t)] = E[N(t)]E[Y_1] = \lambda t E[Y_1]$$

and

$$\mathrm{Var}(X(t)) = \mathrm{Var}(Y_1)E[N(t)] + E[Y_1]^2\,\mathrm{Var}(N(t)) = \lambda t(\mathrm{Var}(Y_1) + E[Y_1]^2) = \lambda t E[Y_1^2].$$

Example 4.8.

Claims received by an insurance company occur according to a Poisson process at a rate of 30 claims per year. Individual claim amounts, which are assumed to be independent, are known to be either \$1,000, \$2,000, or \$3,000. Company records indicate that one year ago, the average total amount paid out was \$56,000 and that the standard deviation of the total amount paid out was \$11,000. Based on this information, how likely was it that an individual claim of \$3,000 took place last year?

Solution. Let $N(t)$ be the # of claims received by time t (in years) which is known to have a POI$(30t)$ distribution.

In addition, let X_i, $i \geq 1$ be the size of the ith individual claim measured in thousands of \$ having pmf:

$$P(X = y) = \begin{cases} \alpha & y = 1 \\ \beta & y = 2 \\ 1 - \alpha - \beta & y = 3 \end{cases}$$

Thus $X(t) = \sum_{i=1}^{N(t)} X_i$ is the total amount paid out measured in thousands of dollars by time t.

We have:

$$\begin{aligned} E[X(t)] &= 30E[X_1] = 56 \\ 30[\alpha + 2\beta + 3(1 - \alpha - \beta)] &= 56 \\ 30\alpha + 60\beta + 90 - 90\alpha - 90\beta &= 56 \\ 30\alpha + 15\beta &= 17 \qquad (4.14) \\ \mathrm{Var}(X(t)) &= 30E[X_1^2] = 11^2 \\ &\Rightarrow 30[\alpha + 4\beta + 9(1 - \alpha - \beta)] = 121 \\ 240\alpha + 150\beta &= 149. \qquad (4.15) \end{aligned}$$

Solving (4.14) and (4.15) gives $\alpha = \frac{7}{20}$. Plugging this value into (4.14) leads to

$$30 \cdot \frac{7}{20} + 15\beta = 17 \Rightarrow \beta = \frac{13}{30}$$

Finally $1 - \alpha - \beta = 1 - \frac{7}{20} - \frac{13}{30} = \frac{13}{60}$.

Let $N_3(t)$ denote the number of \$3000 claims received by time t.

$$N_3(t) \sim \mathrm{POI}\left(30 \cdot \frac{13}{60}\right),$$

$$\begin{aligned} &P(\text{individual claim of \$3000 took place last year}) \\ &\qquad = 1 - P(N_3(1) = 0) \\ &\qquad = 1 - \frac{e^{-13/2}(13/2)^0}{0!} \\ &\qquad = 1 - e^{-13/2} \approx 0.998. \end{aligned}$$

5 Synthesis: Connections Between the Four Chapters

Overview. This final section collects the principal structural relationships developed in the notes. It is intended as a conceptual bridge, not a replacement for the detailed derivations and examples in the chapters.

5.1 Conditioning as the Common Computational Device

The law of total probability and the tower property are discrete and continuous expressions of the same principle: average a conditional law over the distribution of the conditioning information. For a countable random variable Y and an integrable X,

$$\mathbb{E}[X] = \sum_y \mathbb{E}[X \mid Y = y]\mathbb{P}(Y = y),$$
$$\mathbb{P}(A) = \sum_y \mathbb{P}(A \mid Y = y)\mathbb{P}(Y = y).$$

For a continuously distributed Y with density f_Y, analogous formulas are

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \mathbb{E}[X \mid Y = y] f_Y(y)\,\mathrm{d}y,$$
$$\mathbb{P}(A) = \int_{-\infty}^{\infty} \mathbb{P}(A \mid Y = y) f_Y(y)\,\mathrm{d}y.$$

The Chapman–Kolmogorov equations are the same operation with the intervening state of the Markov chain as the conditioning variable.

5.2 Invariance, Recurrence, and Long-Run Behavior

Definition 5.1 (Invariant distribution)**.** For a time-homogeneous Markov chain with transition matrix P, a probability row vector π is *invariant* if $\pi P = \pi$.

If $X_0 \sim \pi$, then for every state j,

$$\begin{aligned}\mathbb{P}(X_1 = j) &= \sum_i \mathbb{P}(X_1 = j \mid X_0 = i)\mathbb{P}(X_0 = i) \\ &= \sum_i \pi_i p_{ij} = (\pi P)_j = \pi_j.\end{aligned}$$

Repeating the same argument shows $X_n \sim \pi$ for each n. An invariant law is therefore a distributional fixed point. In contrast, the convergence of P^n to a matrix with identical rows requires additional recurrence and aperiodicity assumptions; stationarity alone does not imply convergence from every initial state.

5.3 Poisson Counts as a Continuous-Time Analogue of Transitions

For a homogeneous Poisson process at rate $\lambda > 0$, independent and stationary increments imply

$$\mathbb{P}(N(t+s) = j \mid N(t) = i) = \begin{cases} e^{-\lambda s}\dfrac{(\lambda s)^{j-i}}{(j-i)!}, & j \geq i, \\ 0, & j < i. \end{cases}$$

Consequently $N(t)$ is a continuous-time Markov chain on $\mathbb{N}_0$. If $P_s(i,j)$ denotes the displayed transition probability, the independent increment decomposition gives

$$P_{s+t}(i,j) = \sum_{k=i}^{j} P_s(i,k) P_t(k,j),$$

which is the continuous-time version of the Chapman–Kolmogorov relation. The essential distinction is that elapsed time t is continuous rather than an integer step count.

5.4 Scope of These Notes

The chapters and original worked examples retain the core STAT 333 treatment of elementary probability, conditional laws, discrete-time Markov chains, and Poisson-process constructions. The preliminary frameworks and synthesis material clarify assumptions and connect results using the language of measurable random variables and conditional expectations.